

ACOUSTIC CAVITATION

E.A. NEPPIRAS

Consultant, 17, Kinsbourne Avenue, Bournemouth, England



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Abstract:

This article reviews the physics of cavitation generated by acoustic fields, covering the basic theoretical and experimental data needed for a proper understanding of the many effects of cavitation and their practical applications. The dynamics of bubble motion are developed, stressing the relation between stable and transient types of cavitation and their thresholds. Non-radial types of bubble motion are now known to initiate several important effects, including erosive action, and these are dealt with in detail. Direct verification of theories of bubble dynamics is obtained using high-speed cinematography. The most recent techniques and results are described, as are the highly-sophisticated experimental methods now being applied to the bubble-fields and aggregates, including the acoustic emission from stable and transient cavitation fields.

List of symbols

Only symbols that appear frequently are listed. The first column gives the meanings which they usually have.

a	Perturbation amplitude	
c	Velocity of sound	A real number
f	Frequency	
f_r	Resonance frequency	
h	Specific enthalpy $\int dp/\rho$	
k	Wave number $2\pi/\lambda$	An angular frequency Boltzmann's constant
m	Mass	A numeric
n	A numeric	
p	Pressure	Dimensionless pressure, P_A/P_0
$p_{\max}$	Maximum collapse pressure	
q	Heat flux (energy per unit mass per unit time)	
r	Radial distance, radial displacement	Dimensionless radius R/R_0 or R/R_r
r_m	Maximum value of r	
t	Time	
x	Space co-ordinate	
y	Space co-ordinate	
B	Damping coefficient	
C	Concentration of dissolved gas in liquid	Velocity of sound at the bubble wall
C_0	Saturation concentration of dissolved gas in liquid	
C_p	Specific heat at constant pressure (gas)	
C_v	Specific heat at constant volume (gas)	
D	Diffusivity (mass- or thermal) of gas-phase	
E	Energy distribution function	
H	Enthalpy difference between the bubble wall and infinity	An auxiliary function
K	Thermal conductivity	
L	Diffusion length $= (2D/\omega)^{1/2}$	Latent heat of vaporisation; Length
M	Acoustical Mach number $= \dot{R}/c$	
N	The gas constant	Function describing the number-distribution
P	Pressure	

P_A	Acoustic pressure amplitude	
P_m	Liquid pressure at transient collapse	
P_L	Liquid pressure just outside the bubble wall	
P_T	Threshold acoustic pressure	
$P_{\max}$	Maximum gas pressure in collapsing bubble	
Q	Gas pressure in bubble at its maximum size	Quality factor = δ^{-1}
		Quantity of heat
R	Bubble radius	Reflection coefficient
R_m	Maximum bubble radius	R_e acoustic Reynolds No.
$R_{\min}$	Minimum bubble radius	
R_r	Resonance radius of bubble	
R_T	Threshold radius of bubble	
R', R''	Principal radii of curvature	
S_p	Specific heat at constant pressure for liquid	
S_v	Specific heat at constant volume for liquid	
T	Temperature	Acoustic period = $2\pi/\omega$
$T_{\max}$	Maximum gas temperature	
U	Internal energy per unit mass	
V	Particle velocity of liquid	Gas volume
V_0	Volume of cavitation zone	
W	Total power associated with cavitation losses	
Y_n	Spherical harmonic of degree n	
Z	Volume compression ratio	
α	Henry's coefficient for a weak solution	Accommodation coefficient
		$R_0/L_g; 2\sigma/R_0P_0; \omega R/\omega_r R_0$
β	$\omega/\omega_0; (\rho\omega^2 R^2/3\gamma P_0)^{1/2}$	
γ	Ratio of specific heats for gas	
δ	A small quantity $\ll 1$	Damping factor; $(P_0/\rho\omega^2 R_0^2)$
δ_r	Radiation damping factor	
δ_t	Thermal damping factor	
δ_v	Viscous damping factor	
ϵ	A small quantity $\ll 1$	
ξ	Oscillatory amplitude	
ξ_T	Threshold oscillatory amplitude	
λ	Acoustic wave-length	
λ_s	Wave-length of surface waves	
μ	Shear viscosity of liquid	
Π	Period of non-linear vibrations	
ρ	Density of liquid	
σ	Surface tension	
τ	Non-dimensional time = ωt	$(3\gamma P_0/\rho)^{1/2}(t/2\pi R_0)$
ϕ	Velocity potential in liquid	Entropy
ψ	$2\sigma/R_0P_0$	
χ	$4\pi^2/3\gamma$	
ω	Angular frequency	

ω_r Resonant radial angular frequency
 Γ Polytropic index for gas

Suffix 0 refers to the Equilibrium value
 g refers to the Permanent-gas phase
 v refers to the Vapour phase
 T refers to the Total gas-content
 ∞ refers to the conditions at a great distance

Prefix Δ refers to a small increment

1. Introduction

It is difficult to give an adequate brief definition of cavitation. Some would say that cavitation occurs whenever a new surface is created in the body of a liquid. This broad definition would include such phenomena as boiling and mere effervescence. In the presence of a sound field not only expansion, but also contraction, of existing cavities will generally occur. The term “acoustic cavitation” may therefore be restricted to cases where both expansion and contraction phases are present.

In acoustic cavitation, a time-varying, generally sinusoidal, pressure is superimposed on the steady ambient pressure. The effect on the motion of cavities may be violent, or rather gentle. In fact, we easily recognise a rather sharp distinction between two types of cavitation, called *Stable* and *Transient*. Stable cavities are bubbles that oscillate, often non-linearly, around some equilibrium size. They are relatively permanent and may continue oscillating for many cycles of the acoustic pressure. On the other hand, transient cavities generally exist for less than one cycle. During this one cycle, they expand to *at least double*, and often to many times, their original size. Then they collapse violently, often disintegrating into a mass of smaller bubbles.

Either type of cavity may contain either permanent-gas (generally air) or vapour. We may therefore define four simple models for the free cavity: the gas- or vapour-filled stable or transient cavity. In a transient cavity it is usual to assume that there is no time for any mass-flow, by diffusion, of permanent-gas into or out of the bubble, whereas condensation and evaporation of vapour may occur more or less freely. So, for example, a collapsing gaseous transient cavity is assumed to have a constant permanent-gas content over its lifetime, while a vaporous transient cavity contains only vapour which will often remain at or near its constant equilibrium pressure. The collapse of a vaporous transient may therefore be very violent, as there is no residual permanent-gas to cushion the implosion. With any type of stable cavity, the time-scale is typically so long that mass-diffusion of permanent-gas as well as thermal diffusion, with consequent evaporation-condensation of vapour, can occur, resulting in significant long-term effects. If the liquid is entirely gas-free, so that any cavities must be vaporous, it might be thought that stable cavities could not persist; for it can be very easily shown that under a steady ambient pressure, spherical cavities containing only the liquid vapour must be unstable. However, as we shall see later, there is a thermal mechanism whereby vaporous cavities can achieve stability in the presence of a sound field. In fact, under suitable conditions, stable pure-vapour cavities are very plentiful. Therefore, we conclude that all four of the proposed models can exist in practice.

So far, we have defined models for the behaviour of isolated cavities. In practice, cavities are rarely produced in isolation. We have to contend with *bubble fields*, consisting of many bubbles interacting

with one another and undergoing translatory, as well as radial motions in the sound field. In what follows, we shall deal first with isolated spherical cavities, and later with their interactions in the field.

In the early days of cavitation research, workers were mainly preoccupied with what we have defined above as *transient* cavitation. Interest was centred on the spectacular disruptive effects of collapsing transients, effects such as: erosion, emulsification, molecular degradation, sono-luminescence, sono-chemical and biological effects. These are all related to the very high pressures and temperatures developed in the transient implosion. But the stable types of cavitation are now known to be at least as important as transients. They cover a wider range of motions, and some of the side-effects of stable cavitation are very important. These include the initiation of surface oscillations and microstreaming. Also, in any general cavitation field, the great majority of visible bubbles will be oscillating stably, and since stable cavities by their very nature are long-lived, their integrated effect can be substantial. A further relevant fact is that stable cavities often evolve into transients in the course of time. This they do by a process of mass- or heat-transfer which results in bubble-growth.

In practice, the distinction between the stable and transient forms of cavitation is not always quite clear-cut. The transition: stable $\rightarrow$ transient may occur through the mere passage of time or through changes in the acoustical or environmental conditions. The demarcation areas between stable and transient states, sometimes rather vague, are known as *Transient Cavitation Thresholds*. There is also another threshold region, lying below the transient thresholds, which defines the point where stable cavities can grow to become transients. This may be termed the *Stable Cavitation Threshold*. Study of these thresholds and the relation between them, is an active area of research.

Our definition of acoustic cavitation implies that cavities are in radial motion. Surface oscillations and instabilities, and all forms of translatory motion, are strictly excluded from the definition as they do not involve radial motion. However, we now know that these non-radial motions are often initiated by stable cavitation, and therefore closely linked with it. In any case, all of these motions may be important in practical applications, so we will deal with them in some detail.

As soon as cavitation starts in a liquid, its acoustic properties have changed, and it has become, in effect, a different medium, acoustically lossier and more compressible. The properties of the new medium will control the course of subsequent events. The cavitating liquid will generally be appreciably non-linear. The concept of "acoustical impedance" then loses its validity. Nevertheless, it is still useful to refer to an "effective" acoustical impedance for the cavitating medium. We may then say that the onset of cavitation will generally increase the effective acoustic resistance and compliance of the medium.

It is the high compressibility of the gas bubbles relative to the liquid that permits the wide range of motions seen in cavitation fields. Oscillating bubbles are very effective in redistributing and dissipating the acoustic energy. Stable and transient cavities behave differently in this respect. With transient cavities, great concentration of energy occurs towards the end of the implosion. In effect, energy is extracted from the sound beam (that is, from the transmissible form) and rapidly concentrated into small volumes: energy from progressive waves is transformed to spherical shocks emanating from point sources. But despite the large pressures involved, the effective range of action of the shocks is typically only a few radial distances of the collapsed cavity. We may contrast this with stably-oscillating resonant bubbles. The absorption-scattering cross-section for such a bubble is enormously greater than for a liquid or solid particle of comparable size. The sound beam is therefore effectively scattered and little coherent signal is passed through a region of even mild cavitation.

It is interesting to record that until about 15 years ago almost all theoretical work in acoustic cavitation was concerned with the dynamics of isolated cavities. This problem can now be regarded as

completely solved for bubbles that remain spherical. At the same time, most of the experimental work was concerned with natural bubble-aggregates, uncontrolled bubble-fields. Recently, this approach has been largely reversed. Theoretical work is being concentrated on non-spherical bubbles and on conditions for instabilities to form and develop. Experimentally, the dynamics of single bubbles or controlled bubble-fields is now being actively studied, often using high-speed photographic methods.

Published material on acoustic cavitation is now very extensive, and good papers are still appearing. A number of useful reviews are available covering work published up to the late 1960's [1–5]. These also contain excellent reference lists. The present review aims to bring the story up to date. The past decade has seen new developments in bubble dynamics, especially relating to pure-vapour cavities; detailed analyses of bubble instabilities; much very high-speed photographic work; new work relating to thresholds and cyclic cavitation processes; and development of techniques for controlling and studying bubble fields. Partly through lack of space, we omit any discussion of such topics as: the effects, applications and assessment of cavitation; and some specialist and peripheral topics such as: cavitation in cryogenic liquids; nucleation mechanisms, etc. However, adequate references are quoted in all cases.

2. The equations of cavitation bubble dynamics

Interest in cavitation stretches back about 120 years [6] but the important pioneer theoretical paper is that of Rayleigh, 1917 [7] which describes the collapse of a spherical cavity. Until the late 1940's all theoretical work was concerned with hydrodynamically-generated cavities, culminating in the important paper by Plesset [8]. The first systematic treatment of acoustically-generated cavities was by Blake in 1949 [9], followed by Noltingk and Neppiras, 1950–51 [10, 11]. Since then, many groups have become active.

The basic dynamical problem of acoustic cavitation is to determine the pressure and velocity fields in the two-fluid medium, together with the motion of the bubble wall, when under the influence of a time-dependent (acoustic) pressure. A related problem sometimes of interest concerns the temperature field within the bubble.

Even in the most general case, where the cavity is not spherical, but may assume any arbitrary shape, it is not difficult to formulate the set of equations defining the problem [see, for example, refs. 5 and 12]. But the exercise is purely academic, as no progress is possible without making very drastic assumptions concerning an initial shape and the forces responsible for it. In the simplest case, where the cavity is spherical and remains spherical always – which, fortunately, is the one of most practical interest – the problem is greatly simplified. When, later, we examine conditions for surface instability, the approach will be to imagine an originally spherical surface to undergo an arbitrary perturbation, then examine the conditions under which it would grow.

Our model for the cavity is therefore a spherical bubble, isolated in a liquid that extends to infinity. With spherical symmetry, every physical quantity is a function of only one space co-ordinate, the radial distance, r , from the bubble centre, taken as the origin. The acoustic pressure, superimposed on a steady ambient pressure, is applied at a great distance from the bubble. The equations for this simple model can be formulated to take account of non-uniform conditions (that is, variable pressure, velocity and temperature fields) in both the gas and liquid phases.

The problem to be solved is therefore a two-phase hydrodynamical one in which the two phases are coupled through a moving boundary (the bubble wall). It is of the class of free boundary-value problems, as the position of the boundary will not be known until the final solution is obtained. The dynamical problem is more or less complicated depending on the degree of coupling assumed between

the two phases. Apart from the purely mechanical coupling, mass- and heat-transfer may take place across the bubble wall. The mass-flow may comprise two elements: diffusion of permanent-gas and evaporation-condensation of liquid, which will itself depend on the thermal conditions.

As with any hydrodynamical problem, we obtain general relations by applying the laws of conservation of *Mass*, *Momentum* and *Energy*. Conservation of mass, leading to equations of continuity, can be taken to hold separately in the liquid and gas phases and – as a boundary condition – across the bubble wall, since only at the boundary does any mass-exchange occur. Conservation of momentum applies at any point in either phase and at the bubble wall. Energy conservation must be applied to the complete coupled system. Apart from the moving boundary, a boundary condition exists at $t = 0$, $r = \infty$. In addition to the conservation laws, the set of relevant equations will include the physical laws needed to define elements of the energy equation. These are the laws relating to mass- and thermal-diffusion, evaporation-condensation and dissolution. Finally, the set is completed by the Equations of State for the liquid and gas phases, these being relations between pressure (p) and density (ρ) of the form $p = p(\rho)$.

The reader is referred to ref. [5], for example, for the complete set of equations defining this general case. Some numerical solutions have even been obtained, after some simplifying assumptions [13]. However, instances where the flow fields within the bubble are of interest, or need to be evaluated, are quite limited. On the other hand, great simplification is achieved by assuming spatial uniformity of pressure, velocity and temperature within the bubble during the motion. In what follows, we shall therefore assume that the physical quantities relating to the bubble contents are functions of time only. Physically, this amounts to assuming that the bubble wall velocity, $\dot{R}$, is at all times small compared with the rates at which irregularities smooth themselves out within the bubble. But even when this condition ceases to hold, as near the end of a transient collapse, the simplification is still broadly justified when only the flow fields in the liquid and the motion of the bubble wall are of interest. The relevant equations for this general case are reproduced below. These comprise the equations of Continuity, Motion and Energy, deduced from the conservation laws of Mass, Momentum and Energy, as well as the physical laws of mass- and thermal-diffusion in the liquid-phase. The boundary conditions are to be evaluated at the bubble wall, at $r = R$,

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial r}(\rho V) + \frac{2\rho V}{r} = 0 \quad \text{Continuity (liquid)} \quad (1)$$

$$\frac{\partial V}{\partial t} + V \frac{\partial V}{\partial r} = -\frac{1}{\rho} \frac{\partial p}{\partial r} \quad \text{Motion (liquid)} \quad (2)$$

$$\rho \left(\frac{\partial U}{\partial t} + V \frac{\partial U}{\partial r} \right) = -p \left(\frac{\partial V}{\partial r} + \frac{2V}{r} \right) + \frac{4\mu}{3} \left(\frac{\partial V}{\partial r} - \frac{V}{r} \right)^2 + K_L \left(\frac{\partial^2 T}{\partial r^2} + \frac{2}{r} \frac{\partial T}{\partial r} \right) + \rho q \quad \text{Energy (liquid)} \quad (3)$$

$$\frac{\partial C}{\partial t} + V \frac{\partial C}{\partial r} = D \left(\frac{\partial^2 C}{\partial r^2} + \frac{2}{r} \frac{\partial C}{\partial r} \right) \quad \text{Gas-diffusion in liquid} \quad (4)$$

with boundary conditions at $r = R(t)$:

$$D \frac{\partial C}{\partial r} = \frac{1}{4\pi R^2} \frac{d}{dt} (4\pi R^3 \rho_g) \quad (5)$$

$$\left(p + \frac{2\sigma}{R}\right) = p_T + \frac{4\mu}{3} \left(\frac{\partial V}{\partial r} - \frac{V}{r}\right) \quad (6)$$

$$K_L \frac{\partial T}{\partial r} = \frac{1}{4\pi R^2} \frac{d}{dt} \left(\frac{4}{3} \pi R^3 \rho_T \right) \left[L + \frac{4\mu}{3\rho} \left(\frac{\partial V}{\partial r} - \frac{V}{r}\right) \right] + \rho_T \frac{R}{3} \frac{dU_T}{dt} + p_T V \quad (7)$$

$$\text{and at } r = \infty: \quad p = p_\infty(t) = P_0 - P_A \sin \omega t. \quad (8)$$

As might be expected, these equations are still very difficult to manipulate when all forms of coupling are included. However, we can reasonably examine transient and stable cavitation independently by separating out inertial coupling from that arising from diffusion mechanisms. Transient collapse, and any bubble motion involving only a few cycles of oscillation, can reasonably be studied by concentrating on the inertial terms, ignoring diffusion. For truly stable oscillations the diffusion terms must be included, but the problem is not necessarily more complicated, as plausible simplifying assumptions can be made in this case also.

The set of equations (1)–(8) do not include the State Equations for the liquid and gas phases. For the gas, the assumption that the medium remains a perfect gas, obeying the equation $pV = NT$ is realistic in most cases. Thermal diffusion may then be decoupled by taking one or other of the extreme conditions, isothermal or adiabatic. Then, in effect, the thermodynamic energy equation (3) combines with the State Equation to give $pV = \text{constant}$ (isothermal) or $pV^\gamma = \text{constant}$ (adiabatic). For the liquid, use of the simplest equation of state (that is, $\rho = \text{constant}$, the incompressible assumption) leads to a relatively simple formulation and a set of equations that are, in some cases, even analytic. However, we shall find that this assumption cannot be justified, even approximately, for violent transients. Much theoretical work on cavitation in the decade 1950–1960 aimed at using more realistic equations of state for the liquid, introducing compressibility in various ways.

As well as the two major simplifications already mentioned (spatially uniform conditions within a spherical bubble) several other “obvious” assumptions are implicit in the above formulation:

(i) The acoustic wave-length is large compared with the bubble dimensions; otherwise, radial symmetry would be lost and the problem becomes intractable.

(ii) No body-forces are present: that is, we ignore the effect of gravity and any steady forces arising from the sound field.

(iii) The bulk viscosity coefficient, and also interaction between viscosity and compressibility, are small enough to be ignored.

(iv) The density of the liquid is large, and its compressibility small, compared with that of the gas.

In deriving the equations of bubble dynamics in forms suitable for computation, we will follow the historical development. We therefore start with the simplest assumption for the liquid state, that of incompressibility. Inserting the condition $\rho = \rho_0$, a constant in the equations of continuity and motion (1 and 2), integrating from $r = R$ to ∞ and using the boundary condition (8) gives:

$$V = R^2 \dot{R} / r^2 \quad (9)$$

$$p = p_\infty(t) + \rho_0 \left(\frac{R^2 \ddot{R} + 2R\dot{R}^2}{r} - \frac{R^4 \dot{R}^2}{2r^4} \right). \quad (10)$$

The motion of the bubble wall is now obtained by using the boundary condition (6) after substituting for

V from (9):

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \frac{1}{\rho_0} [p_L(R) - p_\infty(t)] \quad (11)$$

where $p_L(R)$ is the liquid pressure just outside the bubble wall, given by $p_L(R) = p_T(R) - 4\mu\dot{R}/R - 2\sigma/R$ and $p_T(R)$ will in general be the sum of contributions from permanent-gas and vapour: $p_T(R) = p_g(R) + p_v(R)$.

We note that viscosity enters only as a boundary condition and appears as a pressure term analogous to the surface tension pressure. Equation (11) can be solved as soon as the two functions $p_\infty(t)$ and $p_T(R)$ are specified. For acoustic excitation, $p_\infty(t) = P_0 - P_A \sin \omega t$ (boundary condition 8), while $p_T(R)$, the total gas pressure within the cavity, can be evaluated from the other defining equations once the initial gas-content (at $R = R_0$) and the equation of state for the gas have been specified. From the defining equations (1–8) we see that considerable complication will arise from any assumed coupling via mass-diffusion and heat-flow with resulting evaporation-condensation at the bubble wall. It is convenient to ignore these in solving for the motion of the bubble wall. Their effect can be included later, if necessary, in an ad hoc fashion.

In the special case where $p_\infty(t) = P_0$, a constant, $p_T(R) = 0$ (empty cavity) and surface tension and viscosity are neglected, eq. (11) reduces to a very simple form:

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = P_0/\rho_0. \quad (12)$$

This is the famous equation first obtained by Rayleigh [7] by a simple energy-balance argument. It is of great historical importance and we shall consider it in more detail later.

First we assume that the bubble is filled with a permanent-gas obeying the perfect gas equation $p(4\pi R^3/3) = NT$. For the sake of defining the motion of the bubble wall over a limited number of cycles we ignore mass- and heat-flow across the interface. We can then use the adiabatic relation $p(4\pi R^3/3)^\gamma = \text{constant}$ and $p_T(R) = p_g(R) = (P_0 + 2\sigma/R_0) (R_0/R)^{3\gamma}$, since at $t = 0$ where $R = R_0$, the gas pressure in the bubble is just $(P_0 + 2\sigma/R_0)$. Equation (11) then becomes:

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \frac{1}{\rho_0} \left[\left(P_0 + \frac{2\sigma}{R_0} \right) \left(\frac{R_0}{R} \right)^{3\gamma} - \frac{4\mu\dot{R}}{R} - \frac{2\sigma}{R} - p_\infty(t) \right]. \quad (13)$$

This equation, with $p_\infty(t) = P_0 - P_A \sin \omega t$ but without the viscous term, was first derived and investigated by Noltingk–Neppiras [10, 11]:

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \frac{1}{\rho_0} \left[\left(P_0 + \frac{2\sigma}{R_0} \right) \left(\frac{R_0}{R} \right)^{3\gamma} - \frac{2\sigma}{R} - (P_0 - P_A \sin \omega t) \right]. \quad (14)$$

The viscous term was later added by Poritsky [14]. Equations (13) and (14) are still valid in the other extreme condition (isothermal, with $\gamma = 1$). Also, no additional complication arises if the cavity contains liquid vapour at its equilibrium pressure in addition to the permanent-gas. We find that eq. (14) accurately describes the motion of the cavity-wall over a limited number of cycles for all types of stable cavitation, and also for transients where the bubble wall velocity never exceeds about 1/5 of the velocity of sound.

But we know that under violent transient conditions the bubble wall velocity may approach, or exceed, the velocity of sound. Our assumption of incompressibility for the liquid then ceases to hold. A first step towards a more realistic treatment is to postulate a finite but constant stiffness for the liquid (constant sound velocity). Its equation of state then becomes $p/\rho = \text{constant}$; $\partial p/\partial \rho = \text{constant} = c^2$, defining the sound velocity. Such a treatment was first carried out by Herring [15] who was concerned with the expansion of vapour cavities in underwater explosions. Later, Trilling [16] extended the theory to gas-filled bubbles, including an evaluation of the pressure and velocity fields in the liquid.

The “Acoustic Approximation” implied by use of the State Equation $\partial p/\partial \rho = c^2$ confines the treatment to cases where the velocity is always small compared with c , that is, the “Acoustical Mach Number”, $M(=\dot{R}/c) \ll 1$. Disturbances propagate in the liquid at the speed of sound, a finite constant unaffected by the motion. The velocity potential, ϕ must therefore satisfy the acoustic equation for diverging spherical waves:

$$\left(\frac{\partial}{\partial t} + c \frac{\partial}{\partial r}\right) r\phi = 0. \quad (15)$$

The equations of continuity and motion (1 and 2) must hold, and in addition to the boundary condition (6) the following relations must apply at the bubble wall:

$$dp_L/dt = \partial p/\partial t + \dot{R} \partial p/\partial r \quad (16)$$

$$d\dot{R}/dt = \partial V/\partial t + \dot{R} \partial V/\partial r \quad (17)$$

with equations of state:

$$p/\rho = \text{constant} \longrightarrow \partial p/\partial \rho = c^2 \quad (\text{liquid}) \quad (18)$$

$$p(\frac{4}{3}\pi R^3) = NT \quad (\text{gas}). \quad (19)$$

For many liquids, viscosity can be neglected, thereby simplifying eq. (6). Using (18), eq. (1) can be written:

$$\frac{1}{\rho c^2} \frac{\partial p}{\partial t} + \frac{V}{\rho c^2} \frac{\partial p}{\partial r} + \frac{\partial V}{\partial r} + \frac{2V}{r} = 0. \quad (20)$$

Equation (2) integrates to:

$$\frac{\partial \phi}{\partial t} = \frac{V^2}{2} + \int_{p_\infty}^p \frac{dp}{\rho}. \quad (21)$$

Combining (21) with (15) gives:

$$rV \frac{\partial V}{\partial t} + \frac{r}{\rho} \frac{\partial p}{\partial t} + \frac{cV^2}{2} + crV \frac{\partial V}{\partial r} + \frac{cr}{\rho} \frac{\partial p}{\partial r} + c \int_{p_\infty}^p \frac{dp}{\rho} = 0. \quad (22)$$

We can now solve the four simultaneous equations (2), (16), (17) and (20) for the four partial derivatives $\partial p/\partial r$, $\partial p/\partial t$, $\partial V/\partial r$ and $\partial V/\partial t$ in terms of quantities referred to the bubble wall. Using these in eq. (22) and retaining only first-order terms in M gives a relation which reduces to:

$$R\ddot{R}(1-2M) + \frac{3}{2}\dot{R}^2(1-4M) = \frac{1}{\rho_0} \left[p_\infty - p_L + \frac{R}{c}(1-M)\frac{dp_L}{dt} \right] \quad (23)$$

or, approximately:

$$R\ddot{R} \frac{d\dot{R}}{dR} (1-2M) + \frac{3}{2}\dot{R}^2(1-4M) = \frac{R\dot{R}}{\rho_0 c} \frac{dp_L}{dR} + \frac{(p_\infty - p_L)}{\rho_0}. \quad (24)$$

For a completely empty cavity, neglecting surface tension, $p_L(R) = 0$ and eq. (23) reduces to:

$$R\ddot{R}(1-2M) + \frac{3}{2}\dot{R}^2(1-4M) = p_\infty/\rho_0. \quad (25)$$

This is the analogue of Rayleigh's equation (12), reducing to it as $M \rightarrow 0$. (24) is the Herring-Trilling equation. The factors multiplying the inertial terms on the left-hand side are linear in the Mach Number. They express the effect of energy-storage in the liquid medium. The term involving dp_L/dR on the right-hand side refers to the energy radiated as sound, and includes radiation damping. Flynn [1] has shown that the simpler relation obtained by omitting the corrections to the inertial terms:

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \frac{1}{\rho_0} \left[p_\infty - p_L + \frac{R}{c}(1-M)\frac{dp_L}{dt} \right] \quad (26)$$

can be used to give reliable information about non-linear stable bubble activity and also the dissipative effects of sound radiation on transient cavities.

The Acoustic Approximation used above is valid only where the liquid velocity remains rather small compared with c . To cope with conditions in rapidly-collapsing transients, a more realistic State Equation is needed for the liquid. For many liquids, including water, a realistic relation is known to be:

$$p = A(\rho/\rho_0)^n - B \quad (27)$$

with

$$\partial p/\partial \rho = c^2 \quad (28)$$

where A and B are constant pressures differing by the steady ambient pressure P_0 and n is an integer. For example, for water $A = (3000 + P_0)$ Bar, $B = 3000$ Bar and $n \approx 7$. In this case also, the equations of continuity and motion (1 and 2) still hold and using equations (27 and 28) these can be re-written in the form:

$$\left[\frac{\partial}{\partial t} + (V + c) \frac{\partial}{\partial r} \right] \left[V + \frac{2c}{n-1} \right] = -\frac{2Vc}{r} \quad (29)$$

$$\left[\frac{\partial}{\partial t} + (V - c) \frac{\partial}{\partial r} \right] \left[V - \frac{2c}{n-1} \right] = \frac{2Vc}{r}. \quad (30)$$

It is convenient to introduce new co-ordinates α and β such that:

$$\frac{\partial r}{\partial \alpha} = (V + c) \frac{\partial t}{\partial \alpha} \quad \text{and} \quad \frac{\partial r}{\partial \beta} = (V - c) \frac{\partial t}{\partial \beta}$$

so that equations (29) and (30) can be re-written:

$$\frac{\partial}{\partial \alpha} \left(V + \frac{2c}{n-1} \right) = - \frac{2Vc}{r} \frac{\partial t}{\partial \alpha} \quad (31)$$

$$\frac{\partial}{\partial \beta} \left(V - \frac{2c}{n-1} \right) = \frac{2Vc}{r} \frac{\partial t}{\partial \beta}. \quad (32)$$

These are the basic equations to be solved for the general compressible liquid, subject to the boundary conditions as before.

Numerical integration of these equations may be carried out by normal procedures. It is to be understood that this general treatment is necessary only for studying the collapse of transients. In general, in the r, t plane, the characteristic curves will be fanned out from the bubble wall as the cavity collapses. Numerical integrations for the case of the empty cavity have been carried out by several investigators: Gilmore [17]; Hunter [18]; and Hickling and Plessett [13]. Hickling and Plessett have also dealt with the case of a collapsing gas-filled bubble.

Gilmore [17] has shown that in certain cases great simplification with, actual analytical solutions can be obtained by making use of the Kirkwood-Bethe hypothesis [19]. This states that, for spherical waves of finite amplitude the quantity $r\phi$ propagates with a velocity equal to the sum of the fluid velocity and the local velocity of sound. This is reasonable and is just a plausible generalisation of a procedure used in acoustical problems. Therefore we assume that $r\phi$ propagates with velocity $(c + V)$ where c is no longer a constant, but depends on the motion. Since $r\phi$ propagates with velocity $(c + V)$, then so must $r \partial \phi / \partial t$. $\partial \phi / \partial t$ may be evaluated from equations (1) and (2) and is found to be just $(h + V^2/2)$ where h is the specific enthalpy, given by $\int_{p_\infty}^p dp/\rho$. The quantity $(h + V^2/2)$ is an energy per unit mass and has been called the Kinetic Enthalpy. Since the quantity $r(h + V^2/2)$ propagates with velocity $(c + V)$, we have the following equation for the diverging spherical wave, analogous to eq. (15):

$$\left[\frac{\partial}{\partial t} + (c + V) \frac{\partial}{\partial r} \right] \left[r \left(h + \frac{V^2}{2} \right) \right] = 0. \quad (33)$$

On evaluation, this becomes:

$$(c + V) \left(h + \frac{V^2}{2} \right) + \frac{r}{\rho} \left[\frac{\partial p}{\partial t} + (c + V) \frac{\partial p}{\partial r} \right] + rV \left[\frac{\partial V}{\partial t} + (c + V) \frac{\partial V}{\partial r} \right] = 0. \quad (34)$$

As before, using (28) eq. (1) can be written:

$$\frac{1}{\rho c^2} \frac{\partial p}{\partial t} + \frac{V}{\rho c^2} \frac{\partial p}{\partial r} + \frac{\partial V}{\partial r} + \frac{2V}{r} = 0. \quad (35)$$

Again, as before, the boundary conditions (16) and (17) hold at the bubble wall and the four partials $\partial p/\partial r$, $\partial p/\partial t$, $\partial V/\partial r$ and $\partial V/\partial t$ can be evaluated from the four equations (1), (16), (17) and (35) and substituted in eq. (34). When evaluated at the bubble wall this gives the Gilmore equation:

$$R\dot{R} \frac{d\dot{R}}{dR} (1 - M) + \frac{3}{2} \dot{R}^2 \left(1 - \frac{M}{3}\right) = H(1 + M) + \frac{R\dot{R}}{C} \frac{dH}{dR} (1 - M). \quad (36)$$

Here, H is the difference in the liquid enthalpy between the bubble wall and infinity. C also refers to the bubble wall and both H and C are therefore functions of R and t .

For a completely empty cavity, $dH/dR = 0$ and eq. (36) reduces to:

$$R\dot{R} \frac{d\dot{R}}{dR} (1 - M) + \frac{3}{2} \dot{R}^2 \left(1 - \frac{M}{3}\right) = H(1 + M). \quad (37)$$

These equations are more complex than they look, and before they can be used, H and C must be evaluated and expressed as functions of P . Using the equation of state for the liquid (27) we obtain:

$$H = \frac{nA^{1/n}}{(n-1)\rho} [(p_L + B)^{(n-1)/n} - (p_\infty + B)^{(n-1)/n}] \quad (38)$$

and

$$C^2 = C_0^2 + (n-1)H \quad (39)$$

which, for the empty cavity reduce to:

$$H = \frac{nA}{(n-1)\rho} \left[\left(\frac{B}{A}\right)^{(n-1)/n} - 1 \right] \quad (40)$$

and

$$\left(\frac{C}{C_0}\right)^2 = \left(\frac{B}{A}\right)^{(n-1)/n} \quad (41)$$

if, as can usually be assumed, the collapse occurs at constant external pressure.

In the general case of the gas-filled bubble $p_L(R)$ is a function of time and no integrals can be found for the Gilmore equation. Numerical methods must be used. On the other hand, for the empty cavity the Gilmore equation becomes an ordinary non-linear differential equation. This is solvable analytically in some special cases; but generally, as before, numerical methods are called for.

3. Stable cavitation

In any general cavitation field, most of the visible bubbles will be oscillating stably. Far removed from the thresholds, oscillations are linear, or nearly so. Below the radial resonance frequency the

radial motion is in phase with the excitation pressure; above it, out of phase. Most stable cavities contain permanent gas, generally air. Only under certain special conditions can a vaporous cavity be maintained in stable oscillation. Strictly speaking, permanent-gas bubbles are also not quite stable, as there is a continual exchange of gas, by diffusion, across the bubble wall. But diffusion operates on a long time-scale, so a relatively stable condition holds.

3.1. Undamped linear oscillations

We assume that the bubble, equilibrium radius R_0 , contains permanent gas in liquid at ambient pressure P_0 . Derivation of the equation of motion for linear oscillations under an impressed sinusoidal pressure is a simple dynamical problem. It may be approached in the usual way, calculating the equivalent mass and stiffness of the system and equating the sum of the inertial and restoring forces to the excitation force. An alternative approach is to linearise any of the general non-linear equations we have already derived, neglecting damping terms if present. For example, using eq. (14) and substituting $R = R_0 + r$, where $r/R_0 > 1$, expanding in powers of $1/R_0$ and retaining only the first-order terms, gives:

$$\ddot{r} + \omega_r^2 r = \frac{P_A}{\rho R_0} \sin \omega t \quad (42)$$

where the resonance frequency, ω_r , is given by:

$$\rho \omega_r^2 R_0^2 = 3\gamma(P_0 + 2\sigma/R_0) - 2\sigma/R_0. \quad (43)$$

For large bubbles, the surface tension term becomes less important and we then obtain:

$$\rho \omega_r^2 R_0^2 = 3\gamma P_0 \quad (\text{adiabatic}) \quad \text{or} \quad \rho \omega_r^2 R_0^2 = 3P_0 \quad (\text{isothermal}). \quad (44)$$

On the other hand, for bubbles so small that surface tension predominates:

$$\rho \omega_r^2 R_0^3 = 2(3\gamma - 1)\sigma \quad (\text{adiabatic}) \quad \text{or} \quad \rho \omega_r^2 R_0^3 = 4\sigma \quad (\text{isothermal}). \quad (45)$$

For large air bubbles in water under normal atmospheric pressure, the relation (44) is approximately: $f_r R_0 \approx 300$ where f_r is the linear frequency in Hz and R_0 is in cm.

The general solution of the linear equation (42) is:

$$r = \frac{P_A}{\rho R_0(\omega_r^2 - \omega^2)} \left[\sin \omega t - \frac{\omega}{\omega_r} \sin \omega_r t \right]. \quad (46)$$

This expression reduces to an indeterminate form at resonance. Further investigation gives, for $\omega = \omega_r$:

$$r = \frac{P_A}{2\rho R_0 \omega^2} (\sin \omega t - \omega t \cos \omega t). \quad (47)$$

The solution shows that the phase of r with respect to P_A changes through resonance. At resonance the solution shows instability, the amplitude increasing with time. Neglecting damping, as we have done, the amplitude can theoretically increase without limit.

It is of some interest to compare the resonance radius with the acoustic wave-length in the liquid (λ). Since $c_g = \gamma P_0 / \rho_g$ and $\lambda = 2\pi c / \omega_r$ using (44) we easily see that

$$\frac{2R_0}{\lambda} = \frac{1}{\pi} \left(\frac{3\rho_g}{\rho} \right)^{1/2} \left(\frac{c_g}{c} \right). \quad (48)$$

For example, for an air bubble in water, taking $\rho_g = 0.0013$; $\rho = 1$; $c_g = 3 \times 10^4$; and $c = 1.5 \times 10^5$ c.g.s. units, $2R_0/\lambda = 0.004$. The resonance diameter is therefore much smaller than the acoustic wave-length. As we shall see later, it becomes difficult to achieve transient cavitation for bubbles above resonance size, while stable bubbles of such sizes are likely to distort into surface vibrational modes. This amounts to justification of our assumption that the acoustic wave-length must be large compared with the bubble size under useful cavitating conditions.

3.2. Damping of stable oscillations

We recognise three sources of damping of the bubble motion:

(i) *Viscous forces* are effective only at the bubble surface, where they exert an excess pressure proportional to the radial velocity. We have seen in Poritsky's equation that the shear viscosity enters as a boundary condition only. The viscous boundary-layer distorts under vibration in opposite directions depending on whether the bubble expands or contracts. On linearising the Poritsky equation (13) we obtain:

$$\ddot{r} + \left(\frac{4\mu}{\rho R_0^2} \right) \dot{r} + \omega_r^2 r = \frac{P_A}{\rho R_0} \sin \omega t. \quad (49)$$

Thus, the damping coefficient due to viscosity is:

$$\delta_v = \frac{4\mu}{\rho \omega_r R_0^2} \quad \text{or} \quad \frac{4\mu \omega_r}{3\gamma P_0}. \quad (50)$$

The viscous damping has only a small effect on the resonance frequency, which is now given by ω'_r where

$$(\omega'_r)^2 = (\omega_r)^2 - \left(\frac{2\mu}{\rho R_0^2} \right)^2. \quad (51)$$

(ii) *Acoustic radiation damping*. In a compressible liquid, an oscillating bubble expends a portion of its energy in radiating spherical waves. That energy must be lost is obvious from the fact that the pressure in the spherical wave at the bubble surface has a component in-phase with the particle velocity. We easily see that if the velocity of the bubble surface is $\dot{R}$, the in-phase pressure component will be given by $\rho c (\omega R_0 / c)^2 \dot{R}$ and the rate at which work is done by the bubble is then $\rho c (\omega R_0 \dot{R} / c)^2$. The time-average of this expression is not zero and the bubble therefore loses energy to the liquid in generating the outgoing wave. The bubble will lose a fraction $(3\rho_g/\rho)^{1/2} (c_g/c)$ of its energy per cycle. At resonance, where $\rho \omega_r^2 R_0^2 = 3P_0$ this loss factor is just $\delta_r = \omega_r R_0 / c = k_r R_0 = (3P_0/\rho_0 c^2)^{1/2}$. This, of course, applies only to large bubbles where surface tension is not a strong controlling influence. Under these conditions therefore resonance damping by sound radiation is independent of frequency.

(iii) *Thermal damping.* As we have seen, damping due to viscous forces and sound re-radiation is easily evaluated. The third form of damping on oscillating bubbles results from thermal coupling, a feature we have so far ignored in formulating equations. It turns out that this is the most important source of damping in most cases, and also the most difficult to assess. We will discuss thermal coupling in detail later. Here we simply review some early history and quote relevant results.

Pioneer theoretical studies of the damping of free and forced oscillations of bubbles were carried out by Pfriem [20] and Saneyoshi [21]. They predicted that thermal coupling would contribute most to the damping of gas bubbles in ordinary low-viscosity liquids at frequencies below a few 100 kHz. This prediction was amply borne out in detailed experimental work by E. Meyer and his group at the University of Göttingen [22–26]. Pfriem [20] showed that the Q -value of air bubbles resonant between 200 and 300 kHz in water would be as low as 6, thermal conduction contributing about 80% of the damping. The best available theoretical treatments of thermal damping of stable cavities are those of Plesset and Hsieh [27] and Hsieh [5]. A useful survey, comparing the three damping processes for air bubbles in water has been published by Devin [28]. The review paper by Flynn [1] also contains a detailed discussion of the three forms of damping applied to both stable and transient conditions.

In the extreme isothermal and adiabatic states the volume and pressure changes are in-phase and there is no dissipation. Between these extremes, the temperature at the bubble wall may still be regarded as constant, as the liquid is an excellent heat sink. But there will be a temperature-gradient in the gas and a phase-difference will exist between the average volume and pressure changes, resulting in loss. The average thermal process can then be regarded as polytropic, with the pressure and volume changes related through an “effective polytropic index”, Γ , which may range between 1 and γ . Then, for large bubbles:

$$p_g(R) = P_0(R_0/R)^{3\Gamma}. \quad (52)$$

Zwicky [29], following Plesset and Zwicky [30], studied the effects of heat-conduction on small-amplitude oscillations of gas bubbles in water, expressing results in terms of Γ defined by eq. (52). His system of curves, reproduced in fig. 1 relate Γ to frequency and R_0 for argon-filled bubbles in water at normal temperature. These results are instructive and revealing. They show, for example, that at low frequencies ~ 20 – 40 kHz, as used in many practical applications, large bubbles with radii above about 10^{-2} cm will behave practically adiabatically, while those with radii below about 10^{-3} cm will behave isothermally. At low mega-Hertz frequencies, as used in medical diagnostics, all bubbles with radii less than about 10^{-4} cm will behave isothermally. By using the resonance relation between ω , R_0 and Γ it is not difficult to deduce the effective Γ to be used in the formula (52) for any bubble size. This is indicated by the broken line in the figure. For example, at 20 kHz the effective Γ at resonance would be about 1.45. The curves suggest that at sufficiently high frequencies all bubbles must behave adiabatically. But of course the results are based on the usual assumption that bubbles are small compared with the acoustic wave-length. Plesset and Hsieh [27] have shown that if this assumption is denied, conditions may become adiabatic again at very high frequencies.

Devin, in his review, followed Pfriem in deriving an expression for the damping due to thermal conduction. The damping constant is evaluated in terms of R_0 , P_0 , γ and α where α is the ratio of R_0 to L_g , the “thermal diffusion length” in the gas, given by $L_g = (2D_g/\omega)^{1/2}$, D_g being the thermal diffusivity in the gas. Devin finally obtained the following result for the thermal damping constant:

$$\delta_t = \frac{(\alpha - 1)}{\alpha + 2\alpha^2/3(\gamma - 1)} \approx \frac{3(\gamma - 1)}{2\alpha} \quad (53)$$

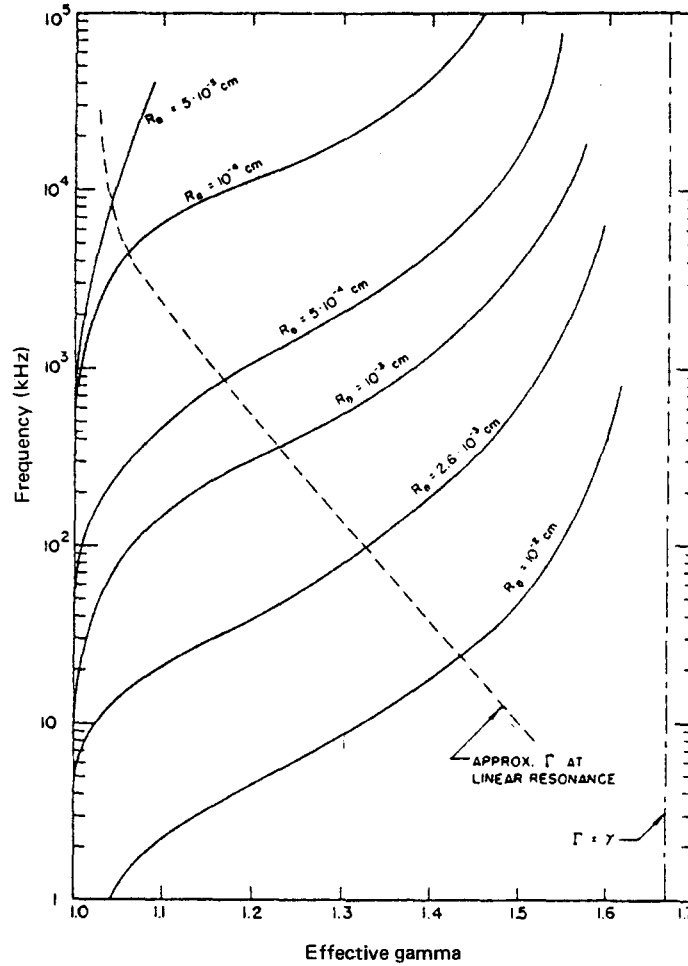


Fig. 1. Effective gamma for stable cavities.

valid for large bubbles where $\alpha \gg$ about 2.5. For small bubbles with $\alpha \leq$ about unity, the result is:

$$\delta_t \approx 2\alpha^2(\gamma - 1)/15\gamma. \quad (54)$$

Very large and very small values of α correspond to adiabatic and isothermal behaviour respectively. Thermal damping vanishes at both extremes, but peaks at a critical value of α between these extremes. Figure 2, plotted from Devin's exact expression for δ_t shows δ_t as a function of α . Thermal conduction also affects the real part of the bubble stiffness, changing the resonance frequency. But the effect is rather small, the difference being no more than about 20% as between the extreme adiabatic to isothermal states. Damping is most important at resonance, and the resonance value of δ_t is obtainable from (53) or (54) by substituting the appropriate value for R_0 contained in α .

The total damping constant δ is simply the sum $\delta_v + \delta_r + \delta_t$. In fig. 3, reproduced from Devin's paper, the three damping constants, together with the total damping, are displayed as functions of frequency for resonant air bubbles in water. Thermal damping is dominant over most of the frequency range from 10 to 1000 kHz. Radiation damping is almost independent of frequency and would dominate

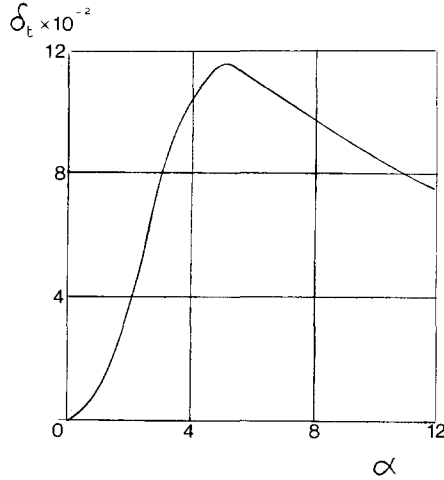


Fig. 2. Thermal damping constant for gas bubbles as a function of $\alpha (=R_0/L_g)$.

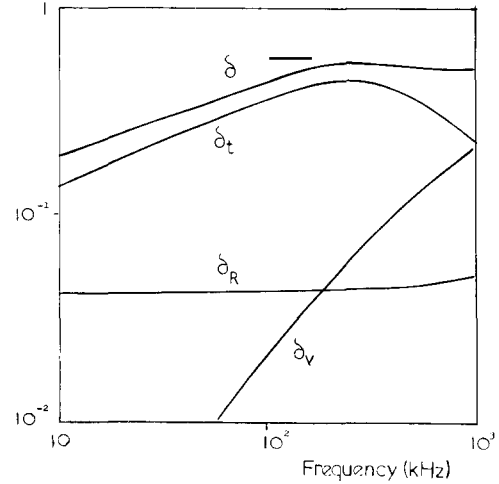


Fig. 3. Frequency-dependence of the theoretical thermal, radiation, viscous and total damping constants for resonant air bubbles in water.

at low frequencies below about 1 kHz. The contribution from viscosity is almost proportional to frequency and becomes very important at MHz frequencies. From fig. 3 we see that at low ultrasonic frequencies, around 20 kHz, the Q -value of resonant air bubbles in water is about 13, falling to about 7 at 1 MHz.

3.3. Coupling by mass-diffusion across the bubble wall

Over long periods of time, mass- and thermal-flows across the bubble wall change the size of the bubble and modify its motion. It is realistic to treat these two effects separately.

Consider first gas-flow by diffusion in the absence of the sound field. If the bubble is filled with a permanent-gas in an undersaturated liquid, it will lose gas at a rate determined by solving the convective diffusion eq. (4). With no sound field, the bubble changes size slowly and it is reasonable to omit the convective term, giving:

$$\frac{\partial C}{\partial t} = D \left(\frac{\partial^2 C}{\partial r^2} + \frac{2}{r} \frac{\partial C}{\partial r} \right). \quad (55)$$

Epstein and Plesset [31] obtained the following solution:

$$\frac{dm}{dt} = 4\pi R_0^2 D (C_\infty - C_0) \left[\frac{1}{R_0} + \frac{1}{(\pi D t)^{1/2}} \right]$$

where C_∞ is the mass-concentration of gas dissolved in the liquid at a great distance from the bubble and C_0 the saturation concentration under the prevailing conditions. This result neglects the effect of surface tension but if this is included, it is easy to see that C_0 must be multiplied by the factor $(1 + 2\sigma/R_0 P_0)$:

$$\frac{dm}{dt} = 4\pi R_0^2 C_0 D \left(\frac{C_\infty}{C_0} - 1 - \frac{2\sigma}{R_0 P_0} \right) \left[\frac{1}{R_0} + \frac{1}{(\pi D t)^{1/2}} \right]. \quad (56)$$

Replacing dm/dt by $4\pi\rho R^2 dR/dt$, the rate of increase in bubble size by diffusion becomes:

$$\frac{dR}{dt} = \frac{C_0 D}{\rho} \left(\frac{C_\infty}{C_0} - 1 - \frac{2\sigma}{R_0 P_0} \right) \left[\frac{1}{R_0} + \frac{1}{(\pi D t)^{1/2}} \right]. \quad (57)$$

Even when the liquid is saturated with dissolved gas ($C_\infty = C_0$) the surface tension term ensures that the bubble will eventually dissolve away. The quantity $(\pi D t)^{-1/2}$ in these equations is large at first, near $t = 0$, but quickly decreases and may generally be ignored where diffusion occurs over long time periods. As numerical examples: an air bubble, 10^{-3} cm radius will dissolve away completely in air-free water in about 1.17 sec; in about 2 sec in half-saturated water; and in 6.63 sec in saturated water.

It is not difficult to show that bubbles in stable oscillation in a sound field may grow by a second-order effect of the field even when the relative concentration C_∞/C_0 is well below unity. This process is usually referred to simply as "rectified diffusion" and may be envisaged physically as follows. On the positive pressure half-cycle the gas in a small bubble will be compressed, while the relative concentration of dissolved gas in the liquid is reduced. Gas then diffuses outwards from the bubble into the liquid. Conversely, during the negative half-cycle of pressure, while the bubble is expanded, gas diffuses from the liquid into the bubble. However, these two rates are not equal as the surface area of the bubble is greater during the negative (tension) half-cycle, and as diffusion rates are proportional to the exposed area, the bubble must gain some gas over a complete cycle.

The first suggestion that gas bubbles should grow by a rectified effect of this sort was made about 40 years ago by Harvey [32]. This was followed by a mathematical treatment by Blake [9] in his famous Harvard report. Blake obtained an approximate solution to the problem by assuming that the bubble wall remains stationary while its area and the gas-concentration vary as if the bubble was oscillating sinusoidally. Murray Rosenberg [33] continued Blake's work, using his results to estimate the pulse-length dependence of the cavitation threshold. Pode [34] further extended Blake's theory by including the effects of motion of the bubble wall as well as its change in area. His result was almost identical with Blake's. The simplification that all these workers were forced to make implied that the convective term in the diffusion equation could be ignored. When, finally, measurements were carried out to assess the effect, the theoretical expressions were found to be seriously in error. In fact, it is not very difficult to see that the convective term in the equation should be important when the diffusion-length in the liquid is small compared with the bubble dimensions.

A more exact treatment, published by Hsieh and Plesset [35] showed that the convective term is indeed very important. They used our simple model of a spherical bubble with uniform interior situated in an infinite liquid. To make the problem more amenable to analysis, the effect of viscosity and thermal exchange were neglected. Of the complete set of equations (1-8), eq. (3) and boundary condition (7) then become redundant, and the analysis is based on equations (1), (2), (4), (5), (6) and (8). The liquid may be assumed saturated with dissolved gas at the pressure existing within the bubble. For small oscillations, we may write:

$$P_g = P_0(1 + \epsilon \sin \omega t); \quad R = R_0(1 - \frac{1}{3}\epsilon \sin \omega t); \quad C = C_0(1 + \epsilon \sin \omega t) \quad (58)$$

where $\epsilon \ll 1$. For the assumed small-amplitude oscillations, the liquid may reasonably be taken as incompressible. The bubble is assumed filled with a perfect isothermal gas. The full set of equations then becomes:

$$V = R^2 \dot{R} / r^2 \quad (9)$$

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \frac{1}{\rho}(p_L - p_\infty) \quad (11)$$

$$\frac{\partial C}{\partial t} + V \frac{C}{\partial r} = D \left(\frac{\partial^2 C}{\partial r^2} + \frac{2}{r} \frac{\partial C}{\partial r} \right) \quad (4)$$

with the boundary condition:

$$4\pi R^2 D \frac{\partial C}{\partial r} = \frac{d}{dt} (4\pi R^3 \rho_g) \quad (5)$$

at $r = R$, along with the linearising relations (58).

The problem is solved by using a system of successive approximations in powers of ϵ . Rectified diffusion appears as a second-order effect (in terms of order ϵ^2). Assuming that the diffusion length in the liquid is small compared with R_0 , the authors find:

$$\langle 4\pi R^2 D (\partial C / \partial r)_R \rangle = \frac{8}{3}\pi D C_0 R_0 \epsilon^2 \quad (59)$$

where the brackets $\langle \ \rangle$ refer to averaging over a complete cycle. Making use of the boundary condition (5) gives the mass flow-rate:

$$dm/dt = \frac{8}{3}\pi D C_0 R_0 (\Delta P / P_0)^2. \quad (60)$$

This result shows, for example, that a small gas bubble would be expected to double its radius in a time:

$$t = 9R_0^2 \rho_g / 4C_0 D \left(\frac{\Delta P}{P_0} \right)^2 \quad (61)$$

if the outward diffusion rate is small enough to be neglected. We must note that the result, eq. (60), is valid only for bubbles large compared with the diffusion length but still small compared with the resonance size. Surface tension has been ignored but may be accounted for by multiplying C_0 by the factor $(1 + 2\sigma/R_0 P_0)$, giving:

$$\frac{dm}{dt} = \frac{8}{3}\pi D C_0 R_0 \left(1 + \frac{2\sigma}{R_0 P_0} \right) \left(\frac{\Delta P}{P_0} \right)^2. \quad (62)$$

As Kapustina [36] and Safar [37] have shown, a correction-factor can be applied to include bubble sizes through resonance. The multiplying factor is:

$$[(1 - \beta^2)^2 + \beta^2 \delta^2]^{-1} \quad (63)$$

where $\beta = \omega/\omega_r$ and δ^{-1} is the Q -value of the resonance.

If the liquid is not over-saturated with dissolved gas, the inflow by rectified diffusion, given by eq. (62) with (63) competes with the steady outflow, given by eq. (56). These two rates then define a threshold pressure or oscillatory amplitude, above which the bubble will grow. Below the threshold, the

bubble will decrease in size and may eventually dissolve away. Adding to zero the mass-flows given by these equations defines the pressure-threshold:

$$\left(\frac{\Delta P}{P_0}\right)^2 = \frac{3}{2} \left(1 + \frac{2\sigma}{R_0 P_0} - \frac{C_\infty}{C_0}\right) \left(1 + \frac{2\sigma}{R_0 P_0}\right)^{-1} [(1 - \beta^2)^2 + \beta^2 \delta^2]. \quad (64)$$

For example, for air bubbles in water well below resonance size, with $R_0 = 6 \times 10^{-4}$ cm and $P_0 = 10^6$ dyne/cm², we find $\Delta P/P_0 \approx 0.55$. For very small bubbles, such that $2\sigma/R_0 P_0 \gg 1$, $\Delta P/P_0 \rightarrow 1.22$. For large bubbles where $2\sigma/R_0 P_0 \ll 1$ but at frequencies still well below resonance, we have $\Delta P/P_0 \rightarrow 0.32$.

We see that the threshold for rectified diffusion falls as the bubble grows towards its resonance size, but the growth rate increases at any fixed value of excitation pressure. Above the resonance size, the threshold increases rapidly until, for bubbles two or three times the resonance size, very high excitation pressures are needed for growth. A practical consequence of rectified diffusion is that it reduces the effective threshold bubble size for transient cavitation to occur and causes the threshold to become time-dependent.

Eller and Flynn [38] avoided the small-amplitude restriction inherent in the Hsieh–Plesset treatment by getting computer solutions of the set of non-linear equations (5), (9) and (11). They also obtained analytical solutions for the asymptotic small-amplitude condition through resonance. Damping was neglected, but they considered both isothermal and adiabatic conditions, using several arbitrary values of γ . Their approximate analytical solution is complex, but resembles eq. (64). When the liquid is saturated with dissolved gas, the approximate result for adiabatic conditions reduces to:

$$\left(\frac{\Delta P}{P_0}\right)^2 = \frac{3\gamma\sigma}{R_0 P_0} \frac{(1 - \beta^2)^2}{(1 - \beta^2/8)} \quad (65)$$

where, as before β is written for $\omega/\omega_r = (\rho\omega^2 R^2/3\gamma P_0)^{1/2}$. This result was published by Eller [39] and may be compared with the modified Hsieh–Plessett threshold, equation (64) after setting $C_\infty = C_0$ and $\delta = 0$.

Threshold measurements by Strasberg [40 and 41] at a frequency of 24.5 kHz; by Eller [39 and 42] at 11 and 26.6 kHz; and by Gould [43] at 20 kHz agree reasonably well with either eq. (64) or (65). In all of these measurements, the bubbles were well below resonance size. Damping was therefore small and $\beta \ll 1$. Eller measured a growth-rate for small air bubbles in water much greater than predicted by the Hsieh–Plesset theory, despite agreement on thresholds. Gould [43] found that growth rates agreed with theory only when precautions were taken to prevent acoustic microstreaming around the bubble. Microstreaming occurs in the acoustic boundary-layer and is associated with the formation of a shear wave due the retarding effect of viscosity. Microstreaming is greatly influenced by any asymmetry, like the presence of solid boundaries or the onset of surface waves. Kapustina [44] also studied the growth rate of a small, 0.02 cm diameter, air bubble in water in sound fields of increasing intensity ($\Delta P = 0$ –1.4 Bar peak). In this case, the bubble was suspended on the measuring microphone and was therefore probably not vibrating symmetrically. As might be expected, the measured growth rates were appreciably greater than predicted by theory.

Microstreaming affects the convective term in the diffusion equation by adding a steady streaming velocity to the oscillatory velocity. Kapustina and Statnikov [45] evolved a theory applicable when the bubble is large compared with the diffusion length in the liquid and where most of the flow velocity is due to streaming. Fig. 4 compares theoretical results for the growth of an air bubble in water, oscillating radially. Curve (1) was plotted from the modified Hsieh–Plesset formula, eqs. (62) and (63), and curve

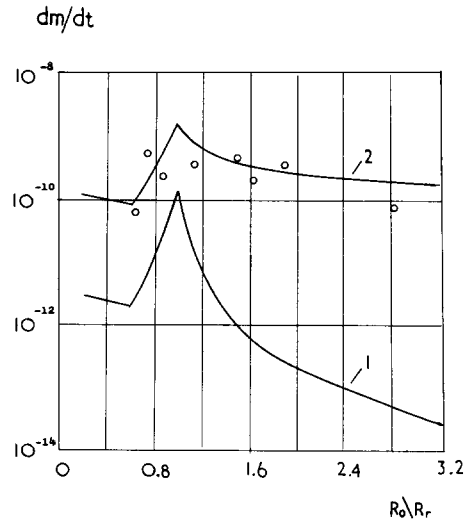


Fig. 4. Rate of inflow of air by diffusion into stably oscillating bubbles in water in terms of the dimensionless ratio R_0/R_r : (1) plotted from the result, eqs. (62, 63); (2) from ref. [45], taking account of acoustic streaming. $P_A = 0.144$ Bar; $f = 26.5$ kHz.

(2) from the authors' theory, assuming that the influx of gas is entirely due to the effect of the stream-flow. The calculations refer to an acoustic pressure amplitude of 0.144 Bar at frequency 26.5 kHz. At such a low pressure microstreaming has a profound effect, the growth rate being about two orders of magnitude greater than predicted from the Hsieh–Plesset theory. The experimental points shown on the graph were obtained with the bubble supported on the measuring microphone. The authors were also able to show that while the steady microstreaming flow is the most important factor at low excitation levels, the oscillatory velocity will assume the major role at sufficiently high levels. Despite reasonable agreement with observations, this theory has been criticised. It seems that unless the liquid was over-saturated with dissolved gas to an extent sufficient to permit growth in the absence of the sound field, microstreaming alone would simply have the effect of increasing the outflow of gas from the bubble. However, as pointed out by Gould [43], the authors interpret their results for $C_\infty = C_0$, under which condition the bubble must be losing gas.

Observations have shown that the bubbles do not continue to grow indefinitely by rectified diffusion. Hsieh and Plesset [35] suggest that this is due to instabilities developing on the bubble surface. Kapustina prefers the explanation that growth will continue only until the liquid has become depleted of dissolved gas. But in any case, as we have seen, the growth rate must slow down after the bubble has reached its radial resonance size.

3.4. Oscillations of stable cavities with thermal coupling

The pioneer papers dealing with forced quasi-linear oscillations of bubbles in a sound field, which include the effects of heat conduction and convection are those of: Plesset and Zwick [30]; Plesset and Hsieh [27]; Hickling and Plesset [13]; and Hsieh [5]. Plesset and Hsieh [27] included thermal conduction in both the liquid and gas-phases. But, as usual, we will first assume spatial uniformity within the bubble. This assumption is equivalent to restricting the analysis to bubbles small compared with both the acoustic wave-length and the thermal diffusion-length in the gas. The set of equations (1–8) then

apply. Plesset and Hsieh also neglected the effects of viscosity, mass-transfer and evaporation-condensation at the bubble wall. Restriction to small-amplitude oscillations justify assuming an incompressible medium.

The governing equations are (1), (2) and (3) with boundary conditions (7) and (8) modified by the above simplifying assumptions. As before, (1) and (2) with the liquid state equation ($\rho = \text{constant}$), evaluated at the bubble wall, give eq. (11), repeated here for convenience:

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = (p_\infty - p_L)/\rho. \quad (11)$$

Equation (3) reduces to:

$$\frac{\partial^2 T}{\partial r^2} + \frac{2}{r} \frac{\partial T}{\partial r} = \frac{1}{D_L} \left(\frac{\partial T}{\partial t} + V \frac{\partial T}{\partial r} \right) \quad (66)$$

and the boundary condition (7) to:

$$3K_L \left(\frac{\partial T}{\partial r} \right)_{r=R} = 3P_g \dot{R} + \rho C_v \dot{T}_g. \quad (67)$$

Here, D_L is the thermal diffusivity in the liquid, given by $K_L/\rho S_v$ where K_L and S_v are the thermal conductivity and specific heat at constant volume for the liquid. C_v is the specific heat at constant volume for the gas. The equation of state for the assumed perfect gas is: $P_g(4\pi R^3/3) = NT_g$ giving:

$$P_L = \left(P_0 + \frac{2\sigma}{R_0} \right) \left(\frac{R_0}{R} \right)^3 - \frac{2\sigma}{R}. \quad (67a)$$

Following Plesset and Hsieh, a linearisation process can be applied to solve the set of four equations (11), (67a), (66) and (67), writing $P = P_0(1 + \epsilon)$ where ϵ is a small quantity $\ll 1$, with similar approximations for R/R_0 , T/T_0 , T_g/T_0 and P_g/P_0 . Solutions of these linear equations can be expressed in terms of error-functions. The formal solutions are not of much practical value, however, and we shall therefore not quote the results here. But the steady-state solution, expressing the average thermodynamic behaviour of the bubble, can readily be found. The main conclusion is that the average behaviour is isothermal or adiabatic depending whether

$$\frac{(\rho C)_g}{(\rho C)_L} \left(\frac{R_0}{L_L} \right)^2 \text{ is } \ll \text{ or } \gg 1.$$

This implies that the behaviour is isothermal if the bubble is small or the frequency low, and vice versa.

For large bubbles and/or high frequencies, the assumption of spatial uniformity within the bubble may not be realistic. Plesset and Hsieh [27] also examined this more complex case and found the asymptotic steady-state solution. The significant practical result is that while conditions are isothermal at low frequencies, they also become isothermal again at very high frequencies if the condition $(K_g C_v / K_L S_v)^{1/2} \ll 1$ holds, as it usually does. As the frequency is increased from low values, the average thermodynamic behaviour changes from isothermal to adiabatic and back again to isothermal at such high frequencies that the acoustic wave-length is small compared with the diffusion length in the gas.

Essentially the same general conclusion can be reached by a simpler argument, which is instructive and worth repeating here (see refs. [27] and [46]). The increase in internal energy in the bubble corresponding to any increment in temperature ΔT is:

$$\Delta Q_2 = \frac{4}{3}\pi R_0^3 \rho_g C_p \Delta T.$$

On the other hand, the loss of heat from the bubble during the compression half-cycle is approximately:

$$\Delta Q_1 = K_L \frac{\Delta T}{L_L} 4\pi R_0^2 \frac{\pi}{\omega}.$$

Writing $D_L = K_L/\rho S_p$ this becomes:

$$\Delta Q_1 = 4\pi R_0^2 \rho S_p (D_L/\omega)^{1/2} \Delta T.$$

Then:

$$\frac{\Delta Q_1}{\Delta Q_2} = 3\pi \frac{\rho S_p}{\rho_g C_p} \frac{(D_L/\omega)^{1/2}}{R_0}.$$

If $\Delta Q_2 \ll \Delta Q_1$ so that

$$3\pi \frac{\rho S_p}{\rho_g C_p} \frac{(D_L/\omega)^{1/2}}{R_0} \gg 1,$$

only a small portion of the energy transferred is available for increasing the internal energy of the gas. In this case, thermal diffusion is so efficient that temperature changes remain small, and the thermodynamic behaviour is essentially isothermal. Conversely, if $\Delta Q_2 \gg \Delta Q_1$ the average behaviour is adiabatic. The detailed analysis agrees with these findings when conditions are spatially uniform within the bubble. But when the temperature field in the bubble can become non-uniform, the high-frequency limit does not lead to adiabatic behaviour. An intuitive picture of the physical situation is still possible by comparing the magnitudes of the three quantities: λ_g (the acoustic wave-length in the gas); L_g (the diffusion length in the gas); and the bubble radius R_0 . Of these three lengths, the first two involve the acoustic frequency. Table 1 summarises the approximate thermodynamic behaviour in the various regimes. The high-frequency transition from adiabatic to isothermal conditions requires frequencies in

Table 1
Thermodynamic behaviour of an oscillating gas bubble in liquid

Frequency range	Comparison of lengths	Relevant criterion	Thermodynamic behaviour
Very high	$\lambda_g \ll L_g \ll R_0$	$L_g \gg \lambda_g$	Isothermal
High	$L_g < \lambda_g < R_0$	$L_g < \lambda_g$	Adiabatic
Moderately high	$L_g < R_0 < \lambda_g$	$L_g < R_0$	Adiabatic
Low	$R_0 < L_g < \lambda_g$	$S_p L_L \gg C_p R_0$	Isothermal

the Giga-Hertz range, and rather large bubbles. Over a very wide range of conditions commonly encountered in practical ultrasonics, the average behaviour is approximately adiabatic.

3.5. *Oscillations of stable cavities with thermal coupling, including evaporation-condensation at the bubble wall*

The dynamical problem becomes more complicated when mass-transfer by evaporation-condensation can occur alongside heat-flow. These processes are coupled and cannot be dealt with in isolation from one another. In some important applications of ultrasonics the vapour pressure is high, with the liquid in a near-boiling condition. Vapour-exchange is then clearly important. The set of equations (1–8) apply when spatially-uniform conditions can be assumed within the bubble.

The first thing to decide is whether the stable oscillations will permit free exchange of vapour or whether the vapour pressure must be regarded as a function of the motion. The criterion is whether the quantity $\alpha P_v(2\pi kT)^{-1/2}$ is $>$ or $< \rho \dot{R}$. Here, α is the “accommodation coefficient”. Applying this condition to the bubble wall velocities that we can expect for low-amplitude stable oscillations, it seems reasonable to assume that P_v remains near to its equilibrium value, thereby simplifying the problem.

As long ago as 1952, Plesset and Zwick [30] carried out a theoretical study of cavity formation in boiling liquids. But the first detailed study of the dynamics of vapour cavities under acoustic excitation is due to Trammell [47]. Trammell was interested in acoustic detection of bubbles in nuclear reactors. As a first step towards quantifying the problem he derived an expression for the dynamic compressibility of pure-vapour bubbles. He made several drastic assumptions, over-simplifying the problem. But he appears to have been the first to appreciate the possibility that rectified heat transfer could occur under acoustic excitation. Soviet workers have also been interested in the acoustical problem, in connection with the operation of the hydrogen bubble chamber (see, e.g. ref. [48]). Shadley [49] and also Wang and Hsieh [50] have independently re-examined the behaviour of a bubble containing a mixture of permanent-gas and vapour in a sound field, including the effects of rectified heat transfer.

In view of the partial nature and discrepancies in previous published work, we decided to develop a linearised theory of the acoustically-excited pure-vapour cavity, taking into account thermal conduction and evaporation-condensation at the bubble wall [51]. The relevant equations are similar to those of Plesset and Hsieh [27] but with an added term to take account of vapour-exchange at the bubble wall. Both the bubble and liquid were assumed free of permanent-gas, viscosity was neglected and spatially-uniform conditions assumed within the bubble. The vapour pressure was related to temperature via data taken from measured vapour-pressure curves. The resulting set of five equations were linearised in the manner already described and an expression obtained for the motion of the bubble wall. By relating this to the excitation pressure, an “effective mechanical impedance” can be cited for the vibrating bubble. The effective stiffness is a complex expression, involving the frequency and bubble-size implicitly.

Detailed study of this expression shows that above a certain critical size the bubble behaves as if filled with permanent-gas. Smaller bubbles will collapse under the influence of surface tension. The critical size increases with decreasing frequency. A resonance condition is inferred by equating the stiffness reactance of the bubble to the mass reactance of the surrounding liquid. The theory was originally developed in an attempt to explain cavitation phenomena observed in low-temperature liquids. In fig. 5 the “effective acoustic admittance” for liquid helium I at 3 K is shown as a function of the bubble size over a range of excitation frequencies. From such computed results, the bubble response can be re-expressed as a resonance frequency – bubble size plot. This is shown for helium at 3 K in fig. 6 and for nitrogen at 77.4 K in fig. 7. The remarkable conclusion to be drawn from these theoretical

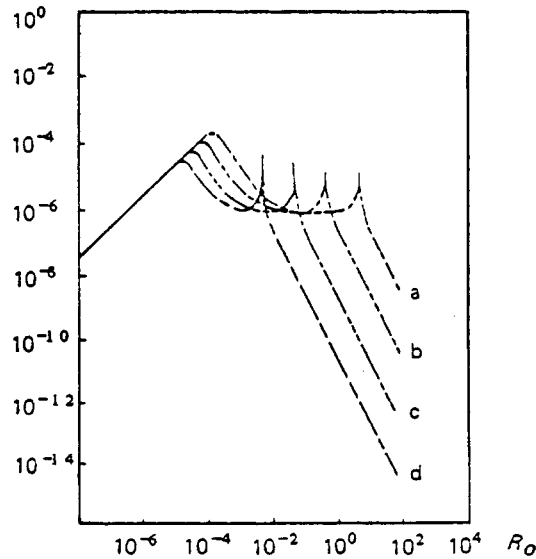


Fig. 5. Effective acoustic admittance for pure-vapour cavities in helium I at 3 K in terms of the equilibrium cavity radius (R_0): (a) $f = 10^2$ Hz; (b) 10^3 Hz; (c) 10^4 Hz; (d) 10^5 Hz.

results is that over a wide range of frequencies, resonance can apparently occur for two widely different bubble sizes. For example, for helium I at 20 kHz the resonance radii are about 2×10^{-2} and 6×10^{-5} cm. The larger size is very near that given by the formula (43), while the other is very small. Although this double resonance phenomenon is difficult to visualise physically, it is in qualitative agreement with experiment [52, 53]. The prediction that a high density of very small and very active bubbles may be present in acoustically-excited volatile liquids is important in a number of practical

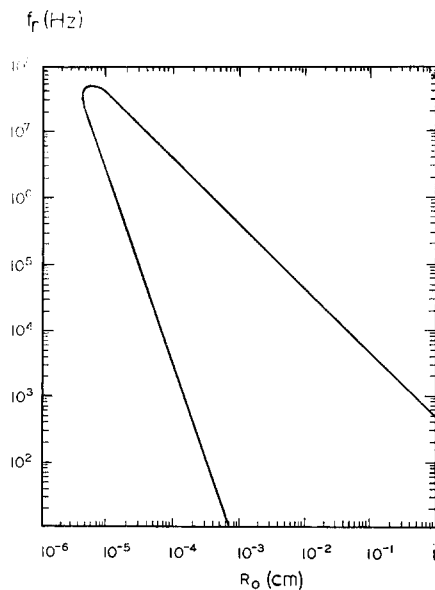


Fig. 6. Resonance frequency (f_r) for pure-vapour cavities in helium I at 3 K as a function of the equilibrium cavity radius (R_0).

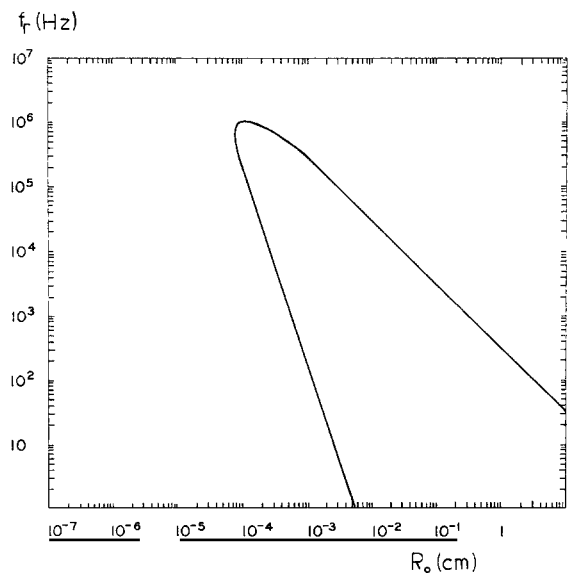


Fig. 7. Resonance frequency (f_r) for pure-vapour cavities in liquid nitrogen at 77.4 K as a function of the equilibrium cavity radius (R_0).

applications, notably ultrasonic cleaning, where low boiling-point fluorocarbon solvents are frequently used.

A more complete study of the dynamics of the acoustically-driven pure vapour cavity has been published by Akulichev and colleagues [54]. These authors show that the motion of the cavity wall can be expressed by the compact equation:

$$R\dot{U} + 2\dot{R}U - \frac{U^2}{2} + \frac{1}{\rho} \left[p_{\infty} - p_L + \frac{2\sigma}{R} \left(1 - \frac{\rho_v}{\rho} \right)^{-1} \right] = 0. \quad (68)$$

Here, the velocity of the bubble wall, $\dot{R}$, is not identified with the particle velocity of the liquid in contact with it (U) because of the coupling by mass-flow due to evaporation-condensation. This eq. (68) combines the equations of motion, continuity and state for the liquid, evaluated at the bubble wall. In the extreme case where $U = \dot{R}$ and $\rho_v \ll \rho$ it reduces to the Noltingk-Neppiras equation (11). Evaporation-condensation is expressed by the equation

$$4\pi R^2 \rho (\dot{R} - U) = dm_v/dt \quad (69)$$

where m_v is the mass of vapour in the bubble. The treatment takes account of the variable temperature and pressure fields within the bubble, so the complete set of governing equations include those of momentum-, mass- and energy-conservation for the vapour-phase as well as the heat-conduction equation and state-equation for the vapour. The complete set of equations is naturally difficult to handle, but numerical solutions were obtained and applied to experimental results. The growth rates of vapour bubbles in nitrogen and hydrogen were found to agree closely with the theory for small bubbles, but beyond a certain size, wide discrepancies appeared. The authors attributed this to the effects of bubble interactions.

It is easy to see that under static conditions a pure vapour cavity cannot remain in stable equilibrium. At any temperature below the boiling-point, bubbles must collapse under the excess inward pressure due to surface tension. The growth and stabilisation of acoustically-excited vapour bubbles predicted by the above theories can be understood physically as follows. On the expansion phase of the oscillation, vapour in the bubble is cooled slightly and heat diffuses into it. On compression, the reverse process occurs. But as the diffusion rate is proportional to the exposed area, and as the area is slightly greater on the expansion phase, the vapour will gain some heat over complete cycles, maintaining the interior of the bubble at a temperature a little above the surrounding liquid, which may be regarded as a constant-temperature heat-sink. The increased temperature implies an increase in the vapour pressure, so that evaporation must occur to restore equilibrium. But evaporation itself is subject to a rectification due to the area effect, accelerating growth. The bubble will therefore grow, in a manner somewhat analogous to the rectified mass-diffusion previously discussed. Just as for mass-diffusion, a threshold acoustic pressure will exist, the growth mechanism competing with condensation impelled by the surface tension excess pressure. Also, as for mass-diffusion, and for the same reasons, radial resonance will be a more or less stable condition. Rectified thermal diffusion is important only for mainly-vapour-filled bubbles. The effect is small for gas bubbles in ordinary liquids at normal temperatures, for which rectified mass-transport predominates.

The threshold and rate of growth by rectified heat flow can readily be obtained from the above theories. For example, fig. 8, reproduced from ref. [55] shows the threshold acoustic pressure (P_A) required to maintain bubbles of radius R_0 in thermal and mechanical equilibrium in helium I at a

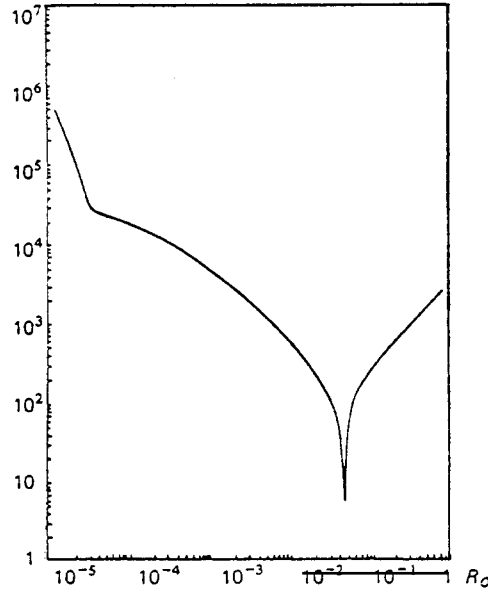


Fig. 8. Theoretical acoustic pressure amplitude (dyne/cm²) required to maintain a vapour cavity at radius R_0 cm in helium I at 3 K and driving frequency 10 kHz.

temperature of 3 K at 10 kHz. At the main resonance, P_A reaches a critically low value of about 7 dyne/cm² peak.

Interest in stable oscillations of pure vapour cavities arose from studies on the low-temperature liquids, nitrogen, hydrogen and helium. These liquids are normally used near their boiling-points, where any bubbles must be mainly, or entirely, vapour-filled. Cavitation studies in liquid hydrogen are relevant to the design of bubble chambers. In the case of helium, acoustic studies have thrown much light on the whole question of nucleation. This is a specialised area which will not be reviewed in detail here. However, the important refs. [56–70] are included in the bibliography to enable the interested reader to pursue the subject.

3.6. Non-linear oscillations of stable cavities

In the above theories, useful results have been obtained by linearising the governing equations. Direct numerical solutions of the non-linear equations often show complex stable oscillations that may never evolve into transients. In all these cases, eq. (14), which is based on the incompressible assumption, can safely be used to predict the motion.

To discuss these cases, it is convenient to re-write eq. (14) in a non-dimensional form:

$$r\ddot{r} + \frac{3}{2}\dot{r}^2 = \delta[r^{-3\gamma} - 1 + p \sin \tau + \alpha(r^{-3\gamma} - r^{-1})]. \quad (70)$$

Here $r = R/R_0$; $p = P_A/P_0$; $\tau = \omega t$; $\delta = P_0/\rho\omega^2 R_0^2$; $\alpha = 2\sigma/R_0 P_0$. For large bubbles, the resonance frequency is given by $\rho\omega_r^2 R_0^2 = 3\gamma P_0$ so that $\delta = \omega_r^2/3\gamma\omega^2$. The coefficient δ is therefore small for large bubbles at frequencies well above resonance. Under these conditions, we can expect a stable non-linear solution when p is also not too large. Large bubbles, well above the resonance size will therefore tend to oscillate in a quasi-linear fashion. For such large bubbles, surface tension is not likely to be a strong

controlling force. Then, with $\alpha \rightarrow 0$, eq. (70) reduces to:

$$r\ddot{r} + \frac{3}{2}\dot{r}^2 = \delta(r^{-3\gamma} - 1 + p \sin \tau). \quad (71)$$

On the other hand, if the bubbles are small enough for surface tension to dominate over the gas-pressure term, we obtain:

$$r\ddot{r} + \frac{3}{2}\dot{r}^2 = \delta[p \sin \tau + \alpha(r^{-3\gamma} - r^{-1})] \quad (72)$$

where α and δ are now both large. This condition may refer to bubbles that are not only well below resonance size but also below the transient threshold size (see section 5).

None of these equations (70–72) is analytic while the time-dependent pressure term is present. If we omit the forcing term in eq. (70) the equation for the free oscillations is:

$$r\ddot{r} + \frac{3}{2}\dot{r}^2 = \frac{1}{2r^2} \frac{d}{dr} (r^3 \dot{r}^2) = \delta[r^{-3\gamma} - 1 + \alpha(r^{-3\gamma} - r^{-1})]. \quad (73)$$

Integration now gives $\dot{r}$ in terms of r , essentially the energy-balance equation:

$$\dot{r}^2 r^3 = \int_{r_m}^r 2r^2 \delta[r^{-3\gamma} - 1 + \alpha(r^{-3\gamma} - r^{-1})] dr. \quad (74)$$

The r - t characteristic can be obtained by a numerical integration. The stationary points occur at $\dot{r} = 0$ and so the period of the non-linear free oscillation is obtained. In the energy equation (74) the term representing the internal energy of the gas becomes significant only in the final stages of compression. If we also neglect surface tension, eq. (74) becomes, on reverting to the original variables:

$$\rho R^3 \dot{R}^2 = \frac{W}{2\pi} - \frac{2}{3} P_0 R^3 \quad \text{where} \quad W = \frac{4\pi R_m^3}{3} \left[P_0 + \frac{P_g}{(\gamma - 1)} \right]. \quad (75)$$

Herring [71] has solved this equation in terms of beta-functions. Approximating $R_{\min}$ to zero, the period Π is given by:

$$\Pi^6 = 2.14(\rho/P_0)^3 (W/P_0)^2. \quad (76)$$

Under strictly linear acoustic excitation, the free oscillation associated with the linear resonance frequency, although present, remains small and quickly dies out. But when conditions become non-linear, coupling exists and can be maintained between the free and forced vibrations. For this non-linear coupling to generate periodic motion, the ratio of forcing to resonance frequency must be a rational fraction, that is, $\omega/\omega_r = n/m$. The coupling is particularly strong when n and m are small integers. In the general case, the vibration will contain both the free and forcing components, with their harmonics and sum and difference frequencies. Many frequency-components therefore arise, and whenever harmonics of the forcing and free vibrations come within each other's bandwidths, energy can transfer from the forcing to the free vibration. As the bandwidth for air bubbles in water is typically large, strong coupling is possible.

According to Stoker [72] the free oscillation appears as a sub-harmonic component if $m = 1$ and $n > 1$ (that is, $\omega = n\omega_r$); as a harmonic component if $n = 1$ and $m > 1$ (that is $\omega = \omega_r/m$); and as an ultra-harmonic component if neither n or $m = 1$. Flynn [1] presents solutions for the two cases: where $\omega/\omega_r = \frac{4}{5}$, reproduced in fig. 9; and where $\omega/\omega_r = \frac{9}{5}$ (fig. 10). In each case, P_A/P_0 was taken to be 0.33, well within the stable regime, and isothermal conditions assumed. In the first case, fig. 9, where the bubble is driven below resonance, the response contains a strong component at frequency $\omega/4$. When the same bubble is driven above resonance, fig. 10, the amplitude-response is noticeably weaker, as expected, with a strong component at $\omega/9$. For these cases where n and m are small integers, the corresponding lines have been observed in the spectrum of noise emitted by the cavitating field.

Many computer solutions of the general eq. (14) have been obtained for stable oscillations over a wide range of non-linear conditions (see refs. [1], [10], [73–79]). Typical examples were published by Borotnikov and Solukin [79]. They computed radius–time curves over several cycles of oscillation for a wide range of values of P_A and R_0 . For small P_A/P_0 and R_0 no greater than R_r the oscillation occurs approximately at the excitation frequency. But with $R_0 > R_r$ the bubble oscillation has a strong component at its own natural resonance frequency. For small bubbles, with $R_0 \ll R_r$, typical transient conditions set in as P_A is increased above P_0 . But if P_A is further increased, a point is reached eventually where the bubble has grown so large during the tension phase that it has no time to collapse completely before the end of the pressure cycle. It will then usually collapse as a transient near the second positive pressure peak. Further increase in P_A will then delay the transient collapse until the third pressure peak; and so on. Eventually, with $P_A/P_0 \gg 1$ the bubble will never be able to collapse as a transient. This means that an upper threshold in P_A has been reached. This behaviour is well illustrated in the R – t curves of fig. 11, reproduced from ref. [79]. These refer to an air bubble in water, radius 10^{-4} cm driven at a frequency of 500 kHz. The bubble was therefore much smaller than its radial resonance size. The numbers attached to the curves refer to the ratio P_A/P_0 and the time-variation of the acoustic pressure is shown below the curves. The change in the form of the curves takes place

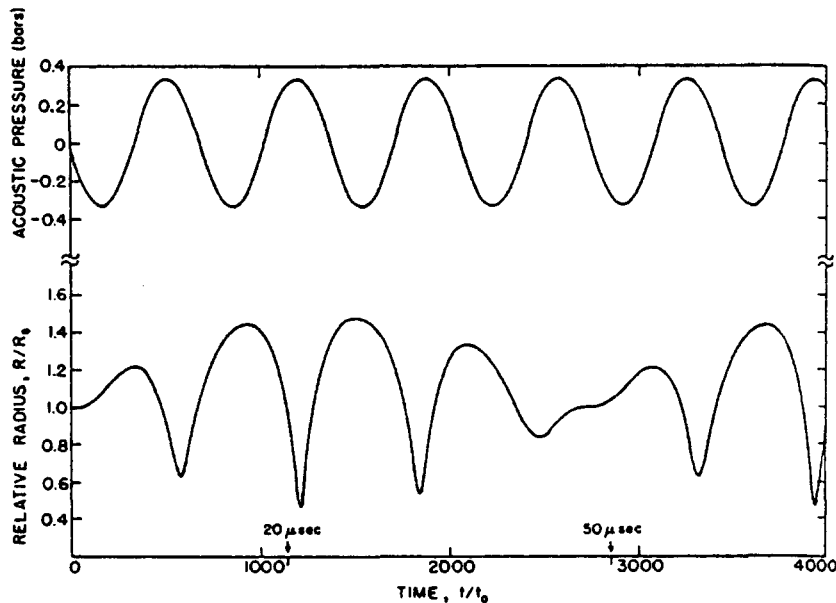


Fig. 9. Radius-time curve for a cavity driven below resonance: $R_0 = 2.6 \times 10^{-3}$ cm; $P_A = 0.333$ Bar; $f = 83.4$ kHz.

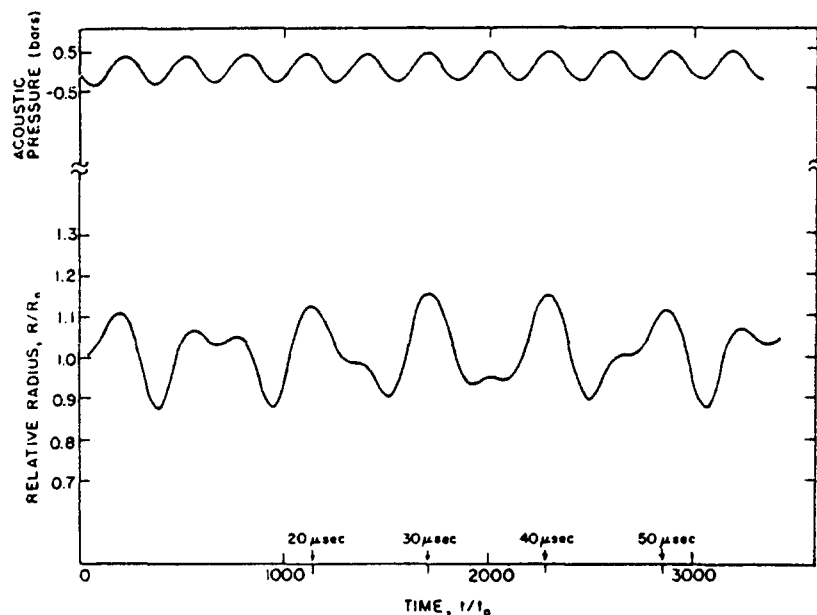


Fig. 10. Radius-time curve for a cavity driven above resonance: $R_0 = 2.6 \times 10^{-3}$ cm; $P_A = 0.333$ Bar; $f = 191.5$ kHz.

suddenly, and the hatched regions of the diagram indicate the regions of instability where a very small increase in P_A will trigger off an additional stable oscillation before the eventual transient collapse. The switch occurs when the theoretical "collapse-time" approaches half the acoustic period. Data of this sort are important in offering an explanation for the strong subharmonic emissions from the cavitation field.

An interesting study of the free oscillations of a gas-filled bubble was carried out by Robinson and Buchanan [78]. They used eq. (14) with P_A zero. If surface tension is also neglected, the non-

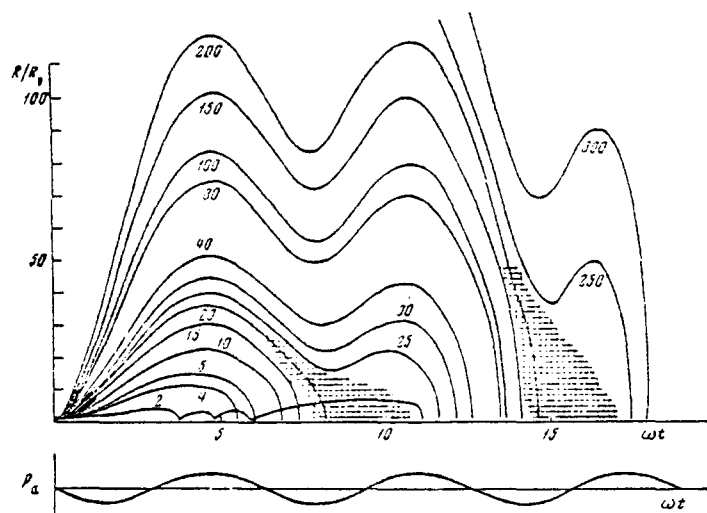


Fig. 11. Radius-time curves for an air bubble in water. $R_0 = 10^{-4}$ cm; $f = 500$ kHz. The numbers on the curves refer to the ratio P_A/P_0 .

dimensional form of eq. (71) becomes simply:

$$r \frac{d^2 r}{d\tau^2} + \frac{3}{2} \left(\frac{dr}{d\tau} \right)^2 = \chi (r^{-3\gamma} - 1). \quad (77)$$

The authors used $\tau = (3\gamma P_0/\rho)^{1/2} t/2\pi R_0$ as the dimensionless time, so that χ is just $4\pi^2/3\gamma$, independent of frequency. In fact, the equation relates only two parameters, r and τ . Equation (77) is analytic, with a first integral:

$$\frac{3\gamma r^3}{8\pi^2} \left(\frac{dr}{d\tau} \right)^2 = \int_r^{r_m} (1 - r^{-3\gamma}) r^2 dr. \quad (78)$$

The oscillations are set off from a bubble in equilibrium initially, with $P_g = P_0$. In the solution we must now retain the term representing the energy of the gas. The R - t curves were obtained by numerical integration. On examining the trajectories over a range of values of γ and bubble-wall velocities, $dr/d\tau$, the authors found that the period of the non-linear oscillations increased steadily with increasing excitation amplitude. Also, as might be expected, increasing γ had the effect of reducing the compression ratio.

Another form of non-linear oscillation, which also illustrates coupling between the free and forcing vibrations, may occur when conditions are very non-linear with bubbles well below their radial resonance size. When P_A is not far below the transient threshold, the first collapse will occur near the peak of the excitation pressure, at phase $3\pi/2$. On collapse, the bubble may not disintegrate, as it would if it were transient, but may oscillate for the remainder of the cycle at its own natural resonance frequency. The sequence may be repeated at the next, and subsequent cycles. This is a case of shock-excitation and of course the resonance frequency of the bubble need not be integrally related to the excitation frequency. The acoustic emission spectrum will then contain a strong component at the bubble resonance frequency. The example shown in fig. 12 was reproduced from ref. [2]. The R - t curve refers to an air bubble in water of equilibrium radius 10^{-3} cm driven at 28 kHz with an acoustic pressure very close to the transient threshold. In this case, the initial phase of expansion and collapse was photographed using a very high-speed camera, and, as can be seen from the figure, measurements of the bubble size taken from the film agree well with the theoretical result.

Soviet workers [80] have shown that eq. (14) can be expressed in a frequency-independent form for a wide range of drive conditions. They illustrate this by writing the equation in the following dimensionless form:

$$\alpha \left(\frac{d^2 \alpha}{d\tau^2} \right) + \frac{3}{2} \left(\frac{d\alpha}{d\tau} \right)^2 = \frac{P_0}{\rho \omega_r^2 R_0^2} \left[\frac{2\sigma}{R_0 P_0} \left(1 - \frac{\omega}{\omega_r \alpha} \right) + p \sin \tau \right] - (1 - \omega^{3\gamma}/\omega_r^{3\gamma} \alpha^{3\gamma}) \quad (79)$$

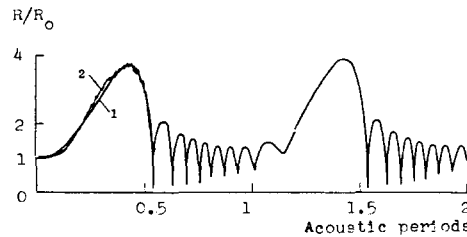


Fig. 12. Radius-time curve for an air bubble in water showing oscillations at the bubble resonance frequency. $R_0 = 10^{-3}$ cm; $f = 28$ kHz; P_A is near the transient threshold. (1) theoretical plot; (2) experimental points.

where: $\alpha = \omega R/\omega_r R_0$; $\tau = \omega t$; $p = P_A/P_0$; and $\omega_r^2 = (P_0 + 2\sigma/R_0)/\rho R_0^2$. If $\omega/\omega_r \ll 1$ or $R/R_0 \gg 1$ this equation reduces to a particularly simple form which does not depend explicitly on the acoustic frequency:

$$\alpha \left(\frac{d^2 \alpha}{d\tau^2} \right) + \frac{3}{2} \left(\frac{d\alpha}{d\tau} \right)^2 = \frac{P_0}{\rho \omega_r^2 R_0^2} \left(\frac{2\sigma}{R_0 P_0} + p \sin \tau \right) - 1. \quad (80)$$

It can easily be seen that the condition $\omega/\omega_r \ll 1$ implies the two conditions $\omega_r/\omega \gg 1$ and $(P_A - P_0) \gg (P_0 + 2\sigma/R_0)$. If these inequalities hold simultaneously, then any solution obtained at one frequency will model that for any other frequency for the same equilibrium bubble size. The authors supply many examples.

So far, our discussions have been based on eq. (14). This equation is suitable for dealing with all forms of stable cavitation over a limited number of cycles where the time-scale is short enough to neglect thermal coupling and mass-diffusion. Theoretical treatments by Plesset and colleagues can be applied to the general problem, but the exact set of equations is not readily handled. Flynn [1, 81] has also researched this problem and succeeded in producing sets of equations that can be programmed, and results obtained relatively quickly, using a modern high-speed computer. To this end, some empirical data, and "plausible" equations of state are used. Thermal coupling and evaporation-condensation can be included, along with acoustic-radiation and viscous damping. In one formulation, use is made of the acoustic approximation, eq. (26), which includes the acoustic radiation term. The effects of thermal conduction are introduced in a simple way: the entropy of the cavity-contents (ϕ) is calculated as a function of time by the use of first-order differential equations:

$$\frac{d\phi}{dt} = \frac{k_1 H}{P_g R^2}; \quad \frac{dH}{dt} = -k_2 R H + P_g R^2 \dot{R} \quad (81)$$

where P_g is given by eq. (82) below. $H(t)$ is an auxiliary function and k_1 and k_2 are constants of the gas involving C_v , N and γ . These equations (81) were derived from the differential equations for heat conduction within the cavity. Uniform pressure within the cavity was assumed and some simplifying assumptions made about the velocity and temperature distributions. The gas pressure in the cavity can then be written:

$$P_g = \exp(\phi/C_v) (P_0 + 2\sigma/R_0) (R_0/R)^{3\gamma}. \quad (82)$$

This can be inserted in the boundary condition at the bubble wall:

$$P_L = P_g + P_v - 2\sigma/R - 4\mu \dot{R}/R. \quad (83)$$

Using eq. (26) then gives:

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \frac{1}{\rho} \left[P_g + P_v + P_A \sin \omega t - P_0 + (1+M) \left(\frac{R}{c} \frac{dP_L}{dt} - \frac{2\sigma}{R} - \frac{4\mu \dot{R}}{R} \right) \right] \quad (84)$$

which with (81) describes the motion of the bubble wall.

Solutions of these equations provide information on the relative importance of thermal, radiation,

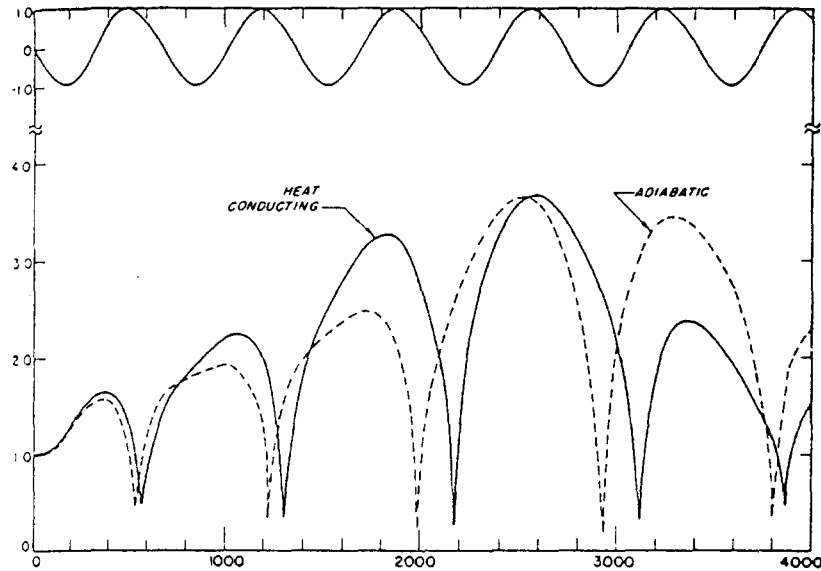


Fig. 13. Radius-time curves for an air bubble in water, comparing adiabatic with heat-conducting conditions. $R_0 = 2.6 \times 10^{-3}$ cm; $f = 83.4$ kHz; $P_A/P_0 \approx 1$.

and viscous damping on the motion of bubbles oscillating non-linearly for long periods. At moderate amplitudes, sound radiation and viscosity have little effect, but thermal conduction is important enough to damp out much of the harmonic content, giving $R-t$ curves with the same period as the excitation field. But under highly non-linear conditions, the effect of the damping becomes more marked. Heat conduction tends to increase the kinetic energy stored in the liquid surrounding a collapsing cavity, the collapse-speed of the heat-conducting cavity being always greater than that of the adiabatic cavity. On the other hand, sound radiation and viscosity must always tend to reduce the violence of the collapse. Clearly also, thermal conduction will always decrease both the maximum temperature and gas pressure within the cavity compared with the adiabatic case. These results are important for explaining the thermal dependence of cavitation effects. Figure 13 reproduced from Flynn's review [1], compares the $R-t$ curves for the heat-conducting and adiabatic cases for an air bubble, radius 2.6×10^{-3} cm, driven just below the transient threshold at a frequency 83.4 kHz. Another instructive result, reproduced in fig. 14 compares the extreme cases of the isothermal and adiabatic motion of the same stably oscillating bubble driven at the same frequency but at much lower excitation pressure ($P_A/P_0 = 0.2$). The $R-t$ curves are substantially different even at this low drive level.

4. The collapse of transient cavities

It is in the final stages of the collapse of transient cavities that the well-known disruptive effects occur. These are the result of the high concentrations of energy and consequent high liquid pressures and velocities. The most important of these effects—metal-erosion—was well known to hydraulic engineers long before acoustically-generated cavitation aroused any interest. Early investigators therefore tended to concentrate on the dynamics of the *collapse* of cavities, without reference to their mode of generation.

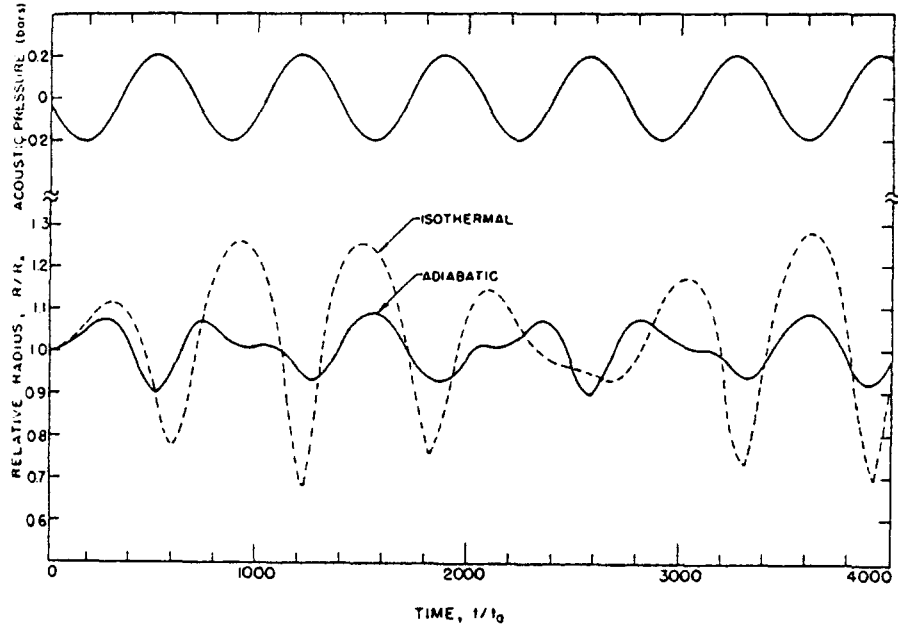


Fig. 14. Radius-time curves for an air bubble in water, comparing adiabatic and isothermal conditions. $R_0 = 2.6 \times 10^{-3}$ cm; $f = 83.4$ kHz; $P_A/P_0 = 0.2$.

The time-scale of an imploding cavity is so short that we can usually ignore any effect of thermal or mass transfer by diffusion. The motion is essentially inertia-controlled. Empty cavities will collapse completely. In a gaseous transient the motion is cushioned, in the final stages, by compression of the residual gas. For a vaporous transient, the implosion will proceed to completion only if vapour has time to condense freely so as to hold the vapour pressure near to its equilibrium value. This possibility needs testing. We here use the formula derived from kinetic theory giving the maximum rate at which vapour can condense at a specified temperature and pressure. Unless the quantity $\alpha P_v (2\pi NT)^{1/2}$ is greater than $\rho_v \dot{R}$, condensation will not occur freely and the vapour pressure must increase. Here, α is the "accommodation coefficient", a number less than unity. We can easily see that for typical transients, some increase in P_v must occur towards the end of the collapse. Nevertheless, it is usual to specify a constant, or zero, vapour pressure over the collapse phase, thereby simplifying the analysis.

In studying collapsing transients, interest is centred mainly on: (i) the velocity field in the liquid, but especially the velocity of the bubble wall; (ii) the pressure field in the liquid; and (iii) the temperature reached by the permanent-gas, if any, in the bubble. These can all be evaluated once we know the three quantities; (i) the initial cavity size from which the collapse started (R_m); (ii) the gas pressure in the cavity at this initial size (Q); and (iii) the ambient liquid pressure responsible for collapsing the cavity (P_m). In an acoustic field, the cavity will have grown from some initial radius R_0 , and R_m and Q will depend on R_0 , P_A , P_0 and ω . Many solutions of the general equations have shown that the collapse generally occupies less than 20% of the acoustic cycle, while the pressure is near its peak. We may therefore regard P_m as nearly constant over the collapse. For the ideal case of a single small bubble in an infinite liquid, P_m must be close to $(P_A + P_0)$; here, we assume that the isolated bubble does not itself interfere with the ambient pressure field, relieving the pressure to any extent. However, in a real cavitation field, with a high density of bubbles, the presence of the bubble cloud must relieve the

ambient pressure to some extent. The acoustic pressure becomes peak-rectified, reducing P_m below the value ($P_A + P_0$). In the extreme case of a very intense cavitation field, measurements have shown [82] that P_m hardly rises above P_0 on the compression half-cycle. On the negative half-cycle also the tension is relieved to the point where the external pressure may not fall much below vapour pressure. The external pressure field can then be represented as a square-wave, with pressure-excursion ($P_0 - P_v$).

4.1. The collapsing empty cavity

In his famous 1917 paper "On the Pressure developed in a Liquid during the Collapse of a Spherical Cavity", Rayleigh [7] quoted Besant's formulation of the problem [6]: "an infinite mass of homogeneous incompressible liquid acted on by no forces is at rest and a spherical portion of the fluid is suddenly annihilated. It is required to find the instantaneous alteration of pressure at any point in the mass, and the time in which the cavity will be filled up, the pressure at an infinite distance being assumed to remain constant".

In this case, the equation of motion of the cavity wall is:

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = P_0/\rho. \quad (12)$$

A first integral gives the bubble wall velocity:

$$\dot{R}^2 = \frac{2P_0}{3\rho} \left[\left(\frac{R_m}{R} \right)^3 - 1 \right] \quad \text{or} \quad M^2 = \frac{2P_0}{3\rho c^2} \left[\left(\frac{R_m}{R} \right)^3 - 1 \right]. \quad (85)$$

This is just the energy equation, which Rayleigh set down immediately by equating the kinetic energy of the liquid mass to the work done by the external pressure, P_0 , in collapsing the cavity. A second integration of the equation gives the collapse time, τ , as the following incomplete β -function:

$$\tau = \left(\frac{3\rho}{2P_0} \right) \int_R^{R_m} \frac{R^{3/2} dR}{(R_m^3 - R^3)^{1/2}}. \quad (86)$$

The time for complete collapse, as $R \rightarrow 0$, is readily obtained:

$$\tau_0 \approx 0.915 R_m (\rho/P_0)^{1/2}. \quad (87)$$

This is the simplest possible model for the collapsing spherical cavity. It is important to notice that *all* transient cavities, whatever additional complications are introduced, *start* their collapse like this. This feature can be included in the definition of a transient; we say that these cavities start their collapse "Rayleigh-like".

Rayleigh examined the distribution of pressure in the liquid using the equation of motion (2). The partial derivatives $\partial V/\partial t$ and $\partial V/\partial r$ can be obtained as functions of r and R by using eq. (85) and the continuity relation $r^2 V = R^2 \dot{R}$. This gives the pressure gradient, which on integration yields:

$$\frac{p}{P_0} = 1 + \frac{R}{3r} (Z - 4) - \frac{R^4}{3r^4} (Z - 1). \quad (88)$$

Here, $Z = (R_m/R)^3$, the volume compression ratio. The distribution of pressure in the liquid, expression (88), is displayed in fig. 15 using Z as parameter. As the collapse proceeds, the pressure peak increases in height and moves closer to the bubble wall. The thickness of the high pressure region also decreases. Eventually, in the final stages of collapse, the cavity is surrounded by a thin shell of highly compressed liquid. At high compression ratios, the maximum liquid pressure, $p_{\max}$, obtained by setting $dp/dr = 0$, is given by:

$$p_{\max}/P_0 = Z/4^{4/3} \quad (89)$$

and this peak occurs at distance $4^{1/3}R$ from the cavity centre.

Rayleigh's analysis loses none of its simplicity if we assume that the cavity is filled with vapour at its constant equilibrium pressure P_v . $(P_0 - P_v)$ then simply replaces P_0 . Also, surface tension may be included with little additional complication, the problem remaining analytic.

Rayleigh's treatment can be applied to the expansion phase of an empty cavity, as well as the collapse. The ambient pressure then has the form of a rectangular wave and we must assume that the cavity is drawn out from a nucleus in the form of an unwetted sphere. It is easy to see that the radial velocity is approximately constant over most of the expansion-phase, at the value $\dot{R} = (2P'_0/3\rho)^{1/2}$ where P'_0 is the constant tension. It is also easy to show that when $P_0 = P'_0$ the collapse time is approximately 3/4 of the expansion time.

For a completely empty cavity the liquid pressures and velocity near the cavity wall reach infinite values as $R \rightarrow 0$. In water, radial velocities reach the velocity of sound at a compression ratio of about 32. Long before this the assumption of incompressibility for the liquid would have become unrealistic. To examine conditions near the end of the collapse, therefore, we must revert to treatments using a more realistic equation of state.

The Herring-Trilling equation (24) assumes a constant velocity of sound, and with a constant

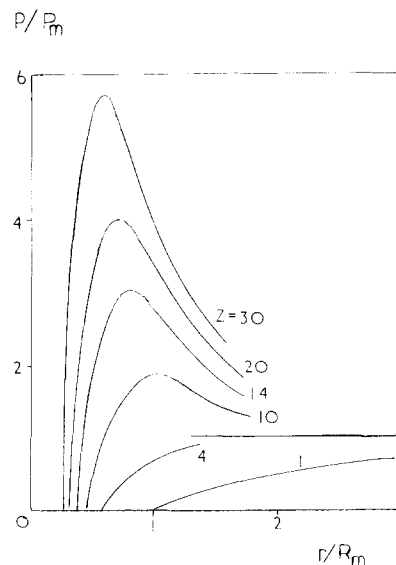


Fig. 15. Pressure developed in the liquid surrounding a collapsing Rayleigh cavity; Z = volume compression ratio.

ambient pressure, reduces to:

$$R\ddot{R}(1 - 2\dot{R}/c) + \frac{3}{2}\dot{R}^2(1 + 4\dot{R}/c) = P_0/\rho. \quad (90)$$

A first integral is:

$$\dot{R}^2 = \frac{2P_0}{3\rho} \left[\left(\frac{R_m}{R} \right)^3 - 1 \right] \left(1 - \frac{4\dot{R}}{3c} \right)^{-1}. \quad (91)$$

These revert to the corresponding Rayleigh equations (12 and 85) as $\dot{R}/c \rightarrow 0$. From (91) $\dot{R}$ can be obtained as a function of R and a numerical integration then gives the $R-t$ characteristic. This will differ from the Rayleigh solution only near the end of the collapse, as we see from plots of R as a function of the radial compression ratio (fig. 16).

The pressure field surrounding the collapsing cavity can be obtained in terms of r and R by a process similar to that used for the Rayleigh case. The result is best expressed as the eliminant of $\dot{R}$ between the expression (91) and:

$$p = \frac{P_0 R}{r} + \frac{R}{r} \left(1 - \frac{R^3}{r^3} \right) \left(\frac{1}{2} \rho \dot{R}^2 \right) + \frac{\dot{R}}{c} \left(1 - \frac{R}{r} \right) (\rho \dot{R}^2 - 2P_0). \quad (92)$$

When the collapse speeds approach the steady-state velocity of sound we are forced to use the Gilmore equation, which, for the empty cavity, reduces to:

$$R\dot{R} \frac{d\dot{R}}{dR} \left(1 - \frac{\dot{R}}{C} \right) + \frac{3\dot{R}^2}{2} \left(1 - \frac{\dot{R}}{3C} \right) = H \left(1 + \frac{\dot{R}}{C} \right). \quad (37)$$

Gilmore obtained exact solutions of this equation for a constant external pressure. A first integral is

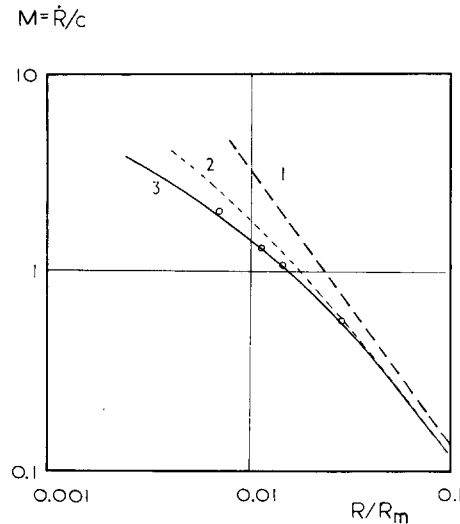


Fig. 16. Comparison of the cavity-wall velocities developed on implosion for three models of the empty cavity: (1) the Rayleigh model; (2) Herring-Trilling model; (3) Gilmore model.

obtained by separation of variables:

$$\ln \frac{R_0}{R} = 2 \int_0^{\dot{R}} \frac{\dot{R}(\dot{R} - C) d\dot{R}}{\dot{R}^2(\dot{R} - 3C) + 2H(\dot{R} + C)} \quad (93)$$

and this can be evaluated numerically or graphically. In most cases, $H \ll C^2$ and eq. (93) then simplifies to:

$$\dot{R}^2 = \frac{2P_0}{3\rho} \left[\left(\frac{R_0}{R} \right)^3 \left(1 - \frac{\dot{R}}{3C} \right)^{-4} - 1 \right] \quad (94)$$

reducing to the Rayleigh formula (85) as $M \rightarrow 0$. In fig. 16 the radial velocities given by eq. (93) are compared with the Rayleigh and Herring–Trilling predictions. The effect of including compressibility is clearly seen in the slowing up of the radial velocity at high Mach numbers. Also shown in fig. 16 are plots obtained by Schneider [83] from a numerical integration using the exact expressions, eqs. (31) and (32). Gilmore's results agree surprisingly well with the exact solutions, even up to $M \approx 2$. The pressure field surrounding the collapsing cavity can be evaluated by a process similar to that used in previous formulations [17].

4.2. The collapsing gas-filled cavity

The presence of permanent-gas in the collapsing cavity prevents the singularity present in the Rayleigh model. We may therefore consider a simple extension of the Rayleigh cavity where the bubble is filled initially with gas at some pressure Q at its maximum radius R_m . As before, we will assume constant ambient pressure P_m over the collapse period. At present, we will not consider how this state was reached. Neglecting surface tension and viscosity and assuming adiabatic compression, eq. (14) for the motion of the bubble wall becomes:

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \frac{1}{\rho} \left[Q \left(\frac{R_m}{R} \right)^{3\gamma} - P_m \right]. \quad (95)$$

The first integral gives the energy equation for the collapse:

$$\frac{3}{2}\rho\dot{R}^2 = P_m(Z - 1) - Q(Z - Z^\gamma)/(1 - \gamma). \quad (96)$$

A numerical integration will now give the R - t curve. This follows the Rayleigh solution for most of its trajectory. Differentiation of eq. (96) gives the acceleration of the bubble wall:

$$\ddot{R} = -\frac{P_m Z}{\rho R} \left[1 - \frac{Q\gamma}{P_m(\gamma - 1)} Z^{1/(\gamma - 1)} \right]. \quad (97)$$

Setting $\dot{R} = 0$ in (96) the minimum radius, $R_{\min}$, reached by the collapsing cavity is given by:

$$\frac{R_{\min}}{R_m} \approx \left[\frac{Q}{P_m(\gamma - 1)} \right]^{1/3(\gamma - 1)} \quad (98)$$

approximately, assuming $Q \ll P_m$ for a typical acoustically-generated cavity. The maximum collapse speed, given by $\ddot{R} = 0$ occurs at:

$$\left(\frac{R}{R_m}\right)^{3(1-\gamma)} \approx \frac{P_m(\gamma-1)}{Q\gamma} \quad \text{or} \quad \left(\frac{R}{R_{\min}}\right)^{3(\gamma-1)} \approx \gamma \quad (99)$$

approximately, and its value is:

$$\dot{R}_{\max}^2 \approx \frac{2P_m(\gamma-1)}{3\rho\gamma} \left[\frac{P_m(\gamma-1)}{Q\gamma} \right]^{1/(\gamma-1)} \quad (100)$$

Under adiabatic compression, we easily find the maximum pressure ($P_{\max}$) and temperature ($T_{\max}$) reached by the gas:

$$P_{\max} \approx Q[P_m(\gamma-1)/Q]^{\gamma/(\gamma-1)} = QZ^\gamma \quad (101)$$

$$T_{\max} \approx T_0 P_m(\gamma-1)/Q = T_0 Z^{(\gamma-1)} \quad (102)$$

to good approximations. The Mach number for the bubble wall motion can be written down from eqs. (96) or (100). Expressed in terms of the radial compression ratio, its maximum value is:

$$M_{\max} \approx (0.015 R_m/R_{\min})^{3/2} (P_m/P_0)^{1/2} \quad (103)$$

if $R_m/R_{\min}$ is greater than about 4. In fig. 17 $M_{\max}$ is displayed as a function of the ratio P_m/Q with P_m as parameter. We know that the incompressible assumption, on which the theory is based, is strictly valid only up to $M \approx 0.2$. From fig. 17 it is clear that the validity of the assumption can also be expressed in terms of P_m or Q . For example, the assumption is justified for Q/P_m as low as 1.5% when

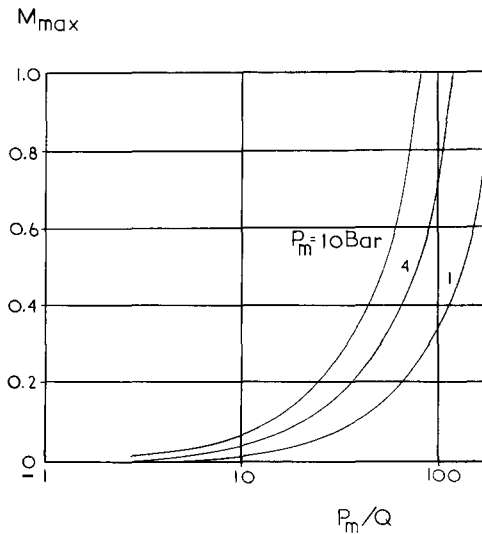


Fig. 17. Maximum Mach number for the bubble-wall velocity for a collapsing gas-filled bubble.

the collapse pressure is 1 Bar, but only for Q/P_m about 4% for a collapse pressure of 10 Bar. Table 2 indicates the range of validity of the simple expression (95) for the collapse by comparing its prediction of the minimum bubble size with theories that take account of the liquid compressibility. The figures confirm that the incompressible theory is reasonably reliable for values of Q/P_m up to about 0.01. As we have seen, this corresponds to maximum Mach numbers up to about 0.2–0.3.

Khoroshev [84] has calculated the collapse time for a gas-filled transient from the formula

$$\tau_0 = \int_{R_m}^{R_{\min}} \frac{dR}{\dot{R}}$$

and finds:

$$\tau_m \approx 0.915 R_m (\rho/P_m)^{1/2} (1 + Q/P) \quad (104)$$

with γ taken to be 4/3. This differs from the corresponding Rayleigh collapse time by the factor $(1 + Q/P_m)$.

The pressure field surrounding the collapsing gas-filled bubble can be evaluated using a procedure similar to that for the empty cavity. In terms of r and R the liquid pressure is given by [10]:

$$p - P_m = -\frac{R}{3r} \left[\frac{QZ^\gamma(3\gamma - 4) + QZ}{(1 - \gamma)} - (Z - 4)P_m \right] - \frac{R^4}{3r^4} \left[P_m(Z - 1) - \frac{Q(Z - Z^\gamma)}{(1 - \gamma)} \right]. \quad (105)$$

This function is shown plotted out in fig. 18 for several values of the volume compression ratio, and with $P_m/Q = 10$. As the collapse proceeds, the highly-compressed shell of liquid follows the bubble wall, the peak getting closer to the boundary, until, at maximum compression, the maximum liquid pressure occurs at the bubble wall, equal to the gas-pressure inside, given by: QZ^γ or $Q[P_m(\gamma - 1)/Q]^{\gamma/(\gamma-1)}$. The very high liquid pressures developed around the imploding bubble are important in certain well-known effects of ultrasonics, such as erosion, dispersion and molecular degradation. The photograph of fig. 19 shows the erosion pitmarks that develop on the radiating surface of a metallic magnetostriction transducer driven for long periods at high intensity. The erosion pattern follows the cavitation streamer activity, the significance of which we will discuss later. The high temperatures developed within the

Table 2
Comparison of compressible and incompressible theories
for collapse of gas-filled cavities ($\gamma = 1.4$)

Q/P_m	$R_m/R_{\min}$	
	Incompressible theory	Compressible theory
		$P_m = 1 \text{ Bar}$ $P_m = 10 \text{ Bar}$
10^{-1}	0.262	0.28
10^{-2}	0.047	0.060 0.074
10^{-3}	0.0069	0.018 0.025
10^{-4}	0.0010	0.006 0.009

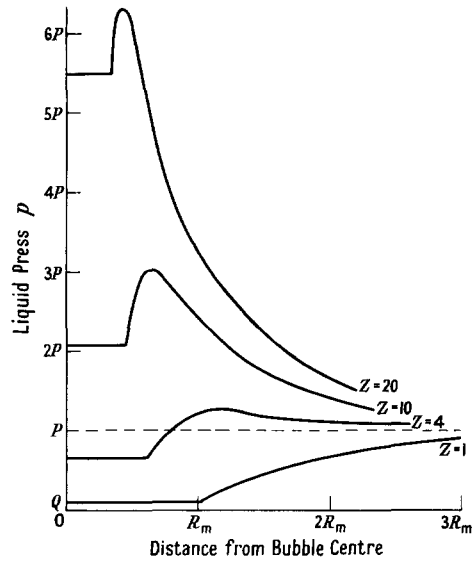


Fig. 18. Pressure developed in the liquid surrounding a collapsing gas filled bubble; Z = volume compression ratio; $P_m/Q = 10$.

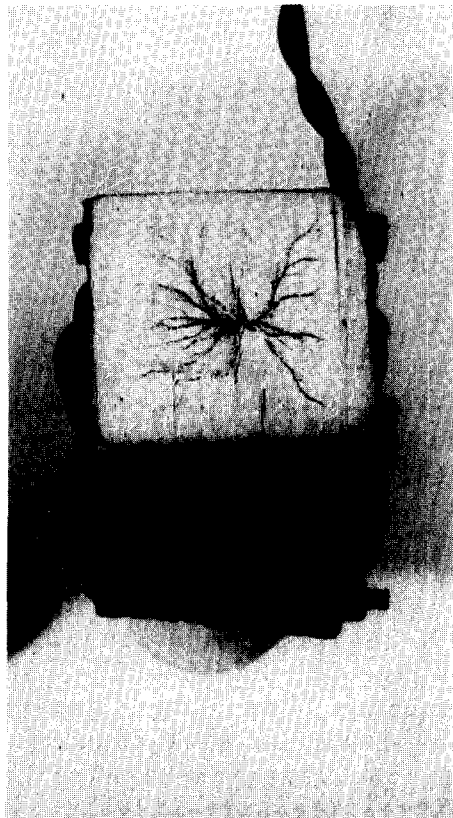


Fig. 19. Erosion pitmarks on the radiating surface of a nickel magnetostriction transducer driven at high intensity in water; $f = 25$ kHz.

collapsing gas-filled transient are responsible for sono-luminescence and sono-chemical effects. The maximum temperature reached is just $T_0 Z^{(\gamma-1)}$ where T_0 is the ambient temperature. Re-expressed in terms of P_m and Q the maximum pressure and temperature are respectively $Q(P_m/3Q)^4$ and $T_0(P_m/3Q)$, where we have taken $\gamma = \frac{4}{3}$. With $P_m = 1$ Bar and $Q = 0.01$ Bar, the maximum pressure is about 1.2×10^4 Bar or 120 kg/mm². An intermittent or alternating pressure of this magnitude would be sufficient to fatigue most metals. With $T_0 = 300$ K, the maximum temperature would be about 10^4 K, sufficient to account for sono-luminescent and sono-chemical effects.

The equation (95) gives reasonably accurate results for the collapse of gas-filled bubbles while fluid velocities remain below about 0.2 M. An even closer approach to reality is obtained by using the Herring–Trilling equation (24). Putting $P_\infty = P_m$ and $P_L = Q(R_m/R)^{3\gamma}$ in (24) gives:

$$R\dot{R} \frac{d\dot{R}}{dR} (1 - 2M) + \frac{3}{2}\dot{R}^2 (1 - 4M) = \frac{P_m}{\rho} - \frac{Q}{\rho} \left(\frac{R_m}{R} \right)^{3\gamma} (1 + 3\gamma M) \quad (106)$$

with $M = \dot{R}/c$. $\dot{R}$ is obtained in terms of R by a numerical integration and a second integration gives the R – t characteristic. Examples have been given by Trilling [16]. Gilmore [17] obtained several very complicated expressions for the pressure field in this case. When the acoustic approximation is used (eq. (26)), the pressure field can be obtained explicitly in terms of $\dot{R}$ and R :

$$p = \frac{R}{r} (P_L - P_m) + \frac{R}{r} \left(1 - \frac{R^3}{r^3} \right)^{\frac{1}{2}} \rho \dot{R}^2 + \frac{\dot{R}}{c} \left(1 - \frac{R}{r} \right) \left[\rho \dot{R}^2 - 2(P_L - P_m) - R \frac{dP_L}{dR} \right] \quad (107)$$

where $P_L = Q(R_m/R)^{3\gamma}$. This equation must be solved simultaneously with (26) to obtain the pressure as a function of r and R .

When M is known to approach or exceed unity, it is necessary to go one step further and make use of the Gilmore approximation (36) or the exact equations (31 and 32). Hickling and Plesset [13] obtained the most complete set of computed results available for the bubble-wall motion and the pressure field in the liquid. They used a combination of the Gilmore theory and solutions of the exact equations. Fig. 20, reproduced from their paper, shows the dependence of the cavity-wall velocity on the radial compression ratio for several values of γ and ambient pressure. The exact solution is compared with that obtained from Gilmore's theory (which uses the Kirkwood–Bethe hypothesis) and with the incompressible case. Gilmore's theory gives results in close agreement with the exact theory, except in the final stages of collapse.

The distribution of pressure in the liquid during the implosion is shown in fig. 21a from the same source. The initial pressure of gas in the bubble was taken as 0.003 Bar. The results are given for successive times, measured from the time of minimum bubble size, expressed as a fraction of the time for complete collapse. These results may be compared with those shown in fig. 18, derived on the basis of the incompressible assumption and where the compression ratio rather than the collapse time is used as the parameter. Also shown, in fig. 21b, is the pressure wave formed on the rebound and its progress as it propagates into the liquid and generates a shock front. The pressure peak forms very close to the bubble wall and attenuates approximately as $1/r$ in travelling outwards from the source. The peak pressures are therefore confined to the immediate vicinity of the collapsed bubble. Extrapolating from the limit to which the computations apply, a peak pressure of about 200 Bar is obtained at $r/R_m = 2$ with $Q = 10^{-3}$ Bar. This is hardly sufficient to cause any severe damage to tough materials.

Ivany and Hammitt [85] carried through computations similar to those of Hickling and Plesset. They

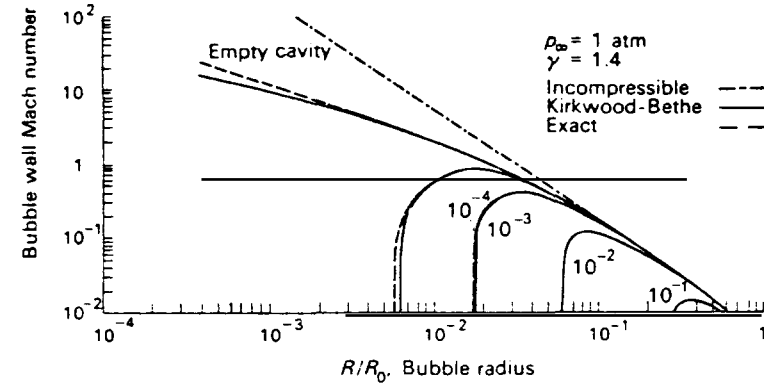
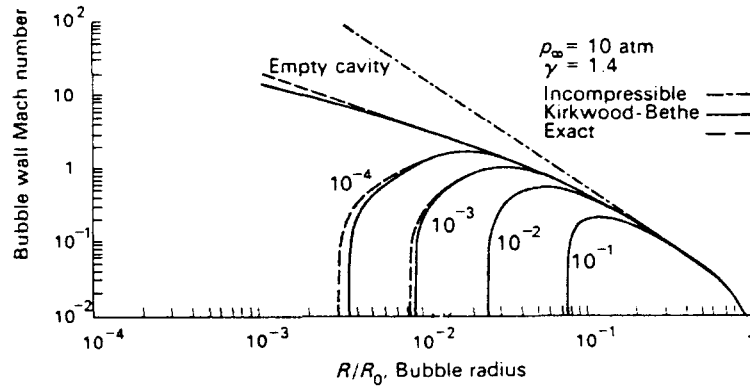
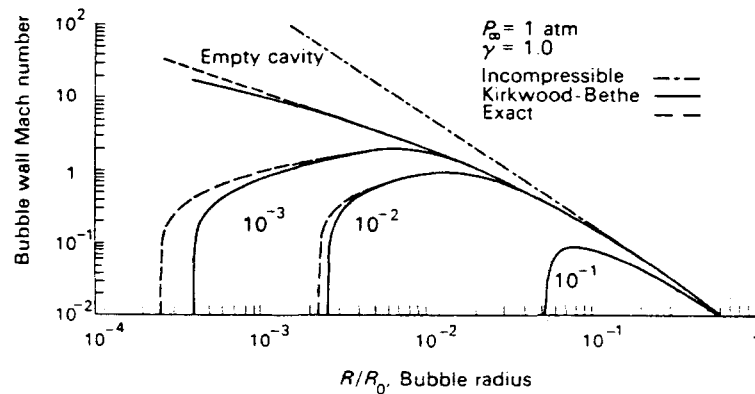
(a) Gas constant $\gamma = 1.4$; ambient pressure $p_\infty = 1$ atm.(b) Gas constant $\gamma = 1.4$; ambient pressure $p_\infty = 10$ atm.(c) Gas constant $\gamma = 1.0$; ambient pressure $p_\infty = 1$ atm.

Fig. 20. Comparison of cavity-wall velocities developed on implosion for three models of empty and gas-filled cavities, and for various values of γ , ambient pressure ($P_\infty = P_0$) and gas-content (Q). The numbers on the curves refer to Q in Bar.

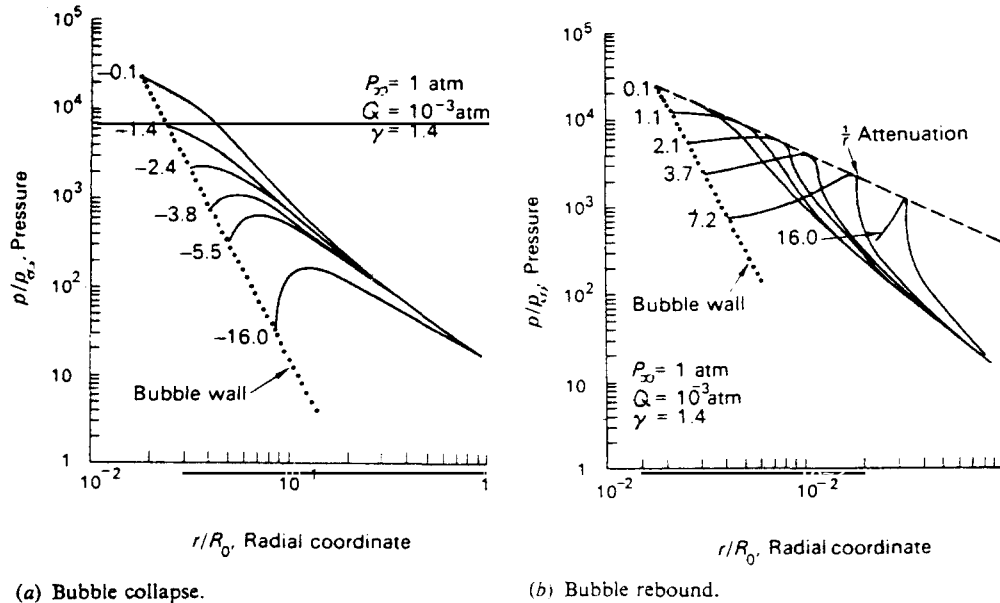


Fig. 21. Pressure developed in the liquid surrounding a gas-filled bubble collapsing and rebounding in water. $P_\infty = P_m = 1$ Bar; $Q = 10^{-3}$ Bar; $\gamma = 1.4$. Numbers on the curves are proportional to the time elapsed from the time at minimum bubble size.

included the effects of surface tension and viscosity. Their results confirm the assumption we have usually made, that neither surface tension nor viscosity affect the general behaviour of the collapsing bubble.

4.3. The generation of shock waves

Following Flynn [1] it is instructive to consider the conditions under which the cavitating bubble acts as a source of sound waves propagating into the liquid as the drive frequency and amplitude are increased: (i) For very slow changes in bubble size, disturbances can be regarded as propagating instantaneously. This is a stable condition and the appropriate model for the liquid would be an incompressible one. The cavity must then be regarded as non-radiating, no work being done on the liquid. All of the energy passed to the liquid during the expansion phase of the bubble oscillation is returned to the bubble on contraction. This is just equivalent to saying that the radiation resistance is negligible when ω is very small; (ii) For more rapid changes in the bubble size, it will eventually become meaningful to say that the pressure developed in the liquid at any point will depend on the motion of the bubble at some earlier time; that is, there is a time-lag between pressure measured in the liquid and the motion of the bubble wall. This time-lag implies energy dissipation, this energy being propagated into the surrounding liquid, which then can be considered to present a finite radiation resistance to the bubble. The appropriate model in this case would be the acoustic approximation; (iii) However, we know that the velocity of sound in the liquid is an increasing function of the pressure. In a transient event, the liquid pressure around the bubble as it implodes and rebounds may increase to the point where the velocity of sound is seriously affected. Disturbances would then tend to overtake those that started earlier. The waves would then start to crowd one another and this is just the condition required for a shock-wave to develop. The appropriate model to use would then be one, like the Gilmore approximation, where the velocity of sound is a function of the motion.

The strength of the cavitation shocks and their range of action have been studied by: Schneider [83], Brand [86], Brook Benjamin [87], Akulichev and colleagues [88], and in detail, by Hickling and Plesset [13]. As we have seen, Hickling and Plesset found radiated pressure waves obeying the geometric attenuation law (amplitude $\propto 1/r$). However, Akulichev and colleagues, basing their computations on the Gilmore approximation, found the attenuation much more rapid than this as the pressure peak moves away from the cavity wall. These authors also show that if a shock wave is to form at all it must develop within about $r = 20R_0$ from the cavity. In any case, as Hickling and Plesset also showed, the shock will remain strong only within a few radial distances from the bubble. Therefore, cavitation effects that depend on shock pressures can take place only in the immediate vicinity of the collapsing bubble. This has been confirmed experimentally.

Brooke Benjamin [87] used a very simple model to study where, on the rebound, the shock would form. He found a criterion in terms of the maximum gas pressure, $P_{\max}$, developed in the collapsing bubble. With $P_{\max}$ in kilo-Bar he found the following simple formula for the distance r at which the shock would form:

$$P_{\max} \approx 13.6 \left(\log \frac{r}{R_{\min}} \right)^{-2/3}. \quad (108)$$

On this basis, when $P_{\max}$ is much less than 2 kilo-Bar the shock can form only in remote parts of the field and will be very weak. But with $P_{\max}$ between 2 and 3 kilo-Bar it is likely that the shock will form at a reasonable distance. The more recent and more exact calculations of Hickling and Plesset give $r/R_{\min}$ about 5–6 when $P_{\max} > 1$ kilo-Bar.

In all the above theory we have assumed that the imploding bubble will remain spherical. Photographic evidence, to be discussed later show that the spherical surface can be expected to distort and the implosion is usually violent enough to fragment the bubble.

4.4. *Acoustically-generated transient cavities*

We have seen that the collapse of gas-filled transients can be analysed in detail once the three parameters P_m , R_m and Q are known. In an acoustic field, the cavity will have grown from some above-threshold size, R_0 , and these parameters will be functions of the four “given” quantities: R_0 , P_A , P_0 and ω . Several research groups have obtained many numerical solutions of eq. (14). Examples of stable, and nearly-stable, solutions are given in fig. 11. The results reproduced in fig. 22 refer to true transients, that is, cavities that collapse completely within a single acoustic cycle. These, and similar, data provide useful information on the dependence of P_m , R_m and Q on the primary variables. For example, we have taken the co-ordinates (R_m, t_m) at the peak of bubble-growth and graphed these as functions of R_0 , P_0 and ω^{-1} . Results are displayed in figures 23–25. Of special interest is fig. 25 showing R_m and t_m roughly proportional to the acoustic period over the range considered. Collapse always occurs while the acoustic pressure is near its peak. This justifies our approximation:

$$P_m \approx (P_A + P_0).$$

Although we have ample evidence that typical gas-filled transients collapse adiabatically, further investigation is needed to establish the growth conditions. But we easily see that nowhere during the expansion phase can $\dot{R}$ exceed $[2(P_g + P_A - P_0)/3\rho]^{1/2}$ or approximately $(2P_A/3\rho)^{1/2}$ for large bubbles.

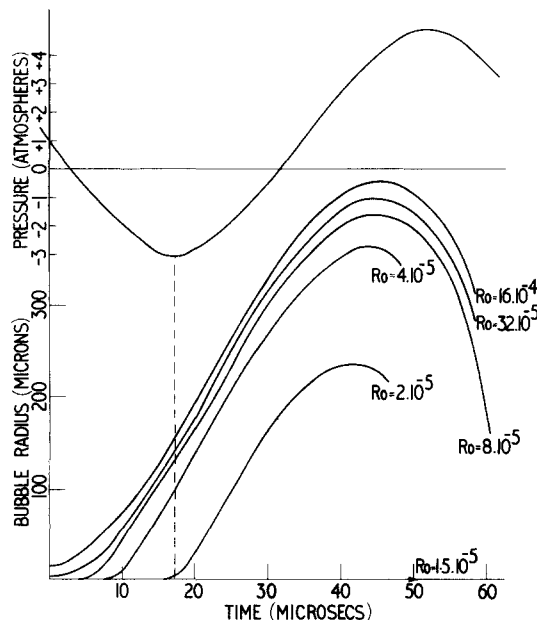


Fig. 22. Radius-time curves showing expansion and collapse of air bubbles in water under transient conditions. $P_A = 4$ Bar; $P_0 = 1$ Bar; $f = 15$ kHz.

For example, for the condition depicted in fig. 22, which corresponds to intense transient cavitation, the Mach number for the expansion-phase cannot exceed about 0.01. Clearly, we are justified in assuming isothermal conditions for the growth-phase, and R_0 , R_m , P_0 and Q are then simply related by the formula $Q = (P_0 + 2\sigma/R_0)(R_0/R_m)^3$. An approximate relation is therefore established between the parameters R_m , P_m , Q and the primary variables R_0 , P_A , P_0 and the collapse conditions can be analysed in terms of these variables.

Flynn [81] has produced a precise mathematical formulation covering the problem of the acoustically-excited cavity, which embraces stable and transient conditions. The treatment includes the effects of thermal conduction, viscosity, surface tension and compressibility of the liquid to a limited extent, using

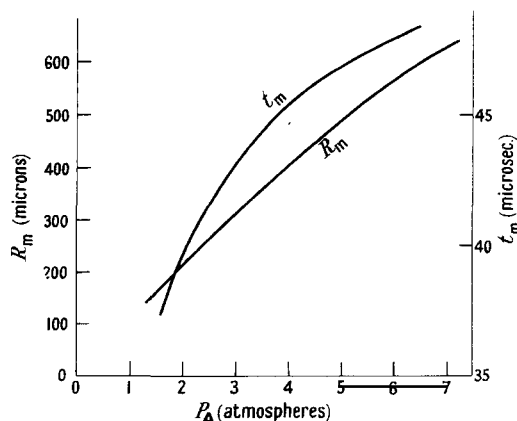


Fig. 23. Dependence of R_m and t_m on R_0 for air bubbles collapsing as transients in water. $P_A = 4$ Bar; $f = 15$ kHz.

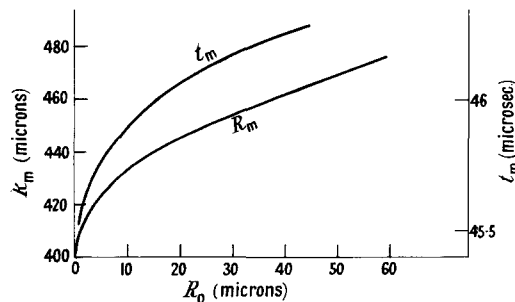


Fig. 24. Dependence of R_m and t_m on P_A for air bubbles collapsing as transients in water. $R_0 = 3.2 \times 10^{-4}$ cm; $f = 15$ kHz.

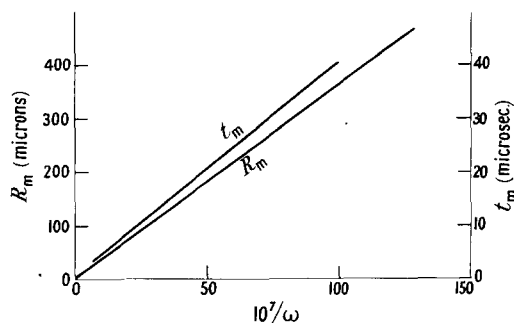


Fig. 25. Dependence of R_m and t_m on ω^{-1} for air bubbles collapsing as transients in water. $R_0 = 3.2 \times 10^{-4}$ cm; $P_A = 4$ Bar.

Gilmore's equation. A comprehensive treatment of this sort necessarily involves assumptions and approximations, some plausible, others less so. Evaporation-condensation at the bubble-wall is neglected and it is assumed that the sound velocity is not affected by the motion. The result is a set of 11 simultaneous equations that can be programmed to give answers in a few minutes using a modern high-speed digital computer. Results are in reasonable agreement with the exact theory of Hickling and Plesset, which relates only to the collapse and rebound of transients. For full details of Flynn's formulation, the reader is referred to ref. [81].

5. Cavitation thresholds

For studying the thresholds, it is sufficient to use the simplest non-linear equation for the motion of the bubble wall (eq. (14)). The thresholds can be located from solutions obtained over a wide range of the parameters P_0 , P_A , R_0 and ω . To illustrate this we may refer back to fig. 11 which shows how the bubble motion can switch from stable to transient and eventually back again to stable, as the excitation pressure is increased. Upper and lower thresholds therefore exist for P_A between which transient conditions hold, while outside the motion is stable. If P_A had been held constant and R_0 used as the variable quantity, the threshold could have been defined in terms of R_0 instead of P_A . Very small bubbles oscillate stably, at a frequency close to the drive frequency, their motion controlled by surface tension. Bubbles too large to collapse as transients in the time available also oscillate stably, at a frequency close to their own natural resonance frequency. Similarly, the thresholds could equally well be expressed in terms of the static ambient pressure, P_0 , if both P_A and R_0 are regarded as constant. Summarising, therefore, the thresholds can be stated in terms of any of the three parameters P_A , P_0 or R_0 , although naturally P_A is generally used, this being the easily-variable quantity.

The transition between stable and transient conditions is often sharply defined. When the term "cavitation threshold" is used without qualification, it is the lower of the two thresholds that is meant. An ability to predict this threshold may be quite important. When transient cavitation would be too damaging or annoying, as in underwater signalling or medical diagnostics or therapy, it will generally be essential to operate within or below the stable regime. On the other hand, as we have seen, transient conditions are desirable in several industrial applications of macrosound. Much experimental work has therefore been directed to locating these thresholds in terms of the acoustic and environmental parameters. Much of this work was carried out in the years 1946–1960 and is well summarised in earlier reviews. We shall therefore not review it here.

5.1. Structural stability of cavitation bubbles

For bubbles below resonance size, yet so large that surface tension is unimportant in controlling the motion, the threshold P_A is very close to P_0 . To illustrate the sharp transition in this important case, fig. 26, from ref. [1], shows solutions of eq. (14) taken over a single acoustic cycle, where P_A/P_0 is increased over the range 0.75–1.0 in small steps. Increasing P_A accelerates the growth of the bubble, while the maximum collapse speed increases over the range 0.02–1.6 M . The non-linear stable oscillation reverts to transient when P_A/P_0 reaches about unity. The liquid is then about to go into tension on the negative half-cycle. At the threshold, the bubble has grown to a maximum size R_m such that $R_m/R_0 \approx 2$, as we might expect intuitively. This is an important threshold condition, deducible from the collapse conditions. At the threshold, the collapse curve is practically identical with the Rayleigh solution over all but the last stages.

Soviet workers [2] and [76] have described a picturesque way of showing the transition from stable to transient conditions. Solutions of eq. (14) are plotted out on the phase-plane. This involves recording dR/dt as a function of R . Stable solutions give closed curves, transients show discontinuities, the curve running away to infinity in some direction. An example is reproduced in fig. 27. The plots refer to an air bubble in water radius $R_0 = 5 \times 10^{-4}$ cm driven at a frequency of 500 kHz. The numbers on the curves refer to the ratio P_A/P_0 and x is written for the dimensionless time, ωt . Conditions are stable for $P_A/P_0 \leq$ about unity; transient for P_A/P_0 between 1 and about 4; and transient, but with a stable component, for P_A/P_0 greater than about 10. Although not shown in this example, for P_A/P_0 very large, the plots must degrade into a series of loops that close up as the oscillations revert to the stable form.

5.2. The transient thresholds

Bubbles below resonance size are stiffness-controlled. In this case, the lower transient threshold is very sharply defined. As it can be expressed in terms of either P_A , P_0 or R_0 it is not surprising that a simple relation should exist between these three quantities, defining the threshold.

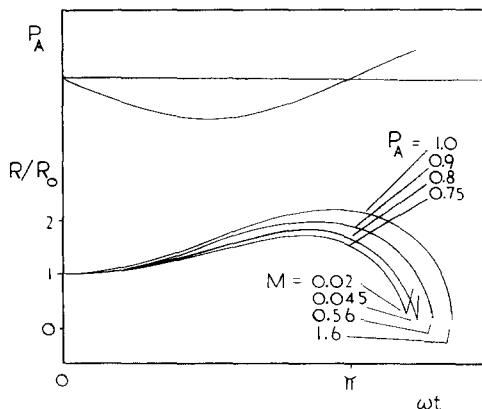


Fig. 26. Radius-time curves for an air bubble in water, plotted for increasing P_A/P_0 in the range 0.75–1.0, showing the development of transient conditions. $R_0 = 2.6 \times 10^{-3}$ cm; $f = 24.5$ kHz.

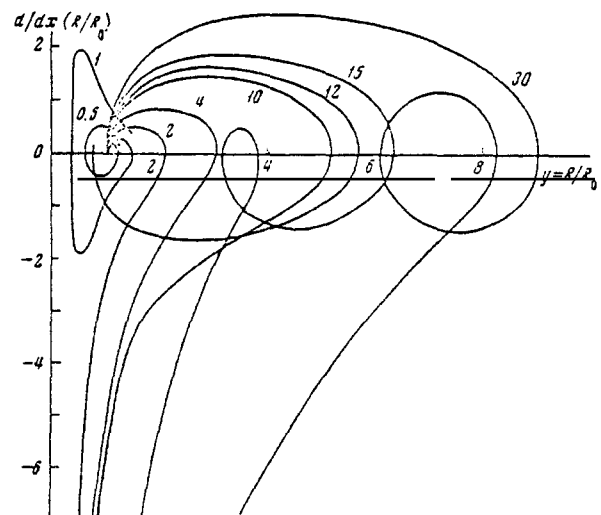


Fig. 27. Solutions of eq. (14) represented in the phase-plane. $R_0 = 5 \times 10^{-4}$ cm; $f = 500$ kHz; $x = \omega t$. The numbers on the curves refer to the ratio P_A/P_0 .

The liquid pressure just outside the bubble: $P_L = (P_0 + 2\sigma/R_0)(R_0/R)^3 - 2\sigma/R$ is a minimum when $dP_L/dR = 0$, that is, where

$$R = R_T = \left[\frac{3R_0^3}{2\sigma} \left(P_0 + \frac{2\sigma}{R_0} \right) \right]^{1/2}.$$

An alternative expression for P_L is $(P_0 - P_A \sin \omega t)$ with a maximum negative value $(P_A - P_0)$. The condition for instability is therefore: $P_T > P_0 + (P_0 + 2\sigma/R_0)(R_0/R_T)^3 - 2\sigma/R_T$, or, substituting for R_T :

$$P_T > P_0 + \frac{4}{3} \left[\frac{2\sigma^3}{3R_0^3(P_0 + 2\sigma/R_0)} \right]^{1/2}. \quad (109)$$

This equation gives the pressure-threshold in terms of the other two relevant parameters, P_0 and R_0 . It can be written: $P_T - P_0 = 4\sigma/3R_T$. If the bubble is so small that the surface tension pressure, $2\sigma/R_0$ overwhelms P_0 , the following simpler result is obtained: $P_T - P_0 \approx 4\sigma/3\sqrt{3}R_0$. Similarly, if P_0 is significantly greater than $2\sigma/R_0$ but with the bubble still below resonance size, we obtain: $(P_T - P_0)^2 \approx 32\sigma^3/27P_0R_0^3$. For $P_0 \gg 2\sigma/R_0$ we obtain $P_T \approx P_0$, as we have already seen.

These threshold conditions apply only to gas-filled bubbles. Similar conditions do not exist for vaporous cavities. However, if the nucleus was an unwetted solid sphere, the threshold tension needed to separate the liquid from the solid would be just: $(P_T - P_0) = 2\sigma/R_0$. This shows that the tension required to expand a gas-filled bubble is less by a factor of about 2.5 than for a spherical non-wetted solid nucleus of the same size.

It is interesting to compare the threshold given by eq. (109) with that obtained by solving the complete equation of motion (14). This is best displayed by plotting out R_0 against R_m taken from the $R-t$ curves. Figure 28 shows the result for a typical case of air bubbles in water with $P_A = 4$ Bar; $P_0 = 1$ Bar, over a wide range of bubble sizes. A sudden increase in R_m is seen as the threshold R_0 is passed. This value agrees closely with that given by eq. (109) on inserting the above values of P_A and P_0 .

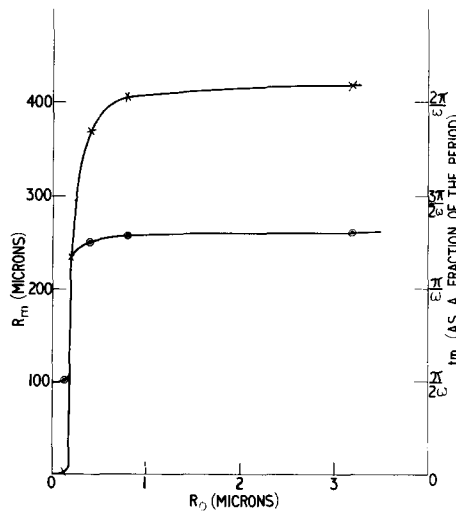


Fig. 28. Dependence of R_m (×) and t_m (•) on R_0 for air bubbles in water, with R_0 variable over the threshold range. $P_A = 4$ Bar; $P_0 = 1$ Bar; $f = 15$ kHz.

It is interesting to note that all bubbles small compared with resonant dimensions grow to approximately the same maximum size.

In deriving the threshold condition (109) it has been possible to neglect the inertial terms, as it relates only to small bubbles, well below resonance size. Therefore, it does not involve frequency as a variable. However, we know that for large bubbles the threshold may become very frequency-dependent, and bubbles above resonance size tend to revert to stable oscillation. For example, the full-line curve of fig. 29 is a solution of eq. (14) for an air bubble in water, radius 8×10^{-5} cm, driven at a frequency of 15 MHz, with $P_A = 4$ Bar and $P_0 = 1$ Bar. The plot is taken over four complete cycles of pressure. In this case, a stable form of oscillation is obtained. The broken-line curve refers to identical conditions, except that the frequency has been reduced to 5 MHz. At this frequency the bubble is below resonance size, and, as expected, the cavitation is transient. Clearly, for any given bubble size, it is always possible to take the frequency high enough to inhibit transient cavitation. In fact, this can be extended to refer to any bubble field with any distribution of bubble sizes. It is a well-known observation that it becomes very difficult to cavitate water at frequencies above about 3 MHz.

Having extracted solutions of the non-linear equation of motion, either one of two simple criteria can be used to identify the transient threshold: $R_m/R_0 \approx 2$ and $\dot{R}_{\max}/c = M \approx 1$. Lauterborn [89] used the latter criterion and produced threshold curves relating P_T to R_0 or ω for the condition $M = 1$. Results agree well with plots of eq. (109) at frequencies below resonance. Through resonance, the Lauterborn thresholds will depend on whatever damping is assumed. More recently, Ieretti and colleagues [90–92] measured transient thresholds through resonance using the onset of the half-order subharmonic emission and sono-luminescence as the threshold criteria. In place of the smooth P_T – R_0 characteristic predicted by Lauterborn, they found many small discontinuities, corresponding to resonances at harmonics and subharmonics of the excitation frequency. This suggested a theoretical study in which the

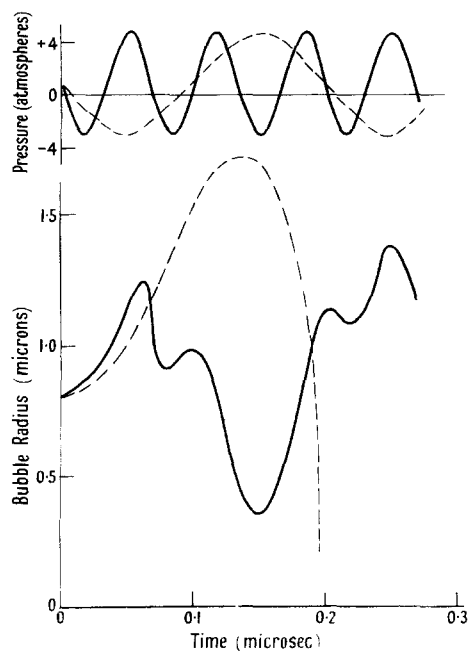


Fig. 29. Radius-time curves for an air bubble in water: $R_0 = 8 \times 10^{-5}$ cm; $P_A = 4$ Bar; $P_0 = 1$ Bar; full-line curve 15 MHz; broken-line curve 5 MHz.

equations (14) and (26) were both used. This confirmed the presence of the discontinuities and the reasons for them. Figure 30 displays a set of results obtained by Creschia et al. [92] (full-line curve) along with the corresponding Lauterborn threshold curve (broken-line) for air bubbles in water at 700 kHz.

The simple threshold condition (109) applies only in the region of stiffness-control. Apfel [93] has shown that a simple approximation to the threshold condition can also be obtained for large bubbles above the resonance size. We first express R_m in terms of R_0 and to this end we examine the bubble growth in two stages: (i) Consider growth while the liquid is still in tension, represented by time t_1 . Then, writing $\Delta P = (P_A - P_0)$, the average bubble wall velocity is approximately $(2\Delta P/3\rho)^{1/2}$ and the radius will increase by an amount $R_1 = t_1(2\Delta P/3\rho)^{1/2}$. Approximate values for t_1 and ΔP can be obtained by considering the second-order approximation to the pressure variation about the phase of maximum tension ($\omega t = \pi/2$). In this way, writing $p = P_A/P_0$ we find:

$$(\omega t_1)^2 = 8(p - 1)/p \quad \text{and} \quad \Delta P = 2P_0(p - 1)/3.$$

Thus, $R_1 = (4/3\omega)(p - 1)(2P_0/\rho)^{1/2}$. (ii) Consider the period from stage (i) until the momentum of the bubble-wall becomes zero, when growth will cease. The further increase in bubble size can be found by equating the initial kinetic energy of the liquid surrounding the bubble to the change in potential energy of its contents while it expands to its maximum size R_m . As the bubble is assumed to be large, we neglect surface tension. We also assume that the ambient pressure remains at P_0 during this period. Then

$$(4\pi R_1^3 \rho) V^2 / 2 = 4\pi(R_m^3 - R_1^3)P_0/3$$

where $V^2 = 2\Delta P/3\rho = 4P_0(p - 1)/9\rho$. This gives $(R_m/R_1)^3 = 1 + 2(p - 1)/3$ and on substituting for R_1 we

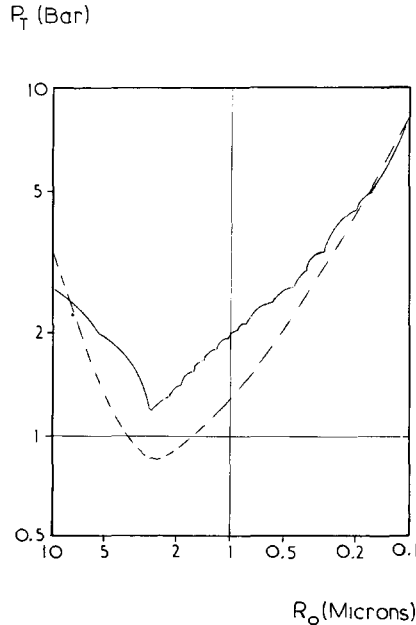


Fig. 30. Transient cavitation threshold for air bubbles in water (ref. [92]).

obtain:

$$R_m = (4/3\omega) (p-1) (2P_0/\rho p)^{1/2} [1 + 2(p-1)/3]^{1/3}.$$

Making use of the resonance condition $\rho\omega^2 R_r^2 = 3P_0$ and using the criterion $R_m = 2R_0$ to define the transient threshold, we get:

$$\frac{R_0}{R_r} = \frac{2}{3}(p-1) (2/3p)^{1/2} [1 + 2(p-1)/3]^{1/3}. \quad (110)$$

This condition is independent of all gas and liquid constants. It can only be even approximately valid for p appreciably > 1 .

In fig. 31 the expressions (109) and (110) are plotted out over the appropriate ranges of bubble sizes for air bubbles in water at the frequency 1 MHz. Over their limited range, these easily-calculated thresholds agree quite well with computed results using either of the two criteria: $R_m/R_0 = 2$ or $M = 1$.

The above theory refers to the lower transient threshold. The upper threshold is of less practical importance. Also, it cannot be expressed in any simple formula, but must be inferred from numerical solutions of the non-linear equations. For example, fig. 32 shows the calculated maximum liquid pressure (p_m) developed in a transient collapse as a function of the ambient static pressure (P_0). p_m was calculated on the assumption that the bubble, with equilibrium radius $R_0 = 1.6 \times 10^{-4}$ cm, contained only vapour at a constant pressure of 10^4 dyne/cm². The acoustic pressure amplitude was taken to be constant at 4 Bar at frequency 15 kHz. The corresponding maximum radius reached by the bubble (R_m) is also indicated. The curve for p_m shows upper and lower thresholds in P_0 outside which transient pressures will not be developed. In this case, the lower threshold for P_A/P_0 is about 1.1 and the upper threshold about 6.0. The peak of p_m , indicating maximum cavitation intensity requires P_0 to be increased well above the normal ambient level. This theoretical prediction of an optimum operating point between

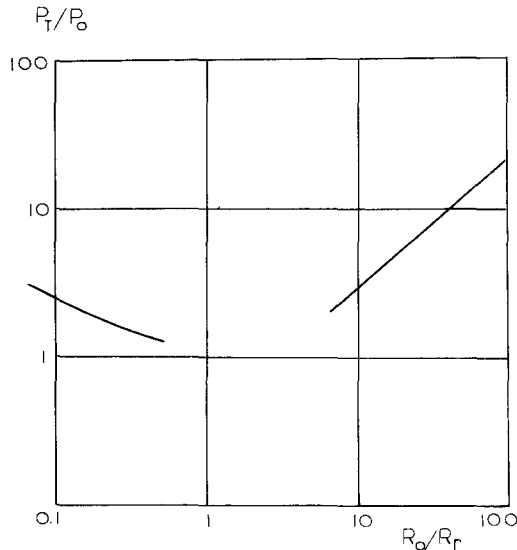


Fig. 31. Transient cavitation threshold for air bubbles in water, plotted from the approximate formulae (109) and (110). $f = 1$ MHz.

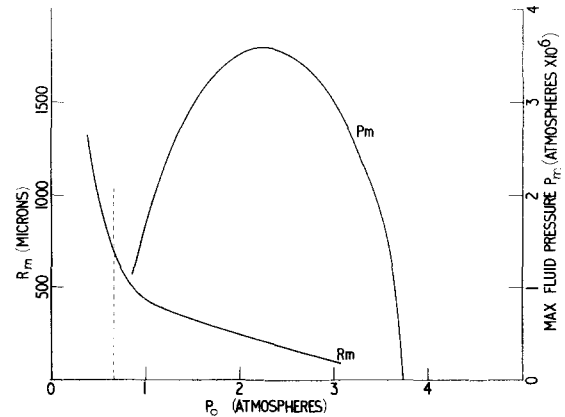


Fig. 32. Maximum liquid pressure (p_m) and maximum radius (R_m) for a collapsing vapour-filled bubble as functions of the ambient pressure.

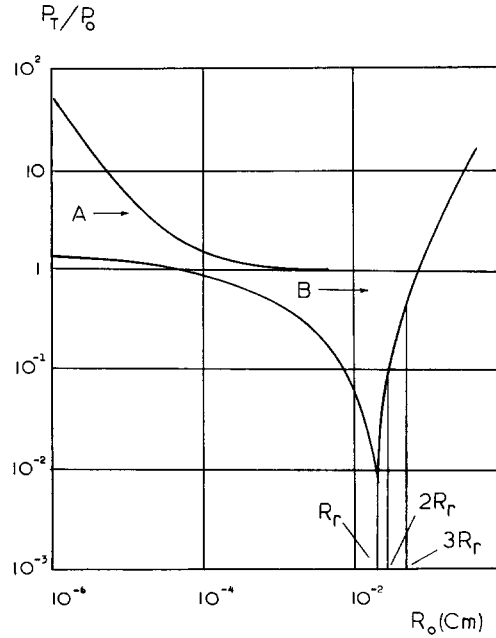


Fig. 33. Stable and transient cavitation thresholds for air-saturated water at $f = 20$ kHz.

the thresholds has been amply demonstrated in practice, and is important in several industrial applications of macrosound.

5.3. The stable cavitation threshold

The threshold for bubble-growth by rectified diffusion, eq. (64), applies over the full range of bubble sizes, through resonance. Above this threshold, bubbles can grow in the sound field to become active. Below it, they will dissolve away or become stabilised against dissolution in some way. At all events, they will cease to be active. For these reasons, the diffusion threshold is frequently referred to as the Stable Cavitation Threshold. The stable and transient thresholds relate the same parameters: P_A , P_0 and R_0 . It is instructive to see them displayed together, as in fig. 33 where the plots refer to air bubbles in air-saturated water at a frequency of 20 kHz.

6. Cyclic cavitation processes

In fig. 33 the Transient Threshold curve has been plotted out from eq. (109). For the Stable Threshold characteristic, eq. (64) is re-formed to display the threshold pressure, P_T , in terms of the bubble radius, R_0 :

$$P_T^2 = \frac{3}{2} [AP_0^2 / (A + R_0)] [(1 - R_0^2 B)^2 + R_0^2 B \delta] \quad (111)$$

where $A = 2\sigma/P_0$ and $B = \omega^2 \rho / 3\gamma P_0$. The curves refer to air-saturated water at normal atmospheric pressure ($\rho = 1$; $\gamma = 1.4$; $P_0 = 10^6$ c.g.s units) at 20 kHz, where the damping factor, δ , for a resonant bubble is 0.0645.

6.1. The gaseous cavitation cycle

Any bubble entering the region below the Stable Threshold cannot grow by diffusion, and will shrink, eventually passing into solution. Bubbles located in the region above the Transient Threshold must immediately expand and collapse as transients. But any bubble finding itself in the region between the thresholds, say at A, in fig. 33, can grow by rectified diffusion. To the right of A the slope of the transient threshold curve is negative, so the growing bubble will eventually reach it and will then immediately expand, implode and disintegrate. The debris will consist of small bubbles, many of which will reach the “dead” region below the stable threshold and disappear. But some may be large enough to re-nucleate the region between the thresholds. These will then grow again and the cycle will repeat. A repetitive cyclic process is therefore set up, with the transient activity located at the threshold. This is called the *Gaseous Cavitation Cycle*. It is easily picked out by examining successive frames of high-speed photographs of cavitation in gassy liquids. However, it is not very regular, as the small bubbles are very variable in size. Neither do they necessarily return to the same pressure region of the field.

6.2. The de-gassing cycle

If the bubble is located in a region of the field where $P_A < P_0$, say at B, it can also grow in the sound field. But it is unlikely to reach the transient threshold. It will continue growing and may eventually become large enough to separate out under gravity. This illustrates the *De-gassing Process*, a well-known effect of stable cavitation. It is used industrially in applications such as: degassing optical glass, metal melts, photographic emulsions, etc. De-gassing may also become intermittent or cyclic. This is seen in the focal regions of ultrasonic concentrators where activity will cease after a time until the region can be replenished by gas diffusing in from surrounding areas. It is clear that de-gassing can only occur within a restricted range of R_0 and P_A and its rate can be maximised by choosing appropriate treatment conditions related to the bubble-size distribution.

6.3. The resonant bubble cycle with emission of micro-bubbles

At sufficiently high frequencies, bubbles will reach their resonance size before separating out under gravity. At radial resonance there is a high probability that instability will set in, encouraging surface modes of vibration. The strong surface vibrations are parametrically excited at half the excitation frequency and are strongly coupled to the radial motion. The equation for the motion of the bubble wall then becomes [94]:

$$R_1 \ddot{R}_0 + 3\dot{R}_1 \dot{R}_0 = [(n-1)\ddot{R}_1 - (n-1)(n+1)(n+2)\sigma/\rho R_1^2]R_0 \quad (112)$$

where R_1 is the radius of curvature of the surface disturbance and n the mode-order of the surface oscillation. The free resonance frequency for modes with symmetry about a diameter ($n \geq 2$) is given by:

$$\omega_1^2 = (n-1)(n+1)(n+2)\sigma/\rho R_1^3. \quad (113)$$

If damping is included, a threshold must exist. This has been evaluated by Sorokin [95] and by Eisenmenger [96] for the case of a plane surface. The threshold oscillatory amplitude, ξ_T , is given by

$\xi_T = 2B/k\omega$ where B is the damping coefficient. For surface vibrations, only viscous damping is important and $B = 2\mu k^2/\rho$ and k is given by the ripple-wave formula: $k^3 = \rho\omega^2/4\sigma$. Then

$$\xi_T = (16\mu/\omega\sigma\rho^2)^{1/3}. \quad (114)$$

Although this applies to a plane surface, the result is also approximately valid for a sphere if the mode-order is high. For large resonant bubbles, the approximate relation between acoustic pressure and oscillatory amplitude is: $P_T/\xi_T \approx 3\gamma\delta P_0/R_0$ and using (114) the threshold pressure for exciting surface waves on a resonant air bubble in water at 20 kHz is found to be about 0.0025 Bar. Clearly, in our example, fig. 33, there is no difficulty in exceeding this threshold. In an intense field, the surface waves may grow to large amplitudes and throw off micro-bubbles from the crests. This can occur very rapidly, the parent bubble apparently exploding, thus accounting for the “disappearing bubbles” that have been reported [97]. These micro-bubbles are all nearly constant in size, radii near $\lambda_s/4$ where λ_s is the wave-length of the surface vibration. This is about 10^{-3} cm in our case. Micro-bubbles can therefore return to the region between the thresholds, re-nucleating it. A cyclic process may then become established, where the bubble sizes at the beginning and end of the cycle are known. In the presence of a strong standing-wave field, the small bubbles will collect at the pressure anti-nodes while they are growing, but will tend to move away on reaching resonance size.

6.4. Bubble-growth above the resonance size

From eqs. (62) and (63) we see that the growth-rate of bubbles increases as resonance is approached. If the acoustic pressure is too weak to disintegrate them, the bubbles can grow through resonance. But above resonance size, the threshold increases, while the growth-rate decreases rapidly. Referring to fig. 33, if R_r is the resonance radius, the threshold pressures for growth by diffusion to radii R_r , $2R_r$ and $3R_r$ are seen to be 0.009, 0.1 and 0.4 Bar respectively. Bubbles at these sizes would resonate at frequencies near ω , $\omega/2$ and $\omega/3$. The probability of bubbles reaching these large sizes is clearly not high, but if they do they can oscillate as stable cavities generating the corresponding sub-harmonic signals. These sub-harmonic emissions have recently been studied energetically, and several authors have predicted thresholds for the $\omega/2$ emission (see section 10). For example, Eller and Flynn [98] give:

$$P_T = 6P_0[(\omega/\omega_r - 2)^2 + \delta^2]^{1/2}. \quad (115)$$

This shows that the threshold is just $6\delta P_0$ when the bubble resonance frequency, ω_r is half the excitation frequency. In our case, with $\delta = 0.0645$ at 20 kHz, the threshold for generating the half-order subharmonic is about 0.4 Bar. As we have seen, the threshold for the *formation* of such bubbles is 0.1 Bar. Further growth to the larger size required for emitting the 1/3 rd order subharmonic is less likely and this line is therefore rarely seen in the emission spectrum.

We should emphasise that this mechanism for generating the sub-harmonics applies only to bubbles in stable oscillation. A very different sequence of events is responsible for the strong signal at the half-order subharmonic that accompanies transient events (see section 10).

6.5. Cavitation relaxation

De Grois and Badilian [99–101] detected a regular cyclic change in a cavitating liquid, due to the medium switching between two states which they associated with “gaseous” and “vaporous” cavitation

conditions. They used a frequency of 1 MHz and acoustic intensities no greater than about 4 watt/cm². They found that in a gassy liquid, with many active bubbles present, on increasing the acoustic intensity a critical level was reached where the large gas bubbles would suddenly disappear. The liquid would then revert to the previous gaseous state and back to the bubble-free state in a regular cyclic fashion on time-constants that could be varied between a few tenths of a second and several minutes depending on the drive level. The gaseous state was characterised by a drop in temperature and reduction in acoustic intensity as measured by thermocouple probes. In this state it was possible to demonstrate well-known effects of transient cavitation, such as sono-luminescence, sono-chemical effects and increased erosive action. However, there was little noise. In the vaporous state, these effects were absent and erosive action reduced. But the noise increased, along with the “fountain” effect and atomisation of the liquid. The authors called this cyclic effect “cavitation relaxation”. It is clearly seen only at high frequencies and cannot be distinguished at the lower frequencies used in industrial applications. The authors’ explanation of this phenomenon is as follows. When gaseous cavitation starts, the medium becomes lossy and less transmissive, the bubbles being good scattering centres, causing most of the energy to be absorbed. The resulting temperature rise evaporates vapour into the bubbles, causing rapid transition to a vaporous state. The properties of the medium have now changed, as vaporous cavities have a low Q -value and do not oscillate and scatter in the same way as gas-filled bubbles. The medium is more transmissive and less energy is absorbed in it. Vapour condenses in the lower-temperature environment and there is a return to gaseous conditions. The cycle repeats on time-constants determined by the heat-exchange mechanisms involved. Several other effects of transient cavitation, other than those noted by the authors, are now thought to correlate with this type of cycling; notably certain electrical effects accompanying bubble activity, studied by Chincholle [102] and the time-dependence of the acoustic emission from cavitating fields observed by Lauterborn [103].

According to the theory presented by the authors [99] the time-constants controlling this cyclic effect depend on the rates of evaporation and condensation and all bubbles involved are operating as true transients. A possible alternative explanation is that the cyclic behaviour is simply an effect of bubbles moving into and out of resonance, the “gaseous” state corresponding to resonance. The scattering cross-section for resonant air bubbles in water is large. Absorption of the scattered energy will increase temperature, evaporating vapour into the bubbles so that they grow beyond resonance size. It is not difficult to calculate the energy needed to evaporate this small quantity of vapour. This fixes one time-constant of the cycle; the other refers to the condensation stage and is just as readily deduced. The effect can be visualised by tracing the changes in bubble size by reference to the diagram, fig. 33.

Figures 34 and 35, reproduced from ref. [101], relate to this relaxation process. Figure 34 shows measured changes in acoustic intensity and temperature as functions of time. In fig. 35 the time-durations, t_1 and t_2 , of the gaseous and vaporous states are shown as functions of the input power to the transducer. The acoustic intensity corresponding to the “cross-over” point was not greater than 3 watt/cm², well below the transient threshold at the 1 MHz frequency used. The cavitation was therefore stable.

6.6. *Cyclic behaviour involving exchange of free gas, aided by micro-streaming*

A useful type of cyclic behaviour, involving only stable cavities, can be observed by trapping a bubble on the surface of a radiator, the dimensions of which are comparable with the resonant bubble diameter [104]. When the surface is set into vibration, small bubbles, moving under radiation forces, collect near the centre of the surface. Growth occurs mainly by accumulating smaller bubbles until the resonance size is reached. Then surface activity sets in, becomes violent and disintegrates the bubble as

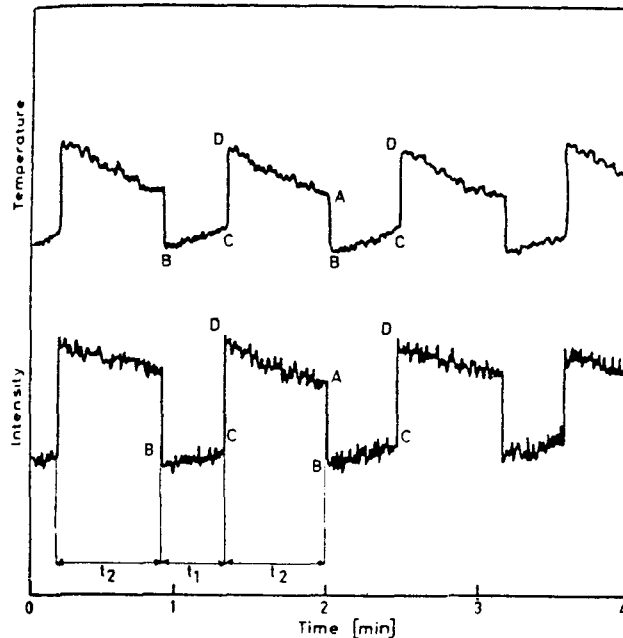


Fig. 34. Temperature and acoustic intensity as functions of time during cavitation relaxation. t_1 duration of gaseous-cavitation; t_2 vaporous.

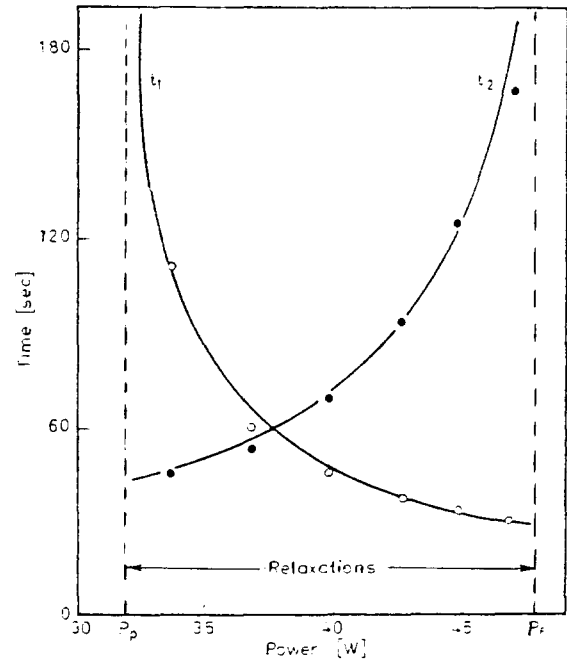


Fig. 35. Durations of the gaseous cavitation regime (t_1), and vaporous cavitation (t_2) as functions of the input electrical power during cavitation relaxation.

already explained. A small bubble remains and this grows again by accumulating micro-bubbles formed from the debris which are recirculated by strong microstreaming flows. Radiation forces are attractive, as all bubbles involved are below the resonance size.

This cyclic process has been studied in detail [104]. It is not only spectacular to watch, but also useful. It suggests a method of generating large numbers of small bubbles of a controlled size from free gas in any form. As we have seen, the emitted micro-bubbles are all near a uniform size related to the surface wave-length. The cycling is very regular, its frequency varying typically between 1–10 per sec, depending on the drive level. It can be generated in most gassy liquids, but is most regular and easily produced in silicon oils, which have a low surface tension and low solubility for air. This permits very small bubbles to remain in suspension for long periods. The photographs of fig. 36, taken at 1/200 sec exposure, show three stages in the catastrophic explosion phase. The micro-bubbles are projected from the parent bubble at a speed of about 1 cm/sec. The surface distorts violently as it approaches resonance size and this accounts for the emission of strong signals at harmonics of the drive frequency. Figure 37 is a c.r.o. trace of the emission at the third harmonic (63 kHz) for two values of the excitation pressure. Here, the total time-base duration is 2 sec. Some white noise has also been recorded, associated with the actual emission of the micro-bubbles. The cycling is very regular, repeatable and permanent. For example, fig. 38 shows the changes observed in the cycling frequency in a typical case over a period of 26 hours of continuous operation.

7. Non-radial bubble motion

For large bubbles in stable oscillation, surface tension is a relatively weak controlling force and the spherical shape is easily lost. In the case of transients, the spherical shape is usually retained over the

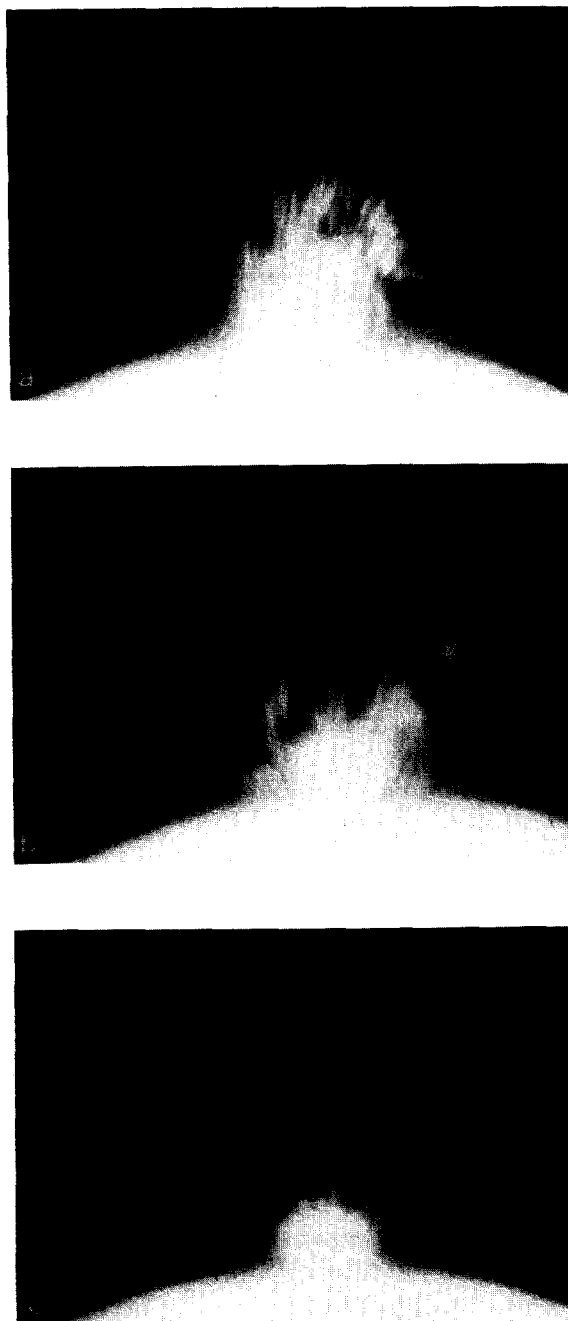


Fig. 36. Bubble distortion and micro-bubble evolution for a bubble of near-resonant size: liquid-silicon oil, viscosity 1 Stoke; exposure time 0.005 sec. (a) near the start of the explosion; (b) and (c) at later stages.

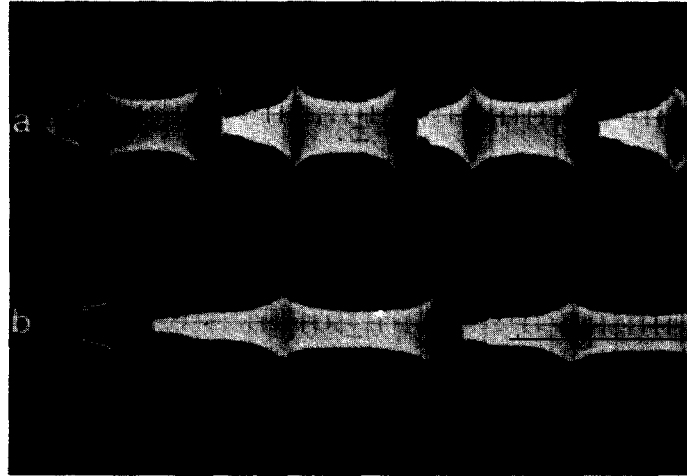


Fig. 37. Acoustic signal at the third harmonic of the excitation frequency (63 kHz). Liquid – silicone oil, viscosity 0.5 Stoke; total time-base duration 2 sec; (a) $P_A = 3.6$ milli-Bar; (b) $P_A = 3.4$ milli-Bar.

growth period, but irregularities often develop towards the end of the collapse. These become more pronounced on the rebound, when the cavity usually breaks up into a mass of micro-bubbles. Surface distortion of an originally spherical bubble must arise from some asymmetry in the environment. Typical asymmetries are: gravity; proximity to boundary walls or other oscillating bubbles; or perhaps just the pressure-gradient across the bubble when its diameter is no longer very small compared with the acoustic wave-length.

Surface distortions have several important consequences. For transients it must set a limit to the maximum pressures and temperatures developed on collapse. With stable cavities, the surface oscillations enhance micro-streaming which has important side-effects. In both cases, micro-bubbles resulting from the final disintegration of the parent bubble re-seed the liquid with fresh nuclei.

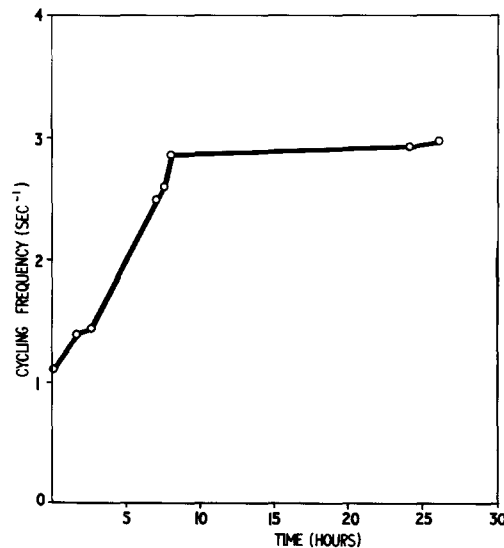


Fig. 38. Time-dependence of the cycling frequency.

7.1. Taylor instability

The theoretical problem of stability of the surface separating two fluids accelerated relative to one another was first treated by Taylor [105]. Here, we follow Plesset's outline of Taylor's solution [106]. The surface is assumed to be plane, accelerated normal to itself. Suppose the interface is represented by $y = H(t)$ and that the fluid in the region $y > H$ is a gas with density negligible compared with the liquid in the region $y < H$. Suppose the interface is perturbed into the form: $y_s = H + a \cos kx$ where $a(t)$ is taken to be a small quantity. The velocity potential ϕ is perturbed to $\phi = \phi_0 + \phi_1$ where $\phi_0 = -Hy$ and $\phi_1 = b \exp k(y + H) \cos kx$. At the boundary $y = H$, we have: $V_s = dy_s/dt = -\partial\phi/\partial y = -(\partial\phi_0/\partial y + \partial\phi_1/\partial y)$ giving $\partial a/\partial t = -kb$, defining b . Using the Bernoulli integral to satisfy the boundary condition at the interface gives: $a \, d^2H/dt^2 - db/dt = 0$. The result is therefore: $d^2a/dt^2 + ka \, d^2H/dt^2 = 0$. The interface is therefore stable when the acceleration is directed from the liquid to the gas-phase and unstable when oppositely directed. Lewis [107] successfully tested this conclusion experimentally. However, observations have shown that for a spherical surface the opposite is true: the interface becomes unstable on collapse, when the acceleration is directed inwards, from liquid to gas.

7.2. Instability at a spherical gas-liquid interface

In attempting to extend Taylor's treatment to the case of a spherical bubble boundary, we note that the essential difference is that when distortion develops in the form of a protuberance extending into the liquid, this extends into a region where not only the density, but also the flow velocity, is changing. The tip and base of the protuberance will be subject to different flow velocities. This complicates the problem to the extent that it becomes impossible to predict, using Taylor's simple argument, when conditions will become unstable for a bubble in radial oscillation.

Plesset [106] has studied this case of the stability of a spherical bubble. We simplify the problem by ignoring viscosity and assuming that both fluids are incompressible. It is then natural to express the perturbation in terms of spherical harmonics, and we may take the equation for the bubble wall to be:

$$r_s = R + aY_n \quad (116)$$

where $R(t)$ is the unperturbed bubble radius, given by eq. (11) with $p_L = (P_g - 2\sigma/R)$ and $p_\infty = P_0$:

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \frac{1}{\rho} \left(P_g - P_0 - \frac{2\sigma}{R} \right). \quad (117)$$

Y_n is a spherical harmonic of degree n and a is the amplitude of the perturbation, with $a \ll R$. To make the problem tractable it is necessary to introduce a further simplification by assuming that the flow configuration is unaffected by the surface instability. The flow velocity is then given by the usual continuity relation: $V = R^2\dot{R}/r^2$.

Plesset then developed his theory by an argument similar to Taylor's. Assuming that the flows in the gas- and liquid-phases can be described by velocity potentials ϕ_1 and ϕ_2 , the kinematic condition at the undisturbed surface is:

$$-(\partial\phi_1/\partial r) = -(\partial\phi_2/\partial r) = \dot{R} + \dot{a}Y_n.$$

The dynamic boundary condition is satisfied by using the Bernoulli equation to evaluate the pressure on either side of the interface, these being connected by the relation $(P_1 - P_2) = \sigma(1/R' + 1/R'')$, where R' and R'' are the principal radii of curvature at the interface. Using eq. (117) and applying a linearisation procedure, the stability condition for $n > 0$ is found to be:

$$\ddot{a} + 3\dot{R}\dot{a}/R + Aa = 0 \quad (118)$$

where:

$$A = \frac{[n(n-1)\rho_L - (n+1)(n+2)\rho_g]\ddot{R} - (n-1)n(n+1)(n+2)\sigma/R^2}{R\{n\rho_L + (n+1)\rho_g\}}.$$

This implies that whereas stability can be correctly inferred, the instability inferred from an increase of a with time is just a reasonable conjecture and not a necessary consequence. For a gas bubble in a liquid, $\rho_g \ll \rho_L$ and after writing ρ for ρ_L , A reduces to:

$$A = (n-1)\ddot{R}/R - (n-1)(n+1)(n+2)\sigma/\rho R^3.$$

If we use the substitution $b/a = R^{2/3}$ eq. (118) transforms to:

$$\ddot{b} + Gb = 0 \quad (119)$$

where:

$$G = (n-1)(n+1)(n+2)\frac{\sigma}{\rho R^3} - \frac{3(R\dot{R})^2}{4} - \frac{(2n+1)\ddot{R}}{2R}.$$

This stability condition can now be investigated for the two cases of interest, that of transient and stable bubble motion.

7.3. Stability of a collapsing transient cavity

Towards the end of a transient collapse we know that $\dot{R} \propto R^{-3/2}$ as $R \rightarrow 0$. Substituting in eq. (119) we see that only the last term is relevant when R is very small and $G \sim c^2/R^5$ as $R \rightarrow 0$, where c is a real number. Asymptotically we find:

$$a = \frac{b}{R^{3/2}} \sim R^{-1/4} \exp\left(\pm ic \int_0^t R^{-5/2} dt\right) \quad (120)$$

as $R \rightarrow 0$. a therefore increases as $R^{-1/4}$ in amplitude and oscillates with increasing frequency as $R \rightarrow 0$. The detailed analysis has been given by Plesset and Mitchell [108]. In fig. 39 the perturbation amplitude is shown as a function of the relative cavity radius for two values of the mode-order. Instability becomes violent for R/R_0 less than about 0.1. Collapsing transients may therefore become unstable before the effects of compressibility, viscosity or surface tension become very great. As the theoretical radial

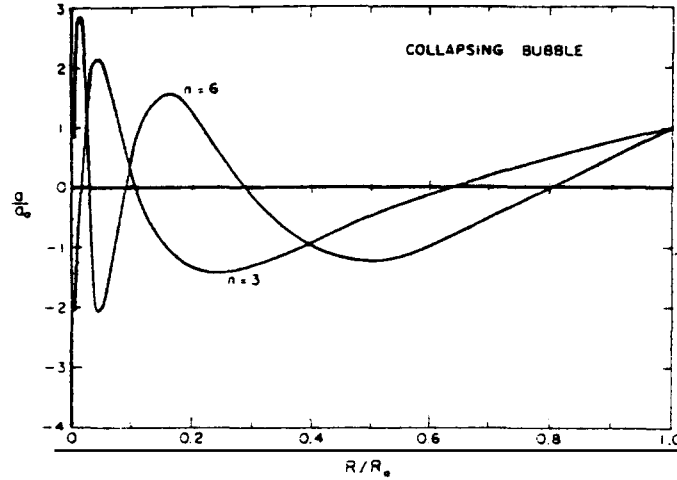


Fig. 39. Perturbation amplitude for a collapsing cavity in terms of the relative radius (R/R_0) for mode orders $n = 3$ and $n = 6$.

compression ratio is often greater than 10, a transient that starts its collapse containing gas at low pressure may be expected to become unstable before reaching its theoretical minimum size. On the other hand, small, relatively-stable bubbles with higher gas pressure may remain stable, the compression not proceeding far enough for instabilities to develop.

We also see from the stability condition (119) that in the expansion phase of the transient cavity, since $\dot{R}^2 \rightarrow 2\Delta P/3\rho$, $a \rightarrow \text{constant}$ as R becomes large, so that $a/R \rightarrow 0$ as $R \rightarrow \infty$. The expansion phase is therefore stable.

In an independent study, Brook Benjamin [87] looked for a criterion for instability in terms of the ratio of the bubble size needed for instabilities to form, to its minimum size. The collapse conditions were assumed adiabatic. He found the rather simple result that a large disturbance of the surface would not be expected until

$$(R/R_{\min})^{3(\gamma-1)} \text{ becomes } < \gamma + (\gamma - 1)/2n \quad (121)$$

where n is the order of the supposed surface spherical harmonic in which the perturbation is represented. With $\gamma = 1.4$ and $n = 2$ this predicts that $R/R_{\min}$ would reach about 1.3 before any surface distortion would grow. This is near the point of maximum deceleration of the bubble wall.

7.4. Stability of an oscillating stable cavity

Historically, Kornfeld and Suvarov [109] were the first to observe surface oscillations on stable cavities. Many workers have subsequently studied these vibrations. Applied to stable conditions, eq. (11) for the motion of the bubble wall can be linearised by writing $P = P_0(1 + \epsilon \sin \omega t)$ where $\epsilon \ll 1$. Using $P_L = (P_0 + 2\sigma/R_0)(R/R_0)^3 - 2\sigma/R$ the linear solution for R can be written: $R = R_0(1 + \delta \sin \omega t)$ where δ is of the same order as ϵ , and a constant phase-difference between P and R has been neglected. The function G in the stability relation (119) can now be evaluated. Following Plesset, retaining only the leading terms, G can be written:

$$G = \alpha + \beta \sin \omega t \quad (122)$$

with

$$\alpha = (n-1)(n+1)(n+2)\sigma/\rho R_0^3$$

and

$$\beta = [(n + \frac{1}{2})\omega^2 - (n-1)(n+1)(n+2)3\sigma/\rho R_0^3]\delta.$$

The stability condition (119) is now just the Mathieu equation. Considering the solution in very simple terms, we can see that it will be essentially unstable if $G < 0$. As stability will increase with decreasing mode-order, we may take $n = 2$, the lowest applicable to distortion modes. The eq. (119) then gives:

$$G = \frac{12\sigma}{\rho R_0^3} - \delta \left[\frac{5\omega^2}{2} - \frac{36\sigma}{\rho R_0^3} \right] \sin \omega t. \quad (123)$$

An order-of-magnitude criterion for stability is therefore:

$$24\sigma = 5\delta\omega^2\rho R_0^3 \quad \text{giving} \quad R_0 \sim (24\sigma/5\rho\delta\omega^2)^{1/3}. \quad (124)$$

For example, for an air bubble in water, taking $\delta = 10^{-2}$ and frequency 20 kHz ($\omega = 1.26 \times 10^5$ per sec), the critical radius is found to be about 2×10^{-2} cm.

Benjamin and Ursell [110] were the first to study the stability of solutions of the Mathieu equation in relation to oscillating bubbles. They found that the surface waves most likely to grow will be parametrically excited at frequency $\omega/2$ where ω is the excitation frequency. The treatment in ref. [110] refers to a plane liquid-gas interface. A theory for the stability of waves on a spherical surface, similar to Plesset's was also independently published by Benjamin and Strasberg [111]. They obtained the same general condition for stability (118), reducing to a Mathieu equation on applying a linearisation procedure.

With linear, non-parametric, excitation, the amplitude of the surface disturbance increases linearly with the drive pressure. However, with parametric excitation, a threshold amplitude related to damping must be reached before the parametric mechanism can be triggered. This is given by eq. (114). Following Nyborg [112] it is instructive to compare this threshold amplitude with a representative value of the bubble dimensions, in particular, the resonant radius of a large bubble, given by (44). We find:

$$\frac{\xi_T}{R_r} \approx \mu\omega \left(\frac{16}{\rho^2\omega\sigma} \right)^{1/3} \left(\frac{P_A}{3\gamma P_0} \right)^{1/2}. \quad (125)$$

For air bubbles in water, this reduces to $\xi_T/R_r \approx 10^{-5}f^{2/3}$ with f in Hz. This ratio increases rather rapidly with frequency. At 20 kHz it is 0.0074, but at 500 kHz it has increased to 0.06. We may express this threshold in terms of the drive pressure via the relation $P_A/P_0 = 3\gamma\delta\xi_T/R_r$. Using appropriate values of γ and δ for air bubbles in water, the threshold pressure for exciting surface waves parametrically on resonant bubbles is found to be only 0.0025 Bar at 20 kHz and 0.037 Bar at 500 kHz.

Surface waves are always easy to excite on bubbles much larger than resonance size. Then the mode-order is large and the equation for free oscillations (113) reduces approximately to:

$$\rho\omega_n^2 R^3 \approx \sigma n^3. \quad (126)$$

In the limit this must revert to the ripple-wave formula $\rho\omega_n^2 = k^3\sigma$ which refers to a plane surface, where k is the wave number for the surface waves. Then $kR = n$ and $2\pi R = n\lambda$.

Equation (113) shows that, for example, an air bubble in water that would be in radial resonance at 1 MHz, of radius 3.5×10^{-4} cm would need to be driven at 0.78 MHz to excite the order-2 surface mode; at 1.43 for order-3; and at 2.15 for order-4. For a bubble, radius 2×10^{-4} cm the frequencies would be 1.67, 3.04 and 4.56 MHz respectively for the same modes. Strong coupling results whenever a surface mode comes within the bandwidth of the radial resonance. Such coupled oscillations are said to be “linearly driven at resonance”. The above figures suggest that there must exist a certain minimum bubble size below which surface modes cannot couple to the radial resonance. The implication is that the surface wavelength is then of the order $2\pi R_r$, that is $kR_r \sim 1$ where k refers to the surface vibration.

Brooke Benjamin [111] has studied in detail the damping to be expected when the bubble surface is distorted in generating surface waves. He finds that in many physical situations the main effect of the film that constitutes the interface is to make the surface incompressible, and increased damping occurs within a boundary-layer in the liquid rather than through quasi-viscous dissipation within the film itself. He finds the simple result that, if α defines the damping, so that $a/a_0 = \rho^{-\alpha t} \sin \omega t$ then $\alpha R_e/\omega = (n+2)^2/2\sqrt{2}n$ where R_e is the acoustic Reynold's number.

7.5. General dynamical problem of the distortion of the surface separating two immiscible fluids

Using a variational approach, Hsieh [113, 114] has tackled the difficult problem of the distortion modes possible in the surface separating any two immiscible fluids, taking non-linearities into account. Among several important results obtained so far is an extension of the Plesset–Mitchell and Benjamin–Strasberg stability relation (118) to the next higher order of approximation:

$$R_0 \ddot{a} + 3\dot{R}_0 \dot{a} + \frac{1}{5} \left[(n-1)(n+2)(n^2+n+9) \frac{\sigma}{\rho R_0^2} - (n^2+n-1)\ddot{R}_0 \right] a = 0. \quad (127)$$

This result is actually identical with (118) for $n = 2$, but of course diverges from it for $n > 2$. Hsieh has also explored the feed-back mechanism whereby the surface modes affect the radial oscillation, with interesting results [113].

7.6. Experimental results

There have been several reported measurements of thresholds for generating surface waves on bubbles excited at low ultrasonic frequencies in the range 20–30 kHz: Strasberg and Benjamin [115]; Eller and Crum [116]; and Gould [117]. In all cases, bubbles were below resonance size, with radii in the range $(15-60) \times 10^{-4}$ cm. The experimental results do agree reasonably well with theory for the larger bubbles, but for those with radii below about 25×10^{-4} cm, the measured thresholds are notably higher than the theoretical predictions. A discrepancy is expected in view of the rather drastic assumptions made in deriving the expressions.

Surface wave activity generates intense micro-streaming which in turn aids thermal- and mass-transfer across the bubble wall. Another physical effect of intense surface-wave activity is the generation of uniform-sized micro-bubbles when the parent bubble disintegrates. This probably has practical applications [104]. It is analogous to the reverse process, i.e., the break-up of liquid films in the “atomisation” of liquids, which has been developed commercially. Here, the atomised liquid droplets

are very uniform in size, with diameters close to $0.34\lambda_s$. We may assume that in our case the micro-bubbles are of comparable size and distribution.

The form taken by the surface distortion that develops when a transient collapses must depend to some extent on the type of asymmetry responsible for it. In the presence of a solid surface the imploding bubbles cause erosion. In this case, we find that the type of distortion is predictable and involves a jetting process. We shall deal with this in detail in the next section.

8. High-speed photographic studies of bubble motion

Confirmation of theories of cavitation bubble dynamics can only be obtained by direct observation of the expansion and collapse. At ultrasonic frequencies, this demands the use of high-speed cameras. Starting in the early 1930's many studies have been made of isolated bubbles oscillating in normal, and gravity-free conditions, and of bubbles collapsing near to solid surfaces and interacting with other bubbles. Three groups of workers have been primarily involved, at: California Institute of Technology (Cal-Tech), U.S.A.; University of Gottingen, West Germany; and at the Acoustics Institute, Academy of Sciences, Moscow, U.S.S.R.

Early photographic studies concerned hydrodynamically-produced cavities. Much of this pioneer work was carried out at Cal-Tech. Professor Knapp and colleagues [118] studied cavitation generated in water flowing past model projectiles and suitably-shaped hydrofoils in a water tunnel. In studies of this sort, experimental conditions differ from those in typical acoustically-generated cavitation. Nevertheless the work is historically important and worth discussing in outline here. The cameras used stroboscopic light sources with flash-duration about 1 micro-second. A battery of 7 synchronised cameras was used, overlapping pictures taken at up to 3000 per second. In this way, effective repetition rates of about 20 000 per second could be achieved. The film was exposed at constant speed. Figure 40, reproduced from ref. [118] is a plot of a typical radius-time curve for the motion of a single bubble. The bubble sizes were taken directly from successive frames of the film and the volume was calculated assuming a

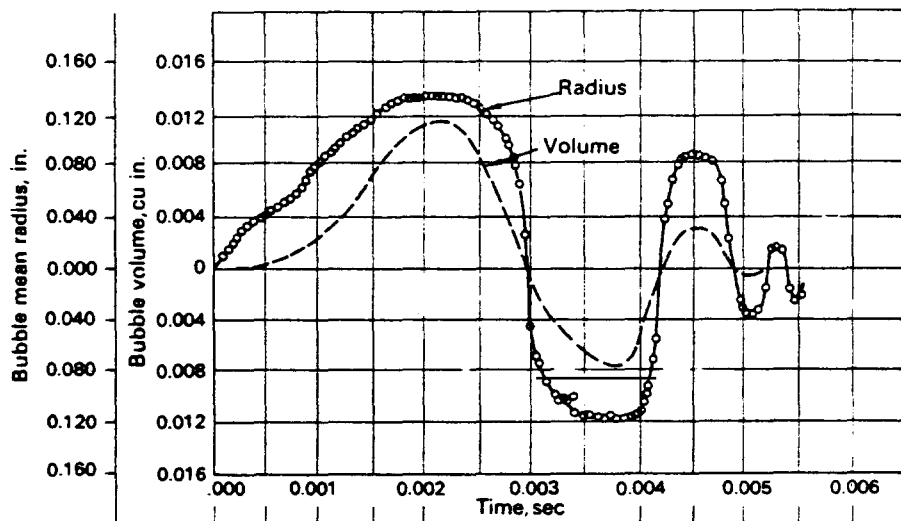


Fig. 40. Radius- and volume-time curves for an oscillating air bubble in water.

spherical shape. The frame-speed was the maximum obtainable, 20 000 per second. The first collapse is rapid by comparison with the initial growth. On the first rebound the bubble recovers to about 65% of its original volume. But after the third rebound it was reduced to near its equilibrium size. The radius-time curve for the first collapse agrees quite closely with Rayleigh's theory. In the figure, the odd-numbered cycles of oscillation are drawn above the axis and even-numbered ones below for clarity.

8.1. Photographic studies at California Institute of Technology

Ellis [119, 120] developed a camera system using a Kerr Cell as the shutter. The Kerr Cell uses a liquid that becomes polarised for light-transmission when placed in a suitable electric field. Combined with polaroid plates the cell can therefore be used as a light switch, triggered by voltages applied to the electrodes. In Ellis's work, the illumination was continuous at very high intensity and the strip of film remained stationary, wound on the inside of a drum. Light from the bubble field was projected along the axis of the drum, and a mirror mounted at the centre and inclined at an angle of 45° to the axis was rotated at high speed. The images, shuttered by the Kerr Cell were projected on to the stationary film. By 1955 Ellis was achieving exposure times of 10^{-7} sec at repetition rates of 10^6 per sec. He later improved the resolution by using laser illumination. The limitation with this type of camera is the small number of frames it can record in a single run. Recently, Ellis [120] and Naude and Ellis [121] have studied cavities collapsing on to solid surfaces. In some cases, the bubbles are seen simply to flatten, but in others the flattening continues to dimpling and penetration of the liquid as a jet into, and through, the bubble. In effect, the bubble turns itself inside out. Near a solid boundary the jet forms on collapse. But if the initial asymmetry is simply a hydrostatic field, for example, gravity, it will form on the rebound before the bubble disintegrates. Initially, the bubble always distorts first by flattening on the high-pressure side, keeping its flattened surface always parallel to a surface of constant pressure in the field. The steeper the pressure gradient the more pronounced the effect. To photograph the shock produced when bubbles imploded on a solid surface, they were made to collapse onto a photoelastic solid illuminated through crossed polaroids. The maximum shock recorded in this way occurred just before the complete collapse of the bubble.

8.2. Photographic studies by Soviet workers

Reports of high-speed photographic studies from U.S.S.R. refer to the motion of compact groups of cavitation bubbles. Using a radial-mode focusing transducer resonant at 15 kHz, Akulichev [122] obtained photographs of the growth and collapse of bubble clouds in water using a camera system similar to Ellis's, that is, using film fixed in a non-rotating drum with rotating mirror sited at the centre of the drum. The cavitating field was illuminated by a focused light beam strobed at the appropriate speed. This system was capable of recording 800 consecutive frames at frame-speeds up to 300 000 per second. The whole cloud of bubbles expanded and collapsed together in-phase. From an estimate of the average size of the bubbles at each phase of the motion, the radius-time curves could be plotted out. For the examples shown in fig. 41 the frame speed was 200 000 per second, giving about 12 frames per cycle. In the figure the measured plots are compared with theoretical computations based on (i) the Herring-Flynn and Gilmore theories (solid lines); and (ii) the simpler Noltingk-Neppiras solution (broken lines). The measurements agree quite well with either theory.

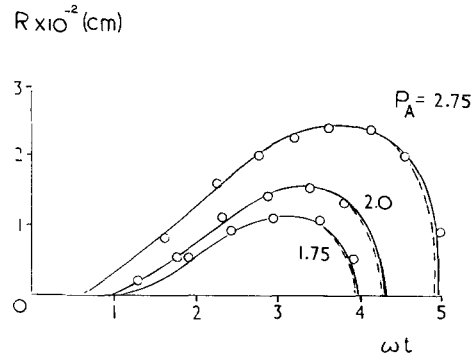


Fig. 41. Measured radius-time curves for air bubbles in water compared with theories; $R_0 = 10^{-4}$ cm; $f = 15$ kHz. The full-line curve was plotted from the Herring-Flynn and Gilmore theories, and the broken-line curve from the Noltingk-Neppiras theory.

8.3. *Photographic studies at the University of Gottingen*

The most recent high-speed photographs, with exceptionally clear pictures, have been published by the team working at the University of Gottingen [123–128]. Photographs have been obtained of isolated stable and transient bubbles, and also of bubble fields. Figure 42 is a schematic diagram of the experimental set-up. The liquid container was a small rectangular tank with transparent sides and the cavities were produced by focusing an intense light beam from a ruby laser, Q-switched by means of a Kerr Cell. The energy involved was in the region of 1 joule with pulse-duration about 30–50 nanoseconds. In this way, existing small air bubbles were expanded to a diameter of a few millimeter and allowed to collapse and rebound under the steady ambient pressure. The cavities therefore contained a mixture of air and vapour. The liquids used included water and various silicone oils with viscosities around 5–10 Poise. Both rotating-mirror and rotating-drum cameras were used to take pictures at frame-speeds up to 900 000 per second.

In water, even when the bubbles were formed in the body of the liquid away from boundaries, they tended to distort after one or two rebounds. But in silicone oils of high enough viscosity, they retain their spherical shape through several rebounds. For the photographs of fig. 43 the bubble was generated in silicone oil of viscosity 4.85 Poise and the pictures were taken at frame speed 75 000 per second. Three rebounds are obtained, after which the bubble has still not lost its spherical shape. The maximum bubble diameter was about 4 mm.

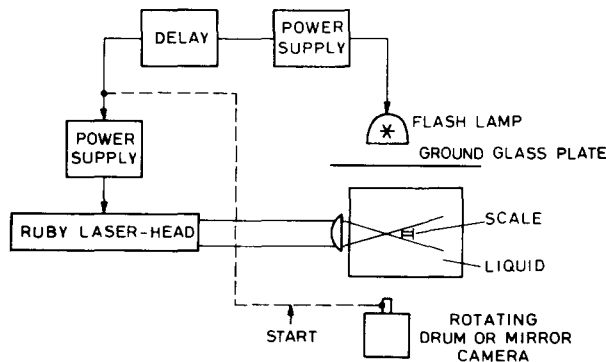


Fig. 42. Experimental arrangement for studying the dynamics of laser-induced cavitation at University of Gottingen.

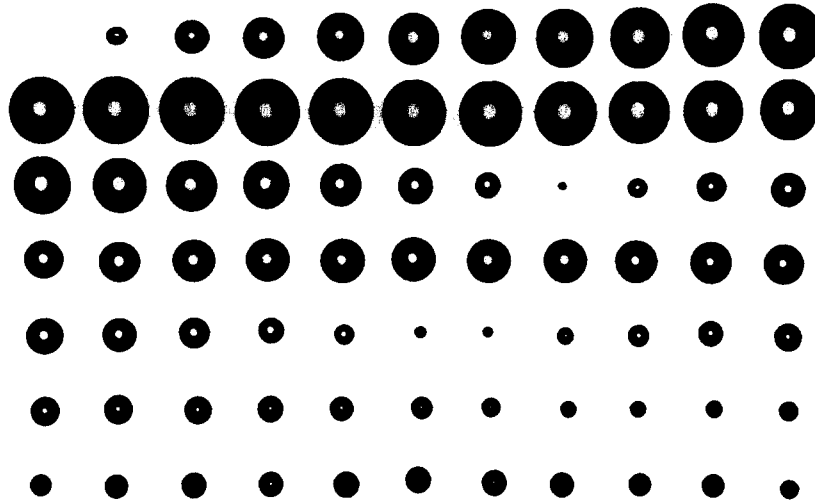


Fig. 43. Oscillation of laser-induced cavitation bubble in silicon oil of viscosity 4.85 Poise. The frame speed was 75 000/sec.

Measurements taken from the film can be compared with theory. The full-line curve in fig. 44 has been plotted from eq. (13) with $P_{\infty} = P_0$. The fit is very good considering that the damping effects of heat conduction and sound radiation are neglected in this equation. In the computation, an arbitrary figure for γ was used.

The study of cavities collapsing near boundaries presents no problems when they are generated by the laser technique. For example, fig. 45, taken at frame-speed 305 000 per second, shows an initially spherical bubble in water collapsing near a rigid boundary. As the bubble approaches to within a few radial distances of the surface a jet develops, projected towards the surface. On rebound, a counter-jet is produced. After that, the bubble invariably disintegrates. This jetting is now accepted as an initiating cause of erosion of surfaces. We show later that theory can predict the initial jet-formation, but there is at present no theory to explain the counter-jet.

When collapsing bubbles of similar sizes are within a few radial distances of one another they will attract and usually coalesce. But bubbles of widely differing sizes will develop jets. They will then not coalesce, but the smaller of the two will collapse first and disintegrate on the rebound as if it were approaching a solid surface.

By noting the effect on small stray bubbles in the vicinity, Lauterborn [126] has been able to estimate the magnitude of the shock pressures developed around the collapsing spherical bubbles. Figure 46 gives an evaluation of the radius-time curve for a small bubble compressed by the shock wave as a large

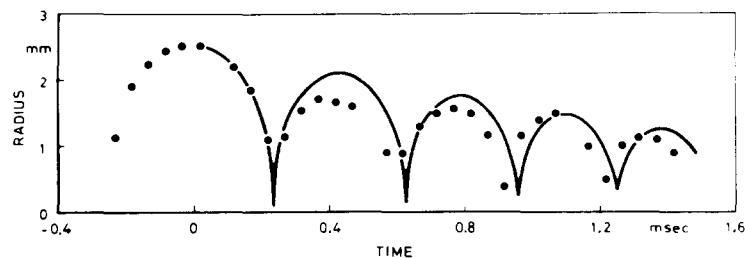


Fig. 44. Decaying non-linear bubble oscillation in silicon oil compared with theory.

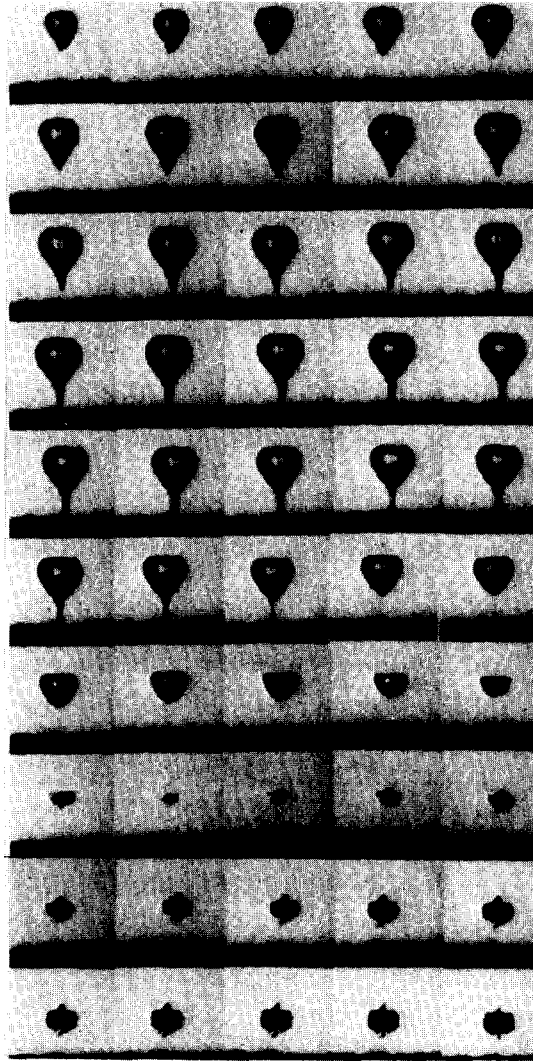


Fig. 45. Dynamics of a spherical cavitation bubble in water near a plane solid boundary, after the first collapse. The bubble was generated at a distance of 2.3 mm from the boundary. The frame speed was 305 000/sec.

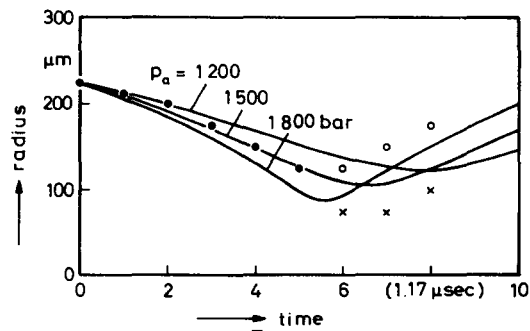


Fig. 46. Radius-time curve for a small air bubble compressed by the shock pulse from an imploding transient. ○ and × refer to the splitting of the value for the radius due to deformation of the bubble by the shock wave.

cavity collapses in water close to a rigid boundary. The curves shown in fig. 46 have been computed from eq. (14), using for P_{∞} a short pulse of sinusoidal shape and 30 nanosec duration. The results for pulses with amplitudes 1200, 1500 and 1800 Bar are shown and these span the measured points. The value of 30 nanosec was chosen for the pulse-duration because of the length of the laser pulse. The true duration of the pressure pulse is unknown, but there is some evidence that it may be as low as 10 nanosec. For pulses as short as this, the peak pressure required to generate any given $R-t$ trajectory will be approximately inversely proportional to the pulse-duration. This means that the true value of the peak pressure in the shock may be as high as 4500 Bar.

In all the work reported above, the steady ambient pressure was used to collapse the bubbles. Good photographs have also been obtained of bubbles forced into oscillation in an acoustic field. Air bubbles formed at the focus of a tubular piezoceramic transducer were driven at a frequency of 13 kHz and photographed using a frame-speed of 100 000 per second. Bubbles were never even approximately spherical but an equivalent radius could be inferred and the measured radius-time curve compared with a theoretical prediction based on eq. (14). After applying an appropriate fitting procedure, the agreement was found to be quite good.

8.4. *Theory relating to bubbles collapsing near solid surfaces*

Plesset and Chapman [129] have shown how the problem of an initially spherical bubble collapsing near a rigid boundary can be solved using numerical methods. Two examples from their results are shown in fig. 47. The liquid is water and the collapse occurs under a steady pressure of 1 Bar. Dimensions of the collapse are scaled in terms of the initial bubble radius R_0 . In case (1) the bubble collapses in contact with a plane surface, while in (2) the bubble surface was initially at a distance $R_0/2$ from the boundary. The computation gives the velocity of the liquid jet as it reaches the boundary as 128 and 170 m/sec in the two cases. Assuming that the characteristic impedance of the liquid $(\rho c)_L$ is small compared with that of the surface material, and taking V as the collapse velocity, the pressure developed on impact is approximately $P \approx (\rho c)_L V$. If now we adopt the acoustic approximation, that is, using a constant velocity of sound at the usual value (1500 m/sec for water), with $V = 130$ m/sec, we obtain $P = 1930$ Bar. Plesset showed that even when a more realistic figure was used for the velocity of sound under the prevailing conditions, the theoretical shock pressure is not increased very much. This stress is effective for a very short time, of the order of the time for the impact signal to traverse the radius of the jet, that is $\sim 10^{-7}$ sec. The time for the jet to propagate completely is just its length divided by its velocity, and for most of this time the pressure will be given by $\frac{1}{2}\rho V^2$ or about 800 Bar in this case. These pressures are hardly great enough to cause erosion in resistant metals.

8.5. *Recent photographic studies using large bubbles excited at low frequencies*

Crum [130] has been able to observe and photograph the jetting phenomena without resorting to high-speed cameras. Large bubbles were excited at their radial resonance frequencies with the ambient pressure reduced to near vapour pressure. This ensured that the bubbles were essentially vapour-filled. The motion of the cavities was then mainly controlled by surface tension. The bubbles were maintained in resonance vibration by forming them on a vibrating platform. Jetting was observed, similar to that obtained in the work already reported. By strobing at the drive frequency, still pictures were taken at various phases of the bubble motion.

Storm [131] seeded large air bubbles, with volumes 0.05–1.0 cm³, in a gel so as to hold them

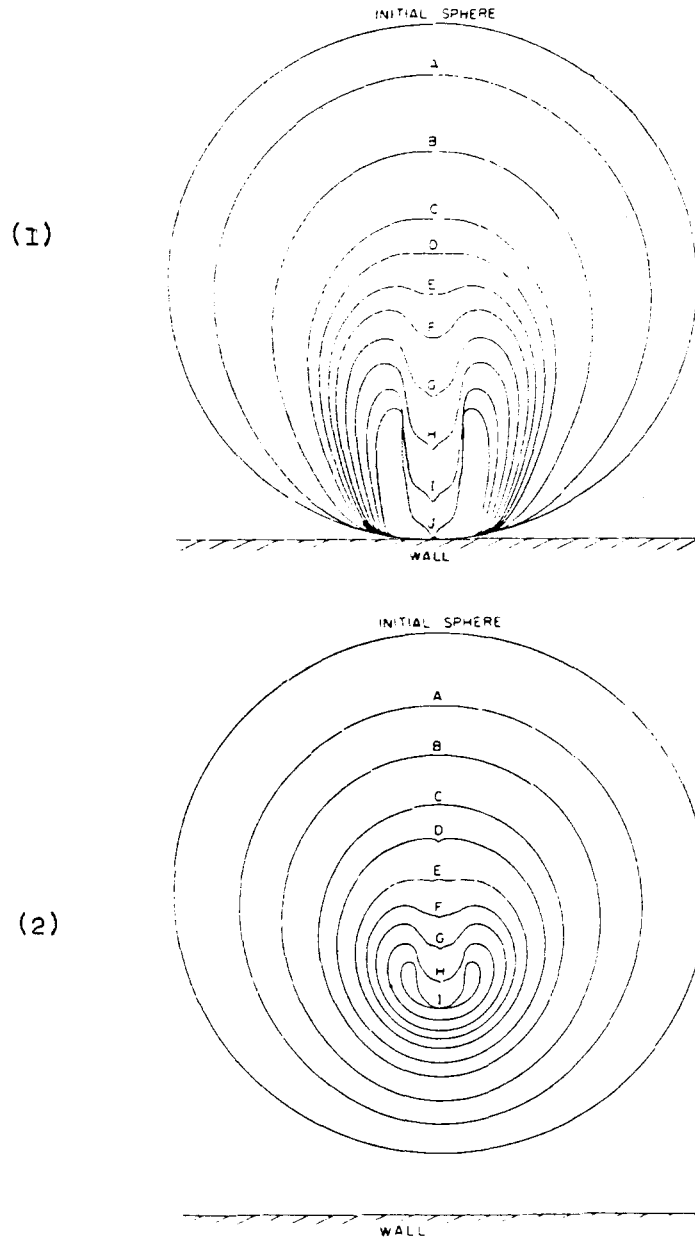


Fig. 47. Surface profiles for bubbles collapsing (1) initially in contact with a plane boundary; (2) initially at distance $R_0/2$ from the boundary. $P_m = 1$ Bar.

stationary. The bubbles were driven at low frequencies in the range 500–1050 Hz. Using a Fastex camera with frame-speeds up to 5000 per second, he was able to examine in close detail the formation of distortion modes and the emission of micro-bubbles. He observed a tendency for originally non-spherical bubbles to become more spherical at low and medium levels of excitation. A notable result was the detection of an acoustic emission from the bubble at the half-order sub-harmonic which could be related to the surface oscillation.

9. Cavitation fields

The study of isolated cavitation bubbles must now be extended to the complex associations of bubbles that make up the bubble fields encountered in practice. The literature contains many qualitative descriptions of what can be seen and inferred when liquids are cavitated under typical conditions, in papers dating back to the 1930's. We have no space to repeat these accounts, but merely point out some of the more useful references [9, 132–136]. The detailed description of the forces controlling the interactions and translatory motions of bubbles in sound fields is also a large subject on its own, relevant to, but rather outside the scope of this review. Here, we merely list the types of forces that may be involved, with some useful references. The chief forces responsible for translatory motion are: gravity; acoustic radiation [2, 112, 137]; acoustic streaming and micro-streaming [2, 112, 138, 139]; and forces due to electric charges carried by small bubbles [102]. The chief interaction forces between bubbles are: Bernoulli attraction; radiation pressure, including the so-called Bjerknes-type forces [112, 140]; and various steady forces due to wave-form distortion. All of these forces are opposed by Stokes-type resistance to motion. Clearly, with so many different forces involved, the interactions of bubbles with the acoustic field, and with themselves, is a theoretical problem of great complexity.

Descriptions of the transducers and instrumentation used in generating and detecting cavitation fields covers a large technological area lying outside the scope of our review. However, the success of experimental studies of cavitation fields often does depend very much on the design of transducers and their associated instrumentation. The emitters usually employed are focused devices, chosen not merely to achieve the required high intensities, but also to ensure that the cavitating region is isolated from interfering boundaries. Again, we have to be content with quoting the useful references: [133, 135, 136, 141]. Detectors and receivers are many and varied; publications covering design and performance will be referenced later.

9.1. Pioneer studies using visual and photographic methods

Before discussing recent research, we must mention two early studies of cavitation fields which are of historical importance and have encouraged later work. Blake, in his famous 1949 Harvard report [9] was the first to distinguish clearly between *stable bubble activity*, *gaseous transients* and *vaporous transients*. He used a focusing transducer resonant at 60 kHz, operated into water. Stable cavitation activity and gaseous transients are observed in gassy water the former as isolated bubbles, or small groups, formed around the focal region, and the latter in the “streamers” or branching structures obtained at higher intensities. On the other hand, vaporous transients are most easily generated in degassed water. They appear as small, short-lived, almost explosive, ruptures, accompanied by a snapping noise. Blake did not observe *stable vaporous bubbles*, as his experimental conditions were never suitable for generating these. To illustrate typical cavitation activity at high intensity in gassy water, fig. 48 shows streamer formations with the generation of micro-bubble clouds in the focal region of a radial-mode magnetostriction transducer resonant at about 18 kHz.

In another historically-important paper, Willard [135] described in detail a particular type of cavitation, obtainable at high frequencies in focused fields. A spherical bowl transducer, resonant at 2.5 MHz was used to generate intense fields in water. Focal intensities up to about 1800 watt/cm², corresponding to pressures of about 70 Bar, were achieved. Willard photographed the cavitation events using a camera operating at a frame-rate that we would now classify as slow-to-medium (24–1000 per sec). He also used a continuous fast-moving film with steady illumination. He concluded that a



Fig. 48. Streamer formation at the focus of an 18 kHz radial-mode magnetostriction transducer.

definite sequence of events was involved in the type of cavitation he observed, building up to a violent explosive event at the focus. First, the liquid must contain suitable nuclei, which are dragged into the focal region by radiation pressure and streaming motion. This is called the “pre-initiation” stage. The second stage, “initiation”, occurs when a suitably weak nucleus streams into the focal region. These initiating nuclei are visible gas-filled bubbles and therefore well above the radial resonance size at the frequency used. The third, or “catastrophic” stage occurs when the initiating nucleus, on entering the focal region, breaks up into a mass of micro-bubbles, accompanied by a loud snapping noise. We now know that in this third stage, the large initiating bubble develops violent surface activity, throwing off micro-bubbles at an accelerating rate, as explained in an earlier section of this review. The second and third stages may occur whatever the air content of the water. The cloud of micro-bubbles from the explosion form a plume-shaped object. The micro-bubble cloud is projected through the focal region very rapidly under radiation forces. This suggests that a large proportion of the micro-bubbles are of near-resonant size. Willard measured translatory speeds up to 10 m/sec through the focus. In aerated water, two further stages termed the “bubble-phase” and the “post-cavitation” condition are recognised. A few relatively large, non-collapsing, bubbles are generated by, and concurrently with, the catastrophic phase. They remain after the Willard event has passed, to be carried off downstream. The catastrophic event is usually over in a few millisec. Fig. 49 shows three frames selected from Willard’s results, illustrating the violent catastrophic phase of the cavitation. For these photographs, degassed water was used. The plume-like shape of the micro-bubble cloud can be seen as it streaks through the focal region. The exposure time was about 22 millisec, but Willard was able to show that the event was completed in less than 4 millisec. As the length of the plume is about 1 cm, this corresponds to a speed through the focus of at least 2.5 m/sec.

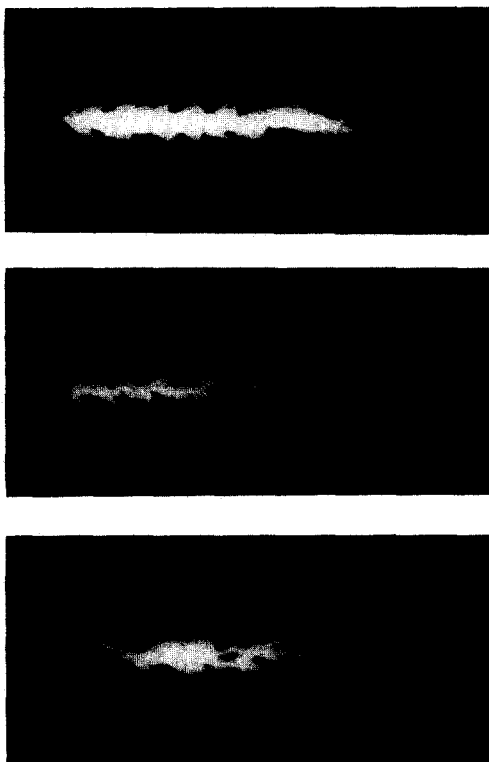


Fig. 49. Typical Willard events in degassed water; $f = 2.5$ MHz.

9.2. Cavitation studies using pulsed holographic techniques

In a detailed study of cavitation fields, we need to follow the paths of individual bubbles in motion under the steady forces of the field. For this purpose, ordinary high-speed photography is not quite suitable. The reason is simply that bubbles tend to move out of focus during successive exposures. This difficulty has been overcome by using holographic recording methods, developed recently at the University of Gottingen [142, 143]. Short pulses of intense illumination are needed and in the work reported in [142] pulses of about 20 nanosec duration were generated by means of a Q-switched laser to form the holograms. The light pulse could be triggered at a pre-selected phase of the bubble motion. Holograms could then be reconstructed for different depths in the field and at any given phase of the motion from a single hologram.

Extending holographic techniques to studying the temporal development of the field requires double- or multiple-pulse equipment. Details of the instruments developed by the Gottingen group are contained in refs. [142, 143] to which the reader is referred. Figure 50 from ref. [142] shows the appearance of a radially-focused cavitation field in gassy water at two instants separated by an interval of 320 microsec. The excitation frequency was 19 kHz, so the interval represents just 6 acoustic cycles. The photographs show that although the basic configuration of the bubble field has remained the same in this short interval, with a common central core of bubbles, the detailed arrangement of bubbles has altered considerably. Hologram reconstructions of this sort are very revealing. They show very clearly that the unaided eye is unable to follow even the translatory motions of bubble-aggregates in cavitating

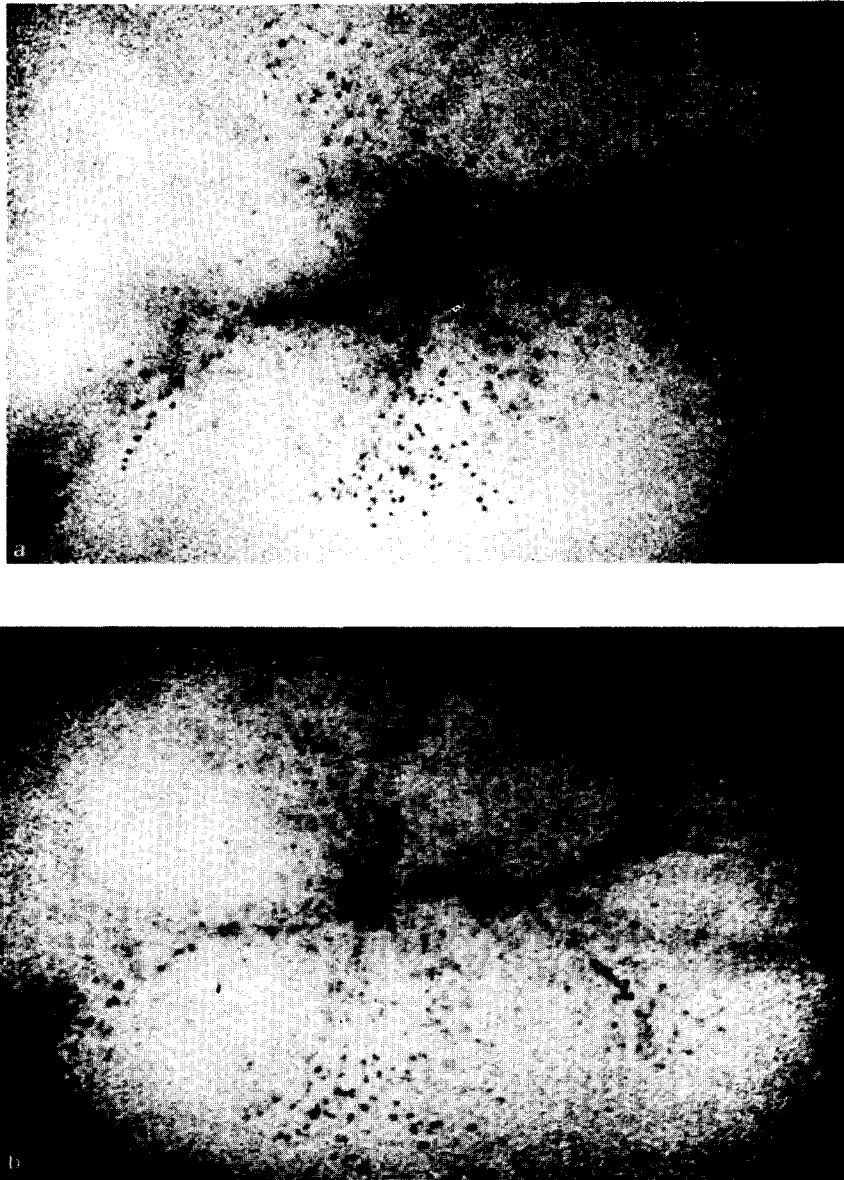


Fig. 50. Temporal development of cavitation bubble configuration. Plate (b) was taken 320 microsec after (a).

fields. The optical equipment developed at Gottingen permits time-separations down to about 10 microsec. But since the holographic plate cannot be moved in such a short time, the images must be superimposed on one another on a single plate. As the images must be separable when reconstructed, an acousto-optical deflection system is used to shift the direction of the laser beam between the exposures. This involves using two switchable beam-splitters with the pulsed laser.

This technique is capable of extension to holo-cinematography, where multiple images are produced at high speed. The authors [144] describe a system for producing up to 8 holograms at 20 kHz speed, all on a single plate, but separated by spatial multiplexing, that is, using a different portion of the plate for

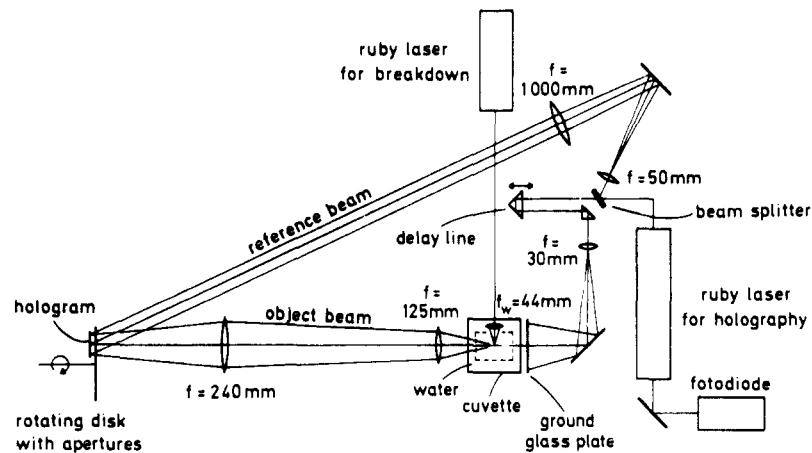


Fig. 51. Experimental set-up for recording holograms using spatial multiplexing.

each exposure. The separation is achieved by using a rotating disc with apertures as a shutter, placed directly in front of the holographic plate. Thus, at each laser pulse a different portion of the plate is exposed. The complete experimental set-up is shown diagrammatically in fig. 51.

9.3. *Acoustic measurements in the cavitating field*

Precautions are necessary when using conventional acoustic instruments to obtain information directly from a cavitating field. Two types of receiving device have been found useful: very small probe-type hydrophones and radiometers.

The value of probe hydrophones is that they measure *instantaneous* values of the acoustic pressure in their vicinity. They pick up the acoustic emission from the field. The complex wave-forms can be analysed into their harmonic, sub-harmonic and white-noise components. These signals are characteristic of the type and intensity of the cavitation and are so important and revealing that section 10 of this review will be devoted to interpreting them. However, it must be emphasised that these probe instruments will not usually respond to the direct shock-waves emitted from collapsing cavities. The reason is that the range of the strong shocks is limited to a few radial distances from the collapsing cavity. Most of the received signal will refer to the lower-frequency components of noise from the more distant environment.

For use in cavitating fields, probe hydrophones must be physically very small, so as to interfere as little as possible with the field. The surface of the probe should be such that bubbles cannot easily collect on the probe itself to interfere with measurements. Precautions must also be taken to ensure that the probe cannot be damaged by cavitation erosive action. These hydrophones are essentially "point-probes" but of course they are also capable of being mechanically scanned to record the field pattern. Detailed information on the construction and use of suitable small hydrophones is given in refs. [133, 145–149].

When interest is centred on the total energy expended in generating cavitation, a different approach is used. In any non-uniform hydrodynamic field, streaming flows are generated by a force which depends on the energy-gradient set up in the medium. In a cavitating liquid, provided the medium itself is not too lossy, the steady streaming can be attributed to the cavitation field. It can be shown that in a

closed region, the sum of the energy-density of the acoustic field and the kinetic energy of the stream flow is constant. This sum is readily measured using a radiometer balance [150]. It is expected that the energy used in generating cavitation gives rise to an equivalent stream flow. Thus, a simple experiment using a radiometer placed downstream from the cavitation zone measures the energy associated with the collapse of cavities intercepted by the beam [151]. It is necessary to separate out the effect of the acoustic field, namely, that portion of the energy due to radiation force. This is done by interposing a thin, acoustically-transparent, film in the path of the beam. This cuts out the stream flow, passing the acoustic component. The difference in the two radiometer readings then refers to the energy carried by the streaming. In a typical case, quoted in ref. [151], about 20% of the total radiated power was used in generating cavitation.

9.4. Measurements taken at the electrical terminals of the transducer

When cavitation starts in a liquid, its physical properties change. It becomes more compressible and more lossy. Its "effective impedance" has changed and one method of studying cavitation is to monitor these changes. In this method it is possible to take all the information needed from the electrical terminals of the drive-transducer. Unlike probe methods, the acoustic field is not interfered with in any way. It is surprising how much useful information can be obtained by measurements of this sort.

The technique is described in detail in references [136] and [152]. The drive transducer is wired up as one arm of an admittance bridge. The bridge is balanced initially at the clamped point of the transducer, that is, with the transducer non-motional, just off-resonance. It can be shown [136] that when the transducer is tuned to resonance the voltage appearing across the detector terminals of the bridge is proportional to, and in phase with, the oscillatory amplitude of the transducer surface. If the transducer is driven from a high-impedance source, then the detector signal will be proportional to the motional impedance, and inversely proportional to the acoustic impedance of the liquid.

Measurements obtained by several groups of workers [153–155] have shown that when cavitation starts, the effective radiation resistance of the medium begins to fall off rapidly. With increasing intensity in degassed water the resistance eventually reaches a low level at approximately one-third of the nominal ρc . The example given in fig. 52 has been reproduced from ref. [154]. In this case, the measured quantities were the acoustic power transmitted to the liquid (W_a) and the particle velocity of

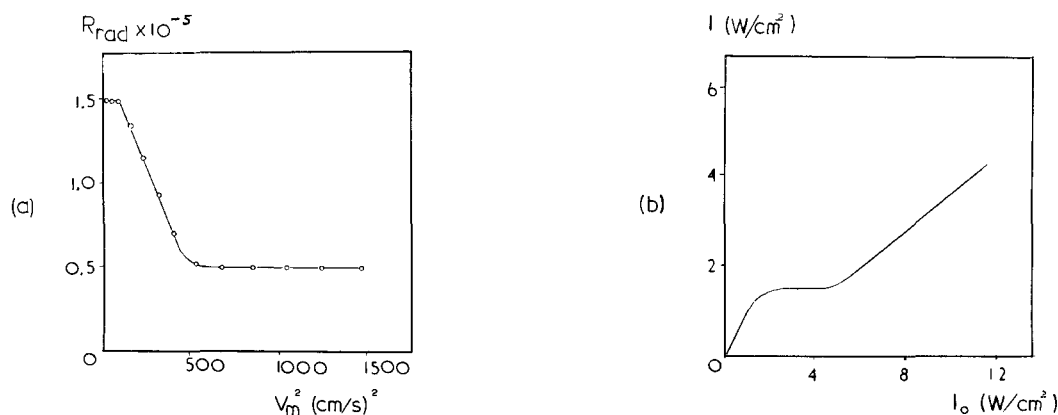


Fig. 52. (a) Effective radiation resistance versus the square of the velocity of the radiator surface; (b) actual radiated intensity versus theoretical intensity under non-cavitating conditions.

the vibrator surface (v_m). The effective radiation resistance per unit area of the vibrator is then $R_{\text{rad}} = 2W_a/v_m^2 S$ where S is the area. In fig. 52a this quantity is shown plotted out as a function of v_m^2 , which would be proportional to the acoustic intensity in the absence of cavitation. Below the cavitation threshold $R_{\text{rad}} \approx 1.5 \times 10^5$ c.g.s. units, as expected, but after cavitation has set in, this drops off rapidly, reaching a constant value of about 5×10^4 at high intensities. Another way of presenting these results is to show the intensity actually radiated into the liquid (W_a/S) as a function of the intensity that would be obtained in the absence of cavitation. This is shown in fig. 52b. In this case, the measured radiated intensity is almost constant at about 1.5 W/cm^2 over a wide range of values of the transducer face velocity.

Under cavitating conditions, the voltage signal received at the detector terminals of the immittance bridge is ripple-modulated, reflecting bubble events occurring at frequencies differing from the drive frequency. The signals due to these cavitation events can be extracted by simply filtering out the fundamental. We have found that this bridge-filter arrangement is a very sensitive indicator of impedance changes due to a wide range of bubble motions. An interesting example, fully explored in ref. [136], relates to the Willard-events referred to above. With suitable filtering of the received signal, the Willard event is recorded as a low-frequency, short-duration voltage pulse. An example of a typical c.r.o. trace is given in fig. 53. The explanation is that many of the micro-bubbles that make up the cloud are in radial resonance. As these bubbles pass through the nodes and anti-nodes of the quasi-standing-wave field they are excited to emit strongly at the pressure anti-nodes, but become quiescent again on passing the nodes. The received pulse therefore consists of a modulated carrier, the wave-length of which corresponds to the half-wave-length separation between the anti-nodes of the standing-wave field. As the acoustic wave-length and c.r.o. time-base speed are known, the c.r.o. signal gives the distance over which the bubbles are active and also the time-duration of the event. In this way, the velocity of the bubble cloud through the focal region was found to be several meter/sec, in substantial agreement with Willard's direct measurements.

The Willard events are spectacular and reflect a large impedance change. But even events so weak as to be undetectable by the naked eye can be recorded in this way. For example, it is easy to detect translatory motion of single resonant bubbles moving through the field. At the 1 MHz frequency used in the work reported in ref. [136] the resonant diameter is only about 6 micron and isolated bubbles

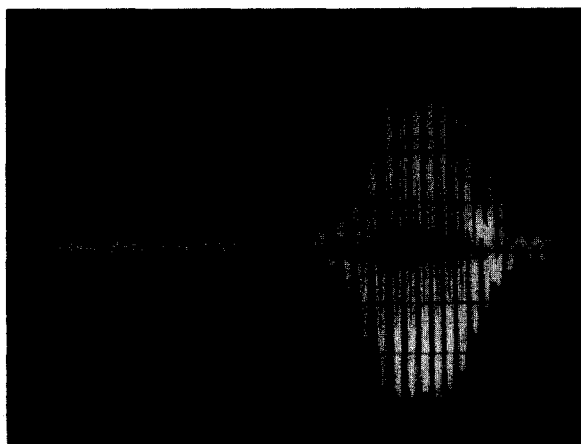


Fig. 53. Acoustic impedance change due to a typical Willard event. Total time-base length = 10 millisecc.

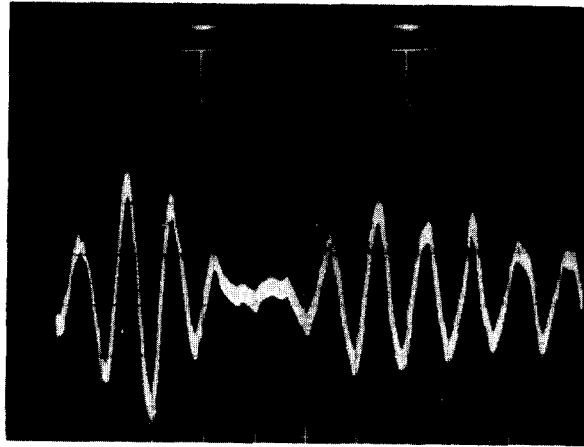


Fig. 54. Acoustic impedance change due to a single resonant bubble moving in the field. Total time-base length = 10 millisecc.

streaking through the field would not normally be detected. Figure 54 shows a typical c.r.o. trace of the signal received from a single bubble moving under radiation forces. The event could not be resolved on photographs taken in synchronism with the c.r.o. time-base.

9.5. *Concerted collapse of cavity clouds*

The acoustic field conditions often encourage cavitation bubbles to collect into groups. When this occurs in the body of the liquid, away from boundaries, the bubbles typically expand and collapse in-phase or approximately so. However, an important practical case is when bubble-clouds collapse on or near a vibrating surface, where erosion may occur. In this case, it has been shown that whereas the bubble may expand approximately in-phase on the tension half-cycle, the collapse will start in the outermost layers of the cloud and propagate inwards, increasing the violence of the implosion.

Morch and colleagues [156] have studied the dynamics of a system of bubbles collapsing as a hemispherical cloud on the surface of an ultrasonic vibrator. The dimensions of the cloud are assumed small compared with the acoustic wave-length. Under these conditions the bubbles will expand uniformly as the vibrating surface is accelerated away from the liquid. But on the positive half-cycle the pressure is relieved by the bubbles to the extent that the acoustic wave is full-wave rectified. The bubbles will then collapse under the ambient static pressure only. The collapse starts in an outer region of the cloud, propagating inwards and triggering the collapse of consecutive layers of cavities. The authors assume that the velocity of sound in the bubbly mixture is small enough for the outer shell to collapse completely before the next is affected, and so on. A convergent spherical pressure-wave is therefore generated, so the final collapse of the bubbles in contact with the surface is greatly amplified. Only in this way, the authors believe, can collapse pressures sufficient to cause erosion be developed.

The authors produced two simplified theories, representing extreme cases, where (i) a definite fraction of the total energy radiated from any symmetrical shell of bubbles is propagated inwards; and (ii) the energy of collapse of each individual bubble in a shell is transferred to the centre of the cloud without attenuating. Case (i) seems the more realistic, and here it can be shown that the energy carried by the implosion on the solid surface may be enhanced by a factor of 10 or more. In case (ii) the boundaries of the cloud are analogous to a spherical acoustic lens focusing towards the centre, and here

also, an appreciable gain is achieved. The authors consider that their results show why cavitation erosion can manifest itself on a vibrating surface right from the initial onset of transient cavitation, whereas otherwise we might expect an "incubation" period, as observed in damage associated with hydrodynamic cavitation.

9.6. *Acoustic intensity-distribution in the presence of cavitation*

In any theoretical approach to the study of bubble fields, we must be content with examining the average properties of the cavitating region. A phenomenological treatment is necessary. Rozenberg [157 and 158] showed how to estimate the space-dependence of the acoustic intensity in a beam generating cavitation in a defined region. Consider first a plane wave. The number of cavitating bubbles in a thin slice, thickness dx normal to the beam, may be written:

$$dx \int_{I_t}^I \frac{dN(I)}{dI} dI$$

where $N(I)$ is a function describing the number-distribution of stationary bubbles at sound intensity I . Assuming that the growth and collapse of bubbles occurs in just one cycle of the field, and writing $E(I)$ for the energy of formation of one bubble, then the total energy lost by the wave in forming all bubbles within the element thickness dx , is:

$$E_n = dx \int_{I_t}^I E(I) \frac{dN(I)}{dI} dI.$$

But we can equate this with $-T dI$ where T is the acoustic period ($2\pi/\omega$). We may then write:

$$-\frac{dI}{dx} = \frac{1}{T} \int_{I_t}^I E(I) \frac{dN(I)}{dI} dI \quad (128)$$

and x can be expressed explicitly in terms of I by further integration. We note that in considering only energy-flow in the x -direction we need take no account of wave-scattering or any detailed behaviour of the wave. The function $N(I)$ can best be obtained by direct counting from high-speed motion-pictures. The function $E(I)$ is not readily obtained, however.

The expression (128) is just the volume-density of the energy lost per unit time in sustaining the cavitation zone in the steady-state. It can be written W/V_0 where W is the total power associated with the cavitation losses and V_0 is the total volume of the cavitating zone. We have previously shown how to measure W using a radiometer probe. Figure 55 gives typical measurements obtained by Rozenberg showing the dependence of W/V_0 on the field intensity I , both expressed in W/cm^2 . The quadratic curve is a good fit to the experimental points, so we can write: $W/V_0 = A(I - I_t)^2 = -dI/dx$ for $I >$ the threshold intensity I_t . Then:

$$x = - \int_{I_0}^I \frac{dI}{A(I - I_t)^2}. \quad (129)$$

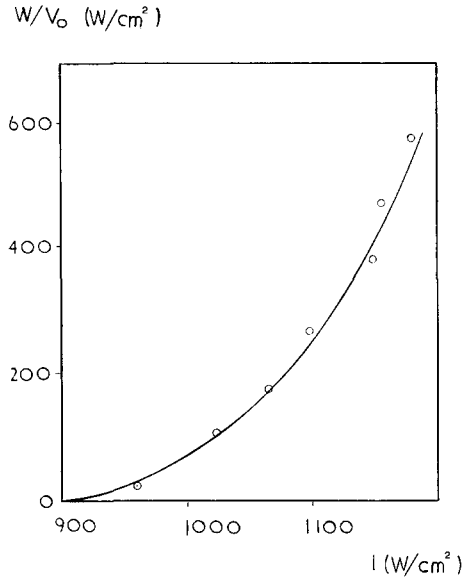


Fig. 55. Power density associated with cavitation losses versus acoustic intensity; $f = 500$ kHz.

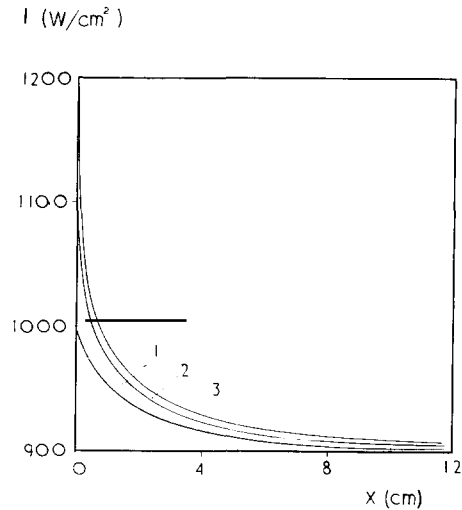


Fig. 56. Intensity-distribution on the axis of a one-dimensional cavitation zone: (1) $I_0 = 1000$ W/cm²; (2) 1100 W/cm²; (3) 1200 W/cm².

This equation is readily solved for I in terms of x . Figure 56 displays the results graphically for the three cases: $I_0 = 1000$, 1000 and 1200 W/cm².

The above discussion applies to plane waves. Rozenberg [158] has extended the treatment to spherical waves and applied it to experimental results obtained using his high-intensity focusing transducer. With a transducer of this sort, the acoustic energy is utilised much more efficiently than in plane-wave systems.

Further relevant experimental results obtained by Rozenberg [158] are displayed in fig. 57. A focusing concentrator was used, resonant at 15 kHz and the cavitation zone was generated at the tip of a wire, to locate the region for photographic purposes. The sound pressure was measured by a miniature hydrophone placed at a suitable distance (about 4–5 mm) from the investigated zone. Figure 57a shows the hydrophone received signal as a function of the transducer drive voltage. The departure from

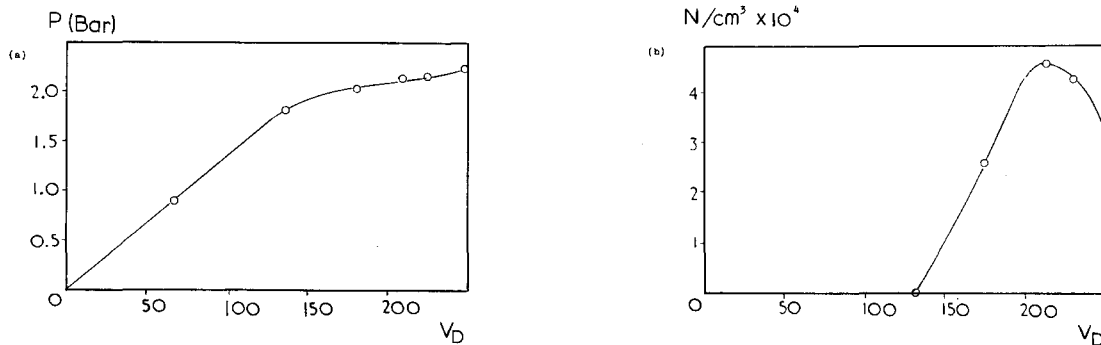


Fig. 57. (a) Acoustic pressure; and (b) Bubble density; at the focus of a radial-mode focusing transducer, as functions of the transducer drive voltage (V_D).

linearity occurs at the cavitation threshold and reflects the loss, by scattering, of the transmitted power when the bubbles first appear. The bubble density (N) as a function of the drive voltage is displayed in fig. 57b. N was obtained by direct counting from photographs, using a high-speed camera. N increases rapidly from the threshold and passes through a maximum which coincides with an upturn in the pressure-voltage characteristic.

9.7. *Controlled bubble fields*

In the work reported so far in this section, cavitation fields have been allowed to develop naturally, that is, from pre-existing nuclei, under the influence of the applied acoustic field. Precise observations of the motion of bubbles is really only possible when bubbles of known size and composition are seeded, or placed, in the liquid at specified points. Some obvious techniques for studying single bubbles have already been mentioned. A successful method of producing an array of bubbles of known small sizes, with diameters down to a few tenths of a micron, has been described by Nyborg and Miller [159, 160]. This method makes use of commercially-available Nuclepore filters [161]. These are sheets of a polycarbonate material about 10 micron thick, containing randomly-spaced holes. The holes are cylindrical and of a very uniform size down to about 0.2 micron diameter. The sheets can be made hydrophobic and in this case, when the filters are simply immersed in an aqueous liquid, the air carried in the holes is trapped, forming an array of small uniformly-sized cylindrical bubbles of known density. Miller, Nyborg and others at the University of Vermont, U.S.A. have reported physical and biological effects occurring near stably-oscillating bubbles trapped in this way. These include: acoustic emission; micro-streaming of liquid around the bubbles; migration of small particles towards the bubbles as a result of radiation forces; and various effects on biological cells. There is evidence that bubbles can be excited into violent non-linear motion before they leave their sites. But, as expected, a limiting drive level exists beyond which the bubbles disintegrate or are displaced from their sites.

It is not difficult to evolve an approximate theory for the dynamics of these cylindrical bubbles. For example, the resonance frequency of a cylindrical bubble, radius R_0 , length L , is found to be:

$$\omega_r = (2\gamma P_0 / \rho R_0 L)^{1/2}. \quad (130)$$

This derivation neglects the effect of surface tension and is analogous to the resonance condition for large spherical bubbles (eq. (44)). As an example, a cylindrical bubble, diameter 5 micron, length 10 micron will resonate at about 0.5 MHz. The resonant amplitude of the oscillation is given by:

$$\xi_0 = LP_A / 2\gamma\delta P_0(1 + R) \quad (131)$$

where P_A is the acoustic pressure, δ the loss angle and R the reflection coefficient from the surface of the bubble.

Reasonable agreement with theory has been found in experimental tests using methods based on measuring the radiation scattered from the bubbles. Modifications to the theory are needed when the bubbles in the array are closely spaced. When the holes are too close, interaction reduces the pressure in the vicinity of each hole. A theory developed by Weston [162] can be used to obtain the necessary correction.

10. Acoustic emission from cavitation fields

Cavitating bubbles act as secondary sources of sound, emitting spherical waves. This acoustic emission can be analysed and used as a means of distinguishing between stable and transient cavitation. The noise may represent a hazard for users of industrial and medical ultrasonic equipment, and is very important in military sonar. For all these reasons, the analysis and interpretation of the sound emitted from cavitation fields has been, and still is, an active area of research.

Historically, the group working at the University of Gottingen was the first to attempt a systematic study of cavitation noise. Esche [163] recorded many noise spectra at various drive frequencies and intensity-levels in gassy water. He was the first to report the sub-harmonic emissions. Strong signals at harmonics of the drive frequency were seen at intensities well below the transient threshold. Lines at sub-multiples of the drive frequency (the sub-harmonics) and a continuum of "white" noise appeared together, apparently at the transient threshold. The noise spectrum shown in fig. 58 is reproduced from Esche's results. It refers to intense transient cavitation in gassy water, generated by a 15 kHz magnetostrictive transducer. The half-order subharmonic is prominent, along with harmonics up to high order, superimposed on a white-noise continuum. Esche's paper was followed by others from the same group [73, 164]. Later, Negeshi [165] also recorded harmonic and sub-harmonic emissions. But neither Esche nor Negeshi could offer explanations of the strong sub-harmonic lines.

Much of the recent research has been concerned with identifying the source of these sub-harmonic emissions. In U.K. these studies have been mainly experimental [166–169], and in U.S.A. both theoretical and experimental [98, 170–173]. Workers in U.S.S.R. have helped to explain the strong sub-harmonic emission and the distribution of white-noise in the transient cavitation spectrum. Recent work by Lauterborn at Gottingen [175, 176] has resolved some remaining mysteries relating to the sub-harmonic and ultra-harmonic emissions.

Very small probe hydrophones are generally used to record the acoustic emission. The received signal is taken through a wide-band amplifier and spectrum-analyser and displayed on a c.r.o. or chart recorder while the frequency is scanned through the appropriate range. The content of the emission depends mainly on: the state of gassification of the liquid; ambient pressure and temperature; the acoustic intensity; and, to a lesser extent, on the drive frequency. It depends very little on the position of the hydrophone in the liquid.

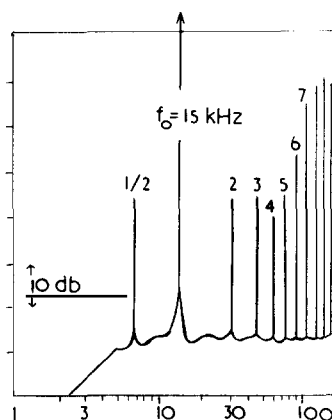


Fig. 58. Intensity–frequency plot of the acoustic emission from strongly cavitating gassy water; resonance frequency, $f_0 = 15 \text{ kHz}$.

10.1. Emission at low intensities: the stable cavitation regime

At very low excitation levels, the received signal consists of the fundamental only. The acoustic wave is partially scattered by any bubbles present, especially those of resonance size. At higher intensities, but still below the transient threshold, the larger bubbles begin to oscillate non-linearly. The spectrum then contains lines at harmonics of the drive frequency, up to high order. The second harmonic component is the most prominent. Its amplitude is proportional to the square of the fundamental, a relation that is predictable theoretically. Some of the harmonic content can usually be attributed to the inherent non-linearity of the medium, but the greater the liquid, the greater the contribution from oscillating bubbles. A low level of white noise is usually present, increasing to an impressive level as the transient threshold is approached. The half-order subharmonic (at $f/2$) may also appear in short bursts separated by long intervals and therefore at low average level. This subharmonic, and also the ultra-harmonics (at frequencies $(2n + 1)f/2$) are more likely at high drive frequencies where the bubbles responsible for them are small enough to remain in suspension in the liquid. Sometimes, lines are seen at frequencies unconnected with the drive frequency. It is thought that these are emissions from large bubbles of arbitrary size, shock-excited into radial vibration [172, 173].

There are three possible explanations for the weak sub-harmonic signals seen in stable cavitation fields:

(i) We have already seen how strong surface vibrations can be excited parametrically at half the drive frequency. These are pure distortion modes, not involving any area-change and therefore weakly coupled to the liquid. Sub-harmonic emissions from this cause have therefore only been detected by hydrophones held very close to the bubble surface [131].

(ii) Above a certain threshold, bubbles of such a size that they would resonate at a frequency close to $f/2$ are capable of emitting a strong signal at their own natural resonance frequency. Three theories have been published defining the threshold for this type of emission. Eller and Flynn [98] obtained the result:

$$P_T = 6P_0 \left[\left(\frac{\omega}{\omega_r} - 2 \right)^2 + \delta^2 \right]^{1/2} \quad (132)$$

giving $P_T = 6P_0\delta$ at $\omega = 2\omega_r$, where δ is the total damping coefficient for the radial motion. Safar [169] published the result for $\omega = 2\omega_r$:

$$P_T = 6\gamma P_0 \delta_\mu. \quad (133)$$

Nayfeh and Saric [171] obtained:

$$P_T = 6P_0 [(\omega/\omega_r - 2)^2 + (kR_r)^2]^{1/2} \quad (134)$$

giving $P_T = 6P_0(kR_r)$ at $\omega = 2\omega_r$, where (kR_r) is just the radiation damping of the bubble at frequency ω_r . Nayfeh and Saric obtained their result directly from the equations describing the natural amplitude and phase of the radial pulsations of the bubble. Eller and Flynn and also Safar derived theirs by examining stability after applying a perturbation to a steady-state solution. Strong emissions at the sub-harmonic frequency have been measured when bubbles of the appropriate size are present. In the work reported in [172, 173] bubbles of the correct size were seeded into the liquid using a mechanical bubble-generator. The bubbles were allowed to rise under gravity in the acoustic field past a small

hydrophone, and the filtered response recorded on either a chart-recorder or c.r.o. time-base. Examples of the emission at the half-order subharmonic for air bubbles in water are reproduced in fig. 59. Here, the drive frequency was 89 kHz and the total time-base duration is about 20 000 cycles (0.22 sec). The excitation level was not known accurately, but could not have been far above the thresholds indicated by the above theories, that is, well below the transient threshold. The photographs show that a single bubble can sometimes give a series of emissions, each lasting many cycles, separated by inactive periods. These traces are quite different in appearance from those obtained when a bubbly liquid is allowed to cavitate naturally. Then the subharmonic response appears as a mass of short overlapping transients. If a bubble of the appropriate size attached itself to the surface of the measuring hydrophone, the emission would continue long enough to record signals over a range of increasing drive levels. Figure 60 records the results of such measurements. In this case the liquid was a 90% glycerol-water mixture and the indicated threshold for the emission is appreciably greater than suggested by the formula (133). Eventually, the bubble detached itself. Flynn [1] has published solutions to the non-linear equation of motion (14) for bubbles satisfying the sub-harmonic condition $\omega = 2\omega_r$. In the example shown in fig. 61 the bubble was driven below, but close to, the transient threshold. The oscillation at the sub-harmonic frequency continues for many cycles, building to large amplitude. In this case, the bubble eventually collapses as a transient when R_m approaches $2R_0$.

(iii) When the liquid medium is sufficiently compressible and non-linear, a signal at $f/2$ may be generated by a parametric amplification process, even in the absence of bubbles [174]. This sometimes occurs in water left standing long enough to remove bubbles of the size required to excite the oscillations described in (ii).

There are two possible explanations for the white-noise signals obtained from stable cavitation:

(i) Under highly non-linear conditions most of the emission consists of discrete pulses containing harmonics of the fundamental, repeated at the excitation frequency. These pulses are emitted during the compression phase of the bubble oscillation. Figure 12 illustrates how, under these conditions, the bubble may be shocked into oscillating for a time at its own natural frequency during each cycle of the acoustic pressure. However, the sequence is never repeated exactly on successive cycles. The received signal will then appear as noise. However, this noise will have a structure. Its frequency-dependence will

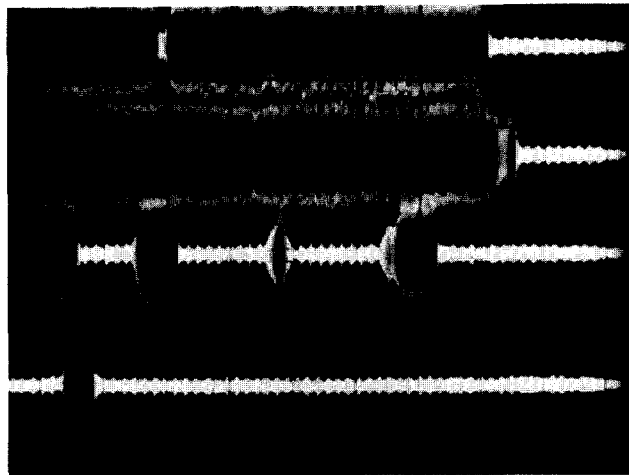


Fig. 59. Signals received at $f/2$ from water seeded with air bubbles of the appropriate size; $f = 89$ kHz; total time-base length = 0.22 sec.

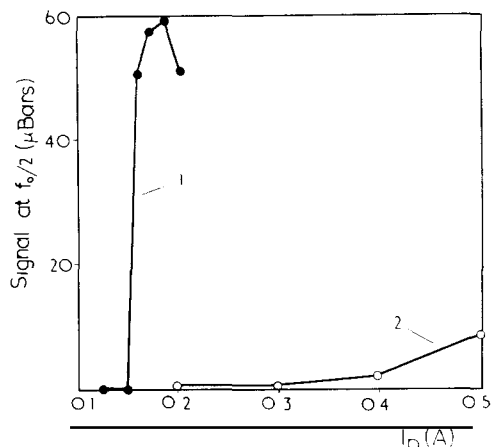


Fig. 60. Signal at $f_0/2$ emitted from a single bubble of the appropriate size caught on the hydrophone, as a function of the transducer excitation current (I_D); $f_0 = 38$ kHz and the liquid is 90% glycerol-water; 1 - Bubble trapped on the hydrophone; 2 - Bubble removed.

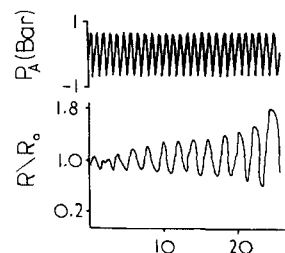


Fig. 61. Unstable motion of a cavity driven at twice its resonance frequency; $R_0 = 2.6 \times 10^{-2}$ cm; $P_A = 0.75$ Bar; $P_0 = 1$ Bar; $f = 24.5$ kHz.

be related to the sizes of the small bubbles responsible for it. The recorded noise-distribution then gives some indication of the size-distribution of bubble-nuclei in the liquid.

(ii) The emission of micro-bubbles during strong surface-wave activity may result in a low level of random noise [104].

10.2. Emission at high intensities: transient cavitation regime

The emission spectrum changes dramatically as soon as transient cavitation starts. There is a sudden increase in intensity of the subharmonics and their harmonics. Many workers have recorded and discussed these emissions [167, 168, 172 and 175]. The intensity of the white noise also increases rapidly with increasing drive level. The harmonic output, as recorded on the hydrophone, is usually weaker, an effect that can be explained in terms of scattering by the developing bubble screen which shields the hydrophone from radiation emanating from stable sources. Measurements reproduced in fig. 62 refer to ordinary tap-water left standing long enough to remove the larger bubbles. Under the conditions of the experiment, the recorded transducer current was almost proportional to the acoustic field pressure. The subharmonic signal comes in very abruptly at the transient threshold. Up to this point the white noise is at a low level, but beyond the threshold increases linearly with increasing excitation pressure. In fig. 63 the second and third harmonic components of the received signal are displayed along with the subharmonic. Again, tap-water was used, freed of large bubbles. The second harmonic is much the strongest signal, increasing approximately as the square of the drive pressure until the transient threshold is reached, when it levels off. For similar conditions, that is, air-saturated water left standing for several hours, the frequency spectrum has the appearance shown in fig. 64. This covers only the range below the excitation frequency, 40 kHz. The excitation pressure was 1.5 times the transient threshold pressure. At this level, the half-order sub-harmonic is always prominent, showing clearly above the noise. This trace also shows a response at the one-third-order subharmonic (at about 13 kHz). This is rarely seen at such a low drive frequency, and must have been due to the chance presence of a large bubble of the appropriate size.

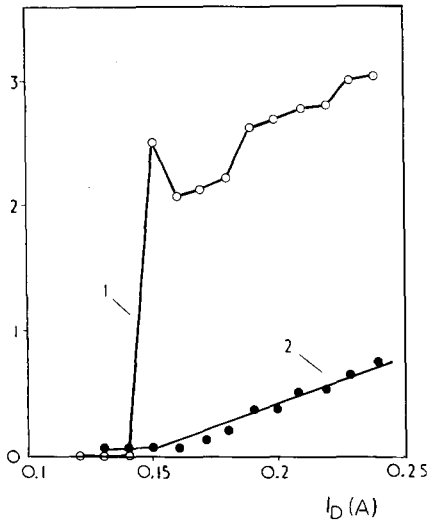


Fig. 62. Emission at: 1—the half-order subharmonic; and 2—white noise as functions of the transducer excitation current (I_D).

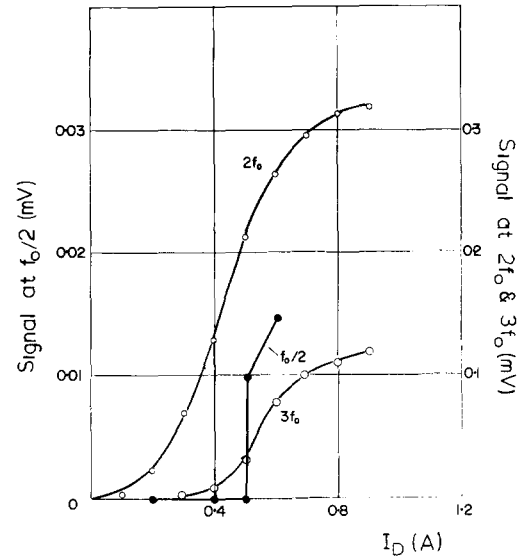


Fig. 63. Emission at $f/2$, $2f$ and $3f$ in air-saturated water; $f = 181$ kHz.

There are two possible explanations of the strong sub-harmonic signal emitted by transient cavities:

(i) We have already seen that situations often occur where bubbles have insufficient time to collapse completely before the end of the acoustic cycle. In such cases, they may complete their collapse as transients at the second, or subsequent, pressure maximum. If the bubble remains intact, this motion will become periodic. This behaviour is seen clearly in the theoretical $R-t$ curves of fig. 11. Even very small bubbles, well below resonance size, will behave in this way. The bubbles eventually collapse as transients and the intensity of the sub-harmonic emission may be comparable with that at the fundamental. Clearly, the probability of exciting any sub-harmonic decreases as the order of the sub-harmonic increases, and this is confirmed from observation.

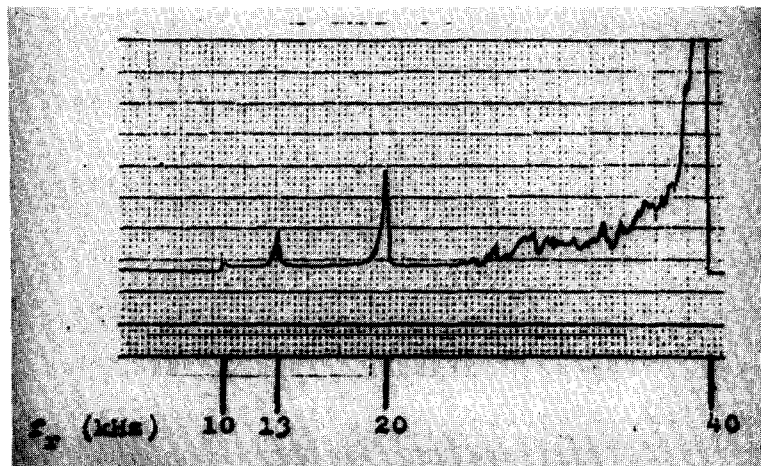


Fig. 64. Frequency spectrum for cavitating air-saturated water. P_A is approximately 1.5 times the transient cavitation threshold; $f = 40$ kHz.

(ii) The type of motion illustrated in fig. 61 can be regarded as a near-threshold condition. The bubble-motion is part stable, part transient. In this case, though, the bubble must be near the appropriate size for resonance at $f/2$. As the sudden increase in response at the half-order subharmonic occurs at, or very near, the transient threshold, it is now generally regarded as the best available threshold indicator [166–168].

The shock wave generated by a collapsing transient and picked up by the hydrophone would in theory be recorded by the wave analyser as an infinite series of harmonic components, decreasing in amplitude as f^{-2} at high frequencies. However, in practice, the higher frequency components will not reach the hydrophone, and with many bubbles emitting, the remainder will contribute to the general noise.

10.3. Recent computations relating to acoustic emission from cavitation fields

By examining many solutions of eq. (14), Lauterborn [175] has concluded that when bubbles are driven below their natural resonance frequencies, the thresholds for the $3f/2$ and $5f/2$ emissions are actually lower than for $f/2$. This suggests that the $3f/2$ and $5f/2$ emissions are more fundamental than $f/2$. In any case, their thresholds being somewhat lower, Lauterborn suggests that it would be logical to use them, rather than the $f/2$ threshold, to indicate transient cavitation.

In another recent paper, Lauterborn and Haussmann [176] have studied the way the cavitation noise spectrum builds up as a function of time. Special attenuation was paid to emissions at the frequencies $(2n + 1)f/2$. Results for aerated water at $f = 20$ kHz show that these emissions are all generated at nearly the same amplitude, with similar time-dependence.

References

- [1] H.G. Flynn, *Physical Acoustics*, Vol. 1B, ed. W.P. Mason (Academic Press, 1964) Chapter 9, pp. 57–172.
- [2] L.D. Rozenberg, *High Intensity Ultrasonic Fields* (Plenum Press, New York, 1971) Parts IV–VI, pp. 203–419.
- [3] D. Ross, *Mechanics of Underwater Noise* (Pergamon Press, New York, 1976).
- [4] R.T. Knapp, J.W. Daly and F.G. Hammitt, *Cavitation* (McGraw Hill, 1970).
- [5] D.-Y. Hsieh, *J. Basic Engineering* 87D (1965) 991.
- [6] W.H. Besant, *Hydrostatics and Hydrodynamics* (C.U.P., London, 1859).
- [7] Lord Rayleigh, *Phil. Mag.* 34 (1917) 94.
- [8] M.S. Plesset, *Trans. A.S.M.E. J. App. Mech.* 16 (1949) 228.
- [9] F.G. Blake Jr, Technical Memo No 12, Acoustics Research Laboratory, Harvard University, U.S.A. (1949).
- [10] B.E. Noltingk and E.A. Neppiras, *Proc. Phys. Soc. B* (London) 63B (1950) 674.
- [11] E.A. Neppiras and B.E. Noltingk, *Proc. Phys. Soc. B* (London) 64B (1951) 1032.
- [12] S. Chapman and T.G. Cowling, *The Mathematical Theory of Non-uniform Gases* (C.U.P., London, 1958) Chapter 8.
- [13] R. Hickling and M.S. Plesset, *The Physics of Fluids* 7 (1964) 7.
- [14] H. Poritsky, *Proc. 1st U.S. National Congress in Applied Mechanics* (A.S.M.E.) (1952) 813.
- [15] C. Herring, NDRC Report C-4-sr 10-010, Columbia University, U.S.A. (1941).
- [16] L. Trilling, *J. App. Phys.* 23 (1952) 14.
- [17] F.R. Gilmore, California Institute of Technology Hydrodynamics Laboratory Report No 26-4 (1952).
- [18] C. Hunter, *J. Fluid Mechanics* 8 (1960) 241.
- [19] J.G. Kirkwood and H.A. Bethe, Office of Science Research and Development Report No 558, U.S.A. (1942).
- [20] H. Pfriem, *Akust. Zhur.* 5 (1940) 202.
- [21] Z. Saneyoshi, *Electro-technical Journal* (Japan) 5 (1941) 49.
- [22] E. Meyer and K. Tamm, *Akust. Zhur.* 4 (1939) 145.
- [23] H. Lauer, *Acustica* 1 (1951) AB 12.
- [24] M.L. Exner, *Acustica* 1 (1952) AB 25.

- [25] M.L. Exner and W. Hampe, *Acustica* 3 (1953) 67.
- [26] H. Haeske, *Acustica* 6 (1956) 266.
- [27] M.S. Plesset and D.-Y. Hsieh, *The Physics of Fluids* 3 (1960) 882.
- [28] C. Devin, *J. Acoust. Soc. Am.* 31 (1959) 1654.
- [29] S.A. Zwick, *J. Mathematics and Physics* 37 (1958) 246.
- [30] M.S. Plesset and S.A. Zwick, *J. App. Phys.* 23 (1952) 95.
- [31] P.S. Epstein and M.S. Plesset, *J. Chem. Phys.* 18 (1950) 1505.
- [32] E.N. Harvey et al., *J. Cellular Comp. Physiol.* 24 (1944) 1.
- [33] M.D. Rosenberg, Technical Memo No 25, Acoustics Research Laboratory, Harvard University, U.S.A. (1951).
- [34] L. Pode, David Taylor Model Basin, Washington, U.S.A. Report No 854 (1953).
- [35] D.-Y. Hsieh and M.S. Plesset, *J. Acoust. Soc. Am.* 33 (1961) 359.
- [36] O.A. Kapustina, *Soviet Physics – Acoustics* 15 (1970) 427.
- [37] M.H. Safar, *J. Acoust. Soc. Am.* 43 (1968) 1188.
- [38] A.I. Eller and H.G. Flynn, *J. Acoust. Soc. Am.* 37 (1965) 493.
- [39] A.I. Eller, *J. Acoust. Soc. Am.* 52 (1972) 1447.
- [40] M. Strasberg, *J. Acoust. Soc. Am.* 31 (1959) 163.
- [41] M. Strasberg, *J. Acoust. Soc. Am.* 33 (1961) 3591.
- [42] A.I. Eller, *J. Acoust. Soc. Am.* 46 (1969) 1246.
- [43] R.K. Gould, *J. Acoust. Soc. Am.* 56 (1974) 1740.
- [44] O.A. Kapustina, *Soviet Physics – Acoustics* 11 (1965) 94.
- [45] O.A. Kapustina and Yu.G. Statnikov, *Soviet Physics – Acoustics* 13 (1968) 327.
- [46] M.S. Plesset, *Cavitation in Real Liquids*, ed. Robert Davies (Elsevier Press, 1964) pp. 1–18.
- [47] G.T. Trammell, *J. App. Phys.* 33 (1962) 1662.
- [48] V.A. Akulichev et al., Communication of the Joint Institute for Nuclear Research, Dubna, U.S.S.R. (1970) Paper No 13-5327.
- [49] J.R. Shadley, University of Houston, Ph.D. Dissertation (1970).
- [50] T. Wang and D.-Y. Hsieh, *J. Acoust. Soc. Am.* 50 (1971) 115A.
- [51] R.D. Finch and E.A. Neppiras, *J. Acoust. Soc. Am.* 53 (1973) 1402.
- [52] E.A. Neppiras and R.D. Finch, *J. Acoust. Soc. Am.* 52 (1972) 335.
- [53] P.L. Marston and D.B. Greene, *J. Acoust. Soc. Am.* 64 (1978) 319.
- [54] V.A. Akulichev et al., *Proc. VI Intern. Symp. on Non-linear Acoustics*, Vol. 2 (Moscow State University, 1976) pp. 16–25.
- [55] R.D. Finch and E.A. Neppiras, *Proc. Ultrasonics Intern. Conf. (I.P.C. Sci. and Tech. Press Ltd, 1973) Paper No 3.1.*
- [56] R.D. Finch et al., *Phys. Rev.* 134 (1964) 1425A.
- [57] R.D. Finch and T.G.J. Wang, *J. Acoust. Soc. Am.* 39 (1966) 511.
- [58] P.D. Jarman and K.J. Taylor, *J. Acoust. Soc. Am.* 39 (1966) 584.
- [59] R.C.A. Brown et al., *Nature* 220 (1968) 1177.
- [60] T. Vroulis, E.A. Neppiras and R.D. Finch, *J. Acoust. Soc. Am.* 59 (1976) 255.
- [61] A. Mosse and R. D. Finch, *J. Acoust. Soc. Am.* 49 (1971) 156.
- [62] R.D. Finch et al., *J. Acoust. Soc. Am.* 40 (1966) 211.
- [63] R.D. Finch, *The Physics of Fluids* 12 (1969) 1775.
- [64] R.D. Finch and M.L. Chu, *Phys. Rev.* 161 (1967) 202.
- [65] P.M. McConnell et al., *Phys. Rev. A* 1 (1970) 411.
- [66] V.I. Ginsberg and L.P. Pitaevsky, *Soviet Physics – JETP* 7 (1958) 858.
- [67] D.-Y. Hsieh, Division of Engineering and Applied Science, California Institute of Technology, U.S.A. Report No 13-5327.
- [68] V.A. Akulichev and Y.Y. Boguslanskii, *Soviet Physics – JETP* 35 (1972) 1012.
- [69] C.W. Smith et al., *J. Acoust. Soc. Am.* 61 (1976) 378.
- [70] R.F. Carey, J.A. Rooney and C.W. Smith, *Phys. Lett.* 65A (1978) 311.
- [71] C. Herring, Office of Naval Research and Development, Washington, U.S.A., Report No 236 (1941).
- [72] J.J. Stoker, *Non-linear Vibrations in Mechanical and Electrical Systems* (Wiley Interscience, New York, 1950) Chapter 1.
- [73] W. Guth, *Acustica* 6 (1956) 532.
- [74] V.A. Akulichev, *High Intensity Ultrasonic Fields* (Plenum Press, 1971) Part IV.
- [75] W. Lauterborn, *Acustica* 20 (1968) 14; 20 (1968) 105; 20 (1968) 370; 22 (1969–70) 48; 22 (1969–70) 35; 31 (1974) 51.
- [76] V.A. Akulichev, *Soviet Physics – Acoustics* 13 (1967) 149.
- [77] M.G. Sirotyuk, *Soviet Physics – Acoustics* 7 (1961) 405.
- [78] R.B. Robinson and R.H. Buchanan, *Proc. Phys. Soc. (London)* B69 (1956) 981.
- [79] M.I. Borotnikov and R.I. Solukhin, *Sov. Phys. – Acoustics* 10 (1964) 28.
- [80] V.A. Akulichev, *Sov. Phys. – Acoustics* 13 (1968) 455.
- [81] H.G. Flynn, *J. Acoust. Soc. Am.* 57 (1975) 1379.
- [82] M.G. Sirotyuk, *High Intensity Ultrasonic Fields*, ed. L.D. Rozenberg (Plenum Press, 1971) pp. 290–294.

- [83] A.J.R. Schneider, California Institute of Technology Ph.D. Thesis (1949).
- [84] G.A. Khoroshev, *Sov. Phys. – Acoustics* 9 (1963) 275.
- [85] R.D. Ivany and F.G. Hammit, *Trans. A.S.M.E., J. Basic Engineering* 87D (1965) 977.
- [86] R.S. Brand, Technical Report No 1, School of Engineering, University of Connecticut, U.S.A. (1962).
- [87] T.B. Benjamin, University of Cambridge, England, Ph.D. Thesis (1954).
- [88] V.A. Akulichev, *High Intensity Ultrasonic Fields*, ed. L.D. Rozenberg (Plenum Press, 1971) pp. 239–257.
- [89] W. Lauterborn, *Acustica* 22 (1970) 189.
- [90] M. Ceschia and G. Iernetti, *Acustica* 29 (1973) 127.
- [91] M. Ceschia, G. Iernetti and R. Nabergoj, *Finite Amplitude Wave Effects in Fluids* (I.P.C. Sci. and Tech. Press Ltd., 1974) pp. 263–267.
- [92] M. Ceschia, G. Iernetti and R. Nabergoj, *Proc. VI Intern. Symp. on Non-linear Acoustics* (Moscow State University, 1976) Vol. 2, pp. 71–79.
- [93] R.E. Apfel, *Acoustic Cavitation in: Ultrasonics* (series: Methods of Experimental Physics) ed. P. Edmonds (1980, in the press).
- [94] M.S. Plesset and T.P. Mitchell, *Q. App. Math.* 12 (1952) 306.
- [95] V.I. Sorokin, *Sov. Phys. – Acoustics* 3 (1957) 281.
- [96] W. Eisenmenger, *Acustica* 9 (1959) 327.
- [97] W.L. Nyborg and D.E. Hughes, *J. Acoust. Soc. Am.* 42 (1967) 891.
- [98] A.I. Eller and H.G. Flynn, *J. Acoust. Soc. Am.* 46 (1969) 722.
- [99] M. DeGrois and B. Badilian, *C.R. Acad. Sci.* 254 (1962) 231 and 257 (1963) 2409.
- [100] M. DeGrois and B. Badilian, *Proc. 4th Intern. Congress on Acoustics, Copenhagen* (1962) Paper No K 44.
- [101] M. DeGrois, *Ultrasonics* 5 (1966) 38.
- [102] L. Chincholle, *Proc. 9th Intern. Congress on Acoustics, Madrid* (1977) Paper No K 56.
- [103] W. Lauterborn and G. Haussmann, *Proc. 9th Intern. Congress on Acoustics, Madrid* (1977) Paper No L 36.
- [104] E.A. Neppiras and E.E. Fill, *J. Acoust. Soc. Am.* 46 (1969) 1264.
- [105] G.I. Taylor, *Proc. Royal Soc. (London)* A201 (1950) 192.
- [106] M.S. Plesset, *J. App. Phys.* 25 (1954) 96.
- [107] D.J. Lewis, *Proc. Royal Soc. (London)* A202 (1950) 81.
- [108] M.S. Plesset and T.P. Mitchell, *Q. App. Math.* 13 (1956) 419.
- [109] M. Kornfeld and L. Suvarov, *J. App. Phys.* 15 (1944) 495.
- [110] T.B. Benjamin and F. Ursell, *Proc. Royal Soc. (London)* A225 (1954) 505.
- [111] T.B. Benjamin, *Cavitation in Real Liquids*, ed. R. Davies (Elsevier Press, 1964) pp. 164–180.
- [112] W.L. Nyborg, *Ultrasound, its Application in Medicine and Biology*, ed. F.J. Fry (Elsevier Press, 1978) Chapter IV.
- [113] D.-Y. Hsieh, *Finite Amplitude Wave Effects in Fluids* (I.P.C. Sci. and Tech. Press Ltd, 1974) pp. 220–226.
- [114] D.-Y. Hsieh, *J. Acoust. Soc. Am.* 56 (1974) 392.
- [115] M. Strasberg and T.B. Benjamin, *J. Acoust. Soc. Am.* 30 (1958) 697A.
- [116] A.I. Eller and L.A. Crum, *J. Acoust. Soc. Am.* 47 (1970) 762.
- [117] R.K. Gould, *J. Acoust. Soc. Am.* 40 (1966) 219.
- [118] R.T. Knapp and A. Hollander, *Trans. A.S.M.E.* 70 (1948) 419.
- [119] A.T. Ellis, California Institute of Technology Hydrodynamics Laboratory Report No 21-12 (1952).
- [120] A.T. Ellis, California Institute of Technology Hydrodynamics Laboratory Report No E-115 (1965).
- [121] C.F. Naude and A.T. Ellis, *Proc. A.S.M.E. J. Basic Engineering* 83D (1961) 648.
- [122] V.A. Akulichev, *High Intensity Ultrasonic Fields*, ed. L.D. Rozenberg (Plenum Press, 1971) pp. 233–237.
- [123] W. Lauterborn, *Acustica* 31 (1974) 51.
- [124] W. Lauterborn et al., *Acustica* 26 (1972) 170.
- [125] K. Hinsch et al., *Finite Amplitude Wave Effects in Fluids* (I.P.C. Sci. and Tech. Press Ltd, 1974) pp. 240–244.
- [126] W. Lauterborn, *App. Phys. Letters* 21 (1972) 27.
- [127] W. Lauterborn, *Optics Communications* 21 (1977) 67.
- [128] K. Hinsch, *Proc. VI Intern. Symp. on Non-linear Acoustics* (Moscow State University, 1976) pp. 26–34.
- [129] M.S. Plesset and R.B. Chapman, *J. Fluid Mechanics* 47 (1971) 283.
- [130] L.A. Crum, *Proc. Conf. on Acoustic Cavitation, Bournemouth, U.K.* (Institute of Acoustics, U.K., 1977) pp. 10–16.
- [131] D.L. Storm, *Finite Amplitude Wave Effects in Fluids* (I.P.C. Sci. and Tech. Press Ltd, 1974) pp. 234–239.
- [132] W.I. Galloway, *J. Acoust. Soc. Am.* 26 (1954) 849.
- [133] J.E. Barger, Technical Memo No 57, Acoustics Research Laboratory, Harvard University (1964).
- [134] D. Lieberman, *Physics of Fluids* 2 (1959) 466.
- [135] G. Willard, *J. Acoust. Soc. Am.* 25 (1957) 667.
- [136] E.A. Neppiras and W.T. Coakley, *J. Sound and Vibration* 45 (1976) 341.
- [137] K. Yosioka and Y. Kawasima, *Acustica* 5 (1955) 167.
- [138] S.A. Elder, *J. Acoust. Soc. Am.* 31 (1959) 54.
- [139] W.L. Nyborg, *J. Acoust. Soc. Am.* 30 (1958) 329.
- [140] W.L. Nyborg and D.L. Storm, *Proc. 8th Intern. Congress on Acoustics, London* (1974) p. 470.

- [141] L.D. Rozenberg, editor, Sources of High-Intensity Ultrasound (Plenum Press, 1969) Vols. I and II.
- [142] W. Lauterborn, K. Hinsch and F. Bader, *Acustica* 26 (1972) 170.
- [143] K. Hinsch, F. Bader and W. Lauterborn, Finite Amplitude Wave Effects in Fluids (I.P.C. Sci. and Tech. Press Ltd, 1974) pp. 240–244.
- [144] K.J. Ebeling and W. Lauterborn, *Optics Communications* 21 (1977) 67.
- [145] E.V. Romanenko, *Akust. Zhur.* 3 (1957) 342.
- [146] I. Koppelman, *Acustica* 2 (1952) 92.
- [147] E.A. Neppiras, *J. Acoust. Soc. Am.* 45 (1969) 587.
- [148] J.P. Weight and A.J. Hayman, The Evaluation and Calibration of Ultrasonic Transducers (I.P.C. Sci. and Tech. Press Ltd, 1978) pp. 24–32.
- [149] A.P.J.B. Walton and R.C. Chivers, The Evaluation and Calibration of Ultrasonic Transducers (I.P.C. Sci. and Tech. Press Ltd, 1978) pp. 96–101.
- [150] F.E. Borgnis, *J. Acoust. Soc. Am.* 25 (1953) 546.
- [151] M.G. Sirotyuk, *Soviet Physics – Acoustics* 10 (1965) 398.
- [152] E.A. Neppiras and W.T. Coakley, Proc. VI Intern. Symp. on Non-linear Acoustics (Moscow State University, 1976) Vol. 2, pp. 80–91.
- [153] Y. Kikuchi and H. Shimizu, *J. Acoust. Soc. Am.* 31 (1959) 1.
- [154] L.D. Rozenberg and M.G. Sirotyuk, *Soviet Physics – Acoustics* 6 (1961) 477.
- [155] E.A. Neppiras, *Soviet Physics – Acoustics* 8 (1962) 4.
- [156] K.A. Morch, Proc. Conf. on Acoustic Cavitation, Bournemouth, U.K. (Institute of Acoustics, 1977) pp. 62–70.
- [157] V.A. Akulichev and L.D. Rozenberg, *Soviet Physics – Acoustics* 11 (1965) 246.
- [158] L.D. Rozenberg, editor, High Intensity Ultrasonic Fields (Plenum Press, 1971) Part VI, pp. 353–356.
- [159] W.L. Nyborg and D.L. Miller, Proc. Conf. on Acoustic Cavitation, Bournemouth, U.K. (Institute of Acoustics, 1977) pp. 16–18.
- [160] D.L. Miller and W.L. Nyborg, *J. Acoust. Soc. Am.* 61 (1977) S93A.
- [161] Nuclepore Corporation, 7035 Commerce Circle, Pleasanton, California 94566, U.S.A.
- [162] D.E. Weston, *J. Acoust. Soc. Am.* 39 (1966) 316.
- [163] R. Esche, *Acustica* 2 (1952) AB208.
- [164] L. Bohn, *Acustica* 7 (1957) 201.
- [165] K. Negeshi, *J. Phys. Soc. Japan* 16 (1961) 1450.
- [166] E.A. Neppiras and J. Parrott, Proc. 5th Intern. Congress on Acoustics, Liege (1965) Paper No D 51.
- [167] E.A. Neppiras, *I.E.E.E. Trans. Sonics and Ultrasonics* SU-15, No. 2 (1968) 81.
- [168] P.W. Vaughan, *J. Sound and Vibration* 7 (1968) 236.
- [169] M.H. Safar, *J. Phys. Dyn. Appl. Phys.* 3 (1970) 635.
- [170] A. Prosperetti, *J. Acoust. Soc. Am.* 56 (1974) 878.
- [171] A.S. Nayfeh and W.S. Saric, Finite Amplitude Wave Effects in Fluids (I.P.C. Sci. and Tech. Press Ltd, 1974) pp. 272–276.
- [172] E.A. Neppiras, *J. Acoust. Soc. Am.* 45 (1969) 587.
- [173] E.A. Neppiras, *J. Sound and Vibration* 10 (1969) 176.
- [174] D.G. Tucker, *J. Sound and Vibration* 2 (1965) 429.
- [175] W. Lauterborn, *J. Acoust. Soc. Am.* 59 (1976) 283.
- [176] W. Lauterborn and G. Haussmann, Proc. 9th Intern. Congress on Acoustics, Madrid (1977) Paper No L36.